Pregeometric Relational Dynamics and Universal Causal Structure

Spectral Constraints, Response Limits, Universal Null Calibration and Functional Consistency of Light

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Abstract

We investigate a programme in which quasilocality, spatial geometry and causal structure are treated as effective properties of a closed relational system. The unified manuscript collects four technical layers. First, finite weighted networks and multicomponent quadratic dynamics are analysed to identify spectral conditions for a common low-energy characteristic cone. Second, controlled response limits on prescribed chains show how properly normalised microscopic tails disappear outside enlarged continuum cones, while coarse-graining, locality and stability of cone universality remain separate requirements. Third, universal null calibration is formulated as a physical common-cone condition rather than a numerical convention for c; reconstruction invariance, spectral length-time correspondence and a quantum continuum limit are treated with explicit hypotheses. Fourth, a Maxwell control sector is used to test which optical observables are linked by one functional structure, including stress-energy, radiation pressure, gauge reduction, commutator support and Hadamard-type state control. The conclusions are conditional: neither a manifold, a time orientation, a global causal order, nor a stable electromagnetic sector is derived from arbitrary neutral relational data.

Keywords: pregeometry; relational dynamics; quasilocality; common cone; response convergence; universal null calibration; Weyl algebra; Maxwell theory; stress-energy; radiation pressure; reconstruction.

Part I. Relational and spectral foundation

1. Problem Statement and Main Results

The starting physical idea is that an object may be a stable mode of organization of an entire system, while geometrical separation may be an effective relation among such modes. The reality of an observed phenomenon does not require its independence from all other degrees of freedom. This process-oriented viewpoint is used here as a guide for choosing variables and questions. Its physical validity must be established by explicit dynamics and observable consequences.

The central question is formulated in terms of causal structure: under what conditions do different physical excitations reproduce one and the same local characteristic cone? The dimensional number c is secondary here. In local coordinates it specifies the relative calibration of spatial and temporal units for a common cone, and its numerical value depends on the chosen units. The substantive problem is to explain the universality of the propagation structure and its stability under changes in the state of the system.

Let R denote a neutral relation of interdependence. It is not assumed in advance to be a causal order. The program seeks to connect R first to stable quasilocal sectors, then to oriented propagation, and, in a suitable limit, to a Lorentzian conformal class [g]. None of these mappings follows from the word “relational” itself. In particular, a finite graph without an embedding does not define physical distance, although it already contains combinatorial adjacency and a chosen set of degrees of freedom. The class studied below is therefore a nonmetric model with explicit structural assumptions.

This work implements a first technical stage of that program. Its core is a spectral analysis of multicomponent dynamics and a test of which mechanisms actually yield a common low-energy cone. We obtain a no-go result for speed equalization by a common network operator alone, a criterion for spectral quasilocality, a universality condition on the massless subspace, an exact two-component dispersion relation, and a closed balance law coupling fields to dynamical relations. All statements are proved within an explicitly specified class of finite models; numerical examples check the formulas and illustrate the size of the corrections.

We do not claim to derive all observed physics from R. A positive kinetic form, Hamiltonian structure, and the existence of a suitable spatial continuum limit remain assumptions behind the corresponding results. The contribution of this paper is to make the original program more precise and more restrictive: conditions for success become testable matrix and spectral conditions, while several natural but insufficient mechanisms are excluded by exact arguments. Standard methods of perturbation theory, elimination of gapped modes, and internal symmetries are used with their roles stated explicitly; no claim of priority for those methods is intended.

Epistemic caution is also appropriate in cosmology. A reconstructed history depends on both the data and the theory used to interpret them; a singular limit of a classical description does not by itself establish the ontology of an initial state. This is not an objection to the well-supported hot cosmological history. No alternative early-Universe history is used here as evidence for the model.

2. Relation to Existing Work

Causal reconstruction theorems concern Lorentzian spacetimes subject to appropriate causal distinguishability and regularity conditions. Malament's result [1] demonstrates the deep relation between causal and topological structures. In causal set theory, order and counting information are taken as primitive data, and their connection to conformal geometry and volume has been developed in detail [2]. An arbitrary partial order, however, need not possess a manifoldlike limit. For the present program there is an additional question: how can an oriented order itself arise from an initially neutral relational structure?

Quantum graphity [3] provides a concrete framework for studying a transition from highly connected graph configurations to states exhibiting locality and low-dimensional behavior. It is therefore an important reference point for network dynamics. It does not replace a proof that an arbitrarily chosen evolution law for the weights generates an appropriate geometric phase, nor does it by itself establish coincidence of the characteristics of arbitrary field sectors.

Interactions are especially important for the universality of limiting speeds. In the models of Anber and Donoghue [4], distinct limiting speeds approach one another under renormalization-group flow. A generic logarithmic approach may be too slow, and faster power-law behavior has also been considered. In strongly coupled holographic models [5], deviations from Lorentzian form may be suppressed by a power of the ratio between infrared and ultraviolet scales. These results demonstrate the feasibility of such mechanisms in specific theories, but the spacetime and field structures of those theories are already given.

Analogue gravity provides another useful control case. A two-component Bose–Einstein condensate [6] can support massless and massive modes with matched effective kinematics under special conditions. Thus the number of microscopic components, the number of massless modes, and the number of characteristic cones must be counted separately. This distinction is central to Section 6.

The relation between entanglement and geometric connectivity in Ref. [7] was obtained in a specific quantum-gravitational and holographic setting. It does not imply that mutual information between an arbitrary pair of subsystems is already a distance. Likewise, Lieb–Robinson-type bounds for harmonic and anharmonic lattices [8] constrain the propagation of disturbances but do not turn an exponentially small response into an exactly vanishing one. For unbounded harmonic variables, the chosen class of observables, for example Weyl operators, is essential.

In a 2026 study [9], spatially inhomogeneous couplings of an XXZ chain are programmed on a quantum processor, producing dynamics consistent with an effective curved metric. This recent example is useful for operational tests of propagation, but the geometric inhomogeneity is engineered through the couplings of the chain. It is not a demonstration that space emerges without prior structure.

Table 1. *Relation of the present program to selected existing approaches.*

Approach	What is assumed	What is obtained	Unresolved step for this program
Causal reconstruction [1, 2]	Causal and regularity conditions	Topological and conformal information	Origin of order from R
Quantum graphity [3]	Graph states and Hamiltonian	Signatures of a geometric phase	Universality of physical cones

Approach	What is assumed	What is obtained	Unresolved step for this program
Renormalization group [4, 5]	Interacting fields and scale dependence	Infrared convergence of limiting speeds	Joint derivation of geometry
Bose–Einstein condensates [6]	Medium and interaction parameters	Effective wave metrics	Protection of parameter matching
Entanglement geometry [7]	Specific quantum-gravitational description	Relation between correlations and connectivity	General reconstruction from correlations
Propagation bounds [8]	Local lattice dynamics	Control of spatial tails	Strict microcausality
Quantum simulation [9]	Programmable XXZ chain	Dynamics in an effective background	Dynamical selection of the background itself

For a subsequent transition toward gravity, the thermodynamic derivation of Einstein's equations [10] and the Brown–York quasilocal charges [11] are relevant. The former uses local causal horizons and thermodynamic assumptions; the latter are defined through the gravitational action and boundary data. Neither result can therefore be directly identified with a sum of vertex energies of a graph. The radiative-stability problem for Lorentz symmetry emphasized in Ref. [12] further prevents one from assuming that all unwanted effects are automatically suppressed at high energy.

3. Nonmetric Network, Change, and Closure

3.1. Variables and Conventions

Let V be a finite set of N vertices and W a symmetric matrix of nonnegative weights with $w_{ii} = 0$. Nonzero weights define undirected connectivity; the vertices are not embedded in physical space. Permutations of their labels must not change physical statements. The edge sum $\langle ij \rangle$ below includes each unordered pair exactly once. The graph Laplacian is

$$L = D - W, \quad D_{ii} = \sum_j w_{ij}, \quad z^T L z = \sum\langle ij\rangle\, w_{ij}(z_i - z_j)^2 \geq 0. \tag{1}$$

For a connected graph, ker L consists of constant vectors. The eigenvalues are ordered as $0 = \nu_0 < \nu_1 \leq \ldots \leq \nu_{n-1}$. The symbol ν always denotes the spectrum of the candidate spatial operator, not a dynamical frequency or a physical momentum.

At each vertex we introduce a real vector of field coordinates $\Phi_i \in \mathbb{R}^m$ with conjugate momentum p_i. Matrices $A = A^T > 0$, $B = B^T > 0$, and $U = U^T \geq 0$ act on component space. A defines the kinetic form, B the intervertex stiffness, and U the intravertex coupling. Their independence of the vertex label is an assumption of the model class considered here. The matrix U should not be confused with the vertex set V.

For a changing network it is convenient to write $w_e = \exp(y_e)$ and introduce a momentum Π_e conjugate to y_e. This parametrization preserves positivity of the weights. Exact zero weights may instead be represented by absent edges; on a complete graph, sufficiently small weights describe weak effective coupling. The choice of the set of possible edges is itself part of the model specification.

3.2. A Unified Hamiltonian

Consider the closed system

$$H = H_p + H_r, \tag{2}$$

$$H_p = \tfrac{1}{2}\sum_i p_i^T A^{-1} p_i + \tfrac{1}{2}\sum\langle ij\rangle\, w_{ij}(\Phi_i - \Phi_j)^T B(\Phi_i - \Phi_j) + \tfrac{1}{2}\sum_i \Phi_i^T U \Phi_i, \tag{3}$$

$$H_r = \sum_e \Pi_e^2/(2I_e) + V_r(y), \quad I_e > 0. \tag{4}$$

The potential V_r is assumed smooth and bounded below. If the full set of edges is allowed, equal I_e and a permutation-invariant V_r ensure independence from vertex labels. For example, a quadratic potential around a common value y_0 gives a well-defined finite dynamics. By itself, however, it does not select a low-dimensional geometry. A more complicated V_r requires an independent physical motivation rather than tuning to a desired metric.

Using the Hamiltonian-flow parameter τ, the equations of motion are

$$\dot{\Phi}_i = A^{-1} p_i, \quad \dot{p}_i = -\sum_j w_{ij} B(\Phi_i - \Phi_j) - U\Phi_i, \tag{5}$$

$$\dot{y}_e = \Pi_e / I_e, \quad \dot{\Pi}_e = -\tfrac{1}{2} w_e (\Delta\Phi_e)^T B \Delta\Phi_e - \partial V_r / \partial y_e. \tag{6}$$

Here $\Delta\Phi_e = \Phi_i - \Phi_j$ for the endpoints of edge e. The fields modify the forces acting on the relations, while the relations modify the propagation operator for the fields. Redistribution and relational restructuring are thus described by a single law. If H has no explicit dependence on τ, the total H is conserved. This follows from Hamiltonian structure, not merely from an ontological assertion of closure.

Positivity of A, B, and U excludes negative quadratic field energies on a fixed background. It does not guarantee stability of every moving background: an arbitrary V_r may possess unstable stationary points, and a nonstationary network may parametrically excite field modes. The dispersion analysis below therefore concerns a fixed background or a regime of controlled slow variation.

3.3. Reparametrization and Internal Clocks

Independence from an arbitrary rate of parametrization can be implemented with the parametrized action

$$S = \int d\lambda\, [\sum_i p_i^T \Phi_i' + \sum_e \Pi_e y_e' - n(\lambda)(H - E_0)]. \tag{7}$$

A prime denotes differentiation with respect to λ, and $n > 0$ is a Lagrange multiplier. Under $\lambda' = f(\lambda)$, $f' > 0$, the coordinates of the same history are relabeled and $n'(\lambda') = n(\lambda)d\lambda/d\lambda'$. Variation with respect to n gives $H = E_0$; the gauge choice $d\tau = n\, d\lambda$ returns Eqs. (5)–(6). The constant E_0 selects an energy hypersurface. This parametrization is not a derivation of physical time and is not equivalent to the local Hamiltonian constraints of general relativity.

For a chosen clock variable T(X) that is monotonic on a given segment of a history, the relational change of an observable O is locally

$$dO/dT = \{O,H\}/\{T,H\}, \quad \{T,H\} \neq 0. \tag{8}$$

A globally monotonic clock need not exist in a finite closed system. The choice of an internal process, its accuracy, its backreaction, and its relation to proper time of an effective metric are separate questions. Formal reparametrization invariance removes the arbitrary calibration of λ, but it does not generate a causal orientation from the symmetric W.

4. Distinguishability and Spectral Quasilocality

4.1. Why Homogeneous Ordering Does Not Yet Create Sectors

Consider a real order parameter ψ_i on a connected graph:

$$F(\psi) = a\sum_i \psi_i^2 + b\sum_i \psi_i^4 + \kappa\sum\langle ij\rangle w_{ij}(\psi_i - \psi_j)^2, \quad b > 0, \kappa > 0. \tag{9}$$

Proposition 1. If $a \geq 0$, the unique global minimum is $\psi = 0$. If $a < 0$, the global minima are $\psi_i = \psi_0$ for all i, where $\psi_0 = \pm\sqrt{(-a/(2b))}$. Thus this functional alone does not generate a stable partition of the vertices into distinct regimes.

Proof. For $a \geq 0$ all terms are nonnegative, and the quartic term vanishes only at $\psi = 0$. For $a < 0$ define $r^2 = -a/(2b)$. Then

$$F(\psi) + Na^2/(4b) = b\sum_i(\psi_i^2 - r^2)^2 + \kappa\sum\langle ij\rangle w_{ij}(\psi_i - \psi_j)^2 \geq 0. \tag{10}$$

The lower bound is attained by the constant configurations ±r. Equality requires identical ψ on every nonzero edge and hence, by connectedness, on the entire graph. For complex ψ, the corresponding set of minima has one common phase. □

A collective transition from zero to nonzero amplitude is therefore possible, but it preserves homogeneity. Structural separation requires additional ingredients: competing couplings, constraints, inhomogeneous stable states, or an independent restructuring of W. The mere presence of a symbol ψ in a Landau functional does not identify it with causal or geometric distinguishability.

4.2. Conditional Criterion for Stable Sectors

Let L_0 be the Laplacian of r disconnected internally connected blocks. Its kernel has dimension r and consists of vectors that are constant on each block. Assume all remaining eigenvalues are at least $\gamma > 0$. Interblock couplings add a positive Laplacian E:

$$L = L_0 + E, \quad E \geq 0, \quad \varepsilon = \|E\| < \gamma/2. \tag{11}$$

Proposition 2. The operator L has exactly r eigenvalues below γ/2; they lie in [0, ε], while all remaining eigenvalues are at least γ. If P_0 projects onto ker L_0 and P projects onto these r low modes of L, then

$$\|P - P_0\| \le \varepsilon/(\gamma - \varepsilon). \tag{12}$$

A proof is given in Appendix A. The condition produces a separation of mixing times: relaxation within the blocks occurs for $s \gtrsim \gamma^{-1}$, whereas interblock differences persist for $s \ll \varepsilon^{-1}$. Here s is the parameter of an auxiliary heat kernel, not a physical clock. Over the intermediate range, block averages are stable effective variables.

The robust object is the low-dimensional subspace itself, not an arbitrarily selected basis of its eigenvectors. For degenerate low eigenvalues that basis may rotate. Estimate (12) does not imply uniqueness of any clustering algorithm. Moreover, the block form of L_0 is an assumption of the theorem, not a dynamically derived outcome of Eqs. (5)–(6). The dynamical emergence of such a spectral separation remains a problem for the choice and analysis of V_r.

For two complete graphs of n vertices each, with internal weight 1 and uniform interblock weights η/n, the spectrum is exact:

$$\mathrm{Spec}\, L = \{0, 2\eta, n + \eta\}, \quad \text{the multiplicity of } n + \eta \text{ is } 2n - 2. \tag{13}$$

For n = 8 and η = 0.04 this gives 0, 0.08, and 8.04, with multiplicity 14 for the last value. The network is connected and has a well-separated intersector mode, but it is not a model of three-dimensional space.

4.3. Spectral Dimension and Limits of the Diagnostic

Define

$$P(s) = N^{-1}\,\mathrm{Tr}\exp(-sL), \quad d_s(s) = 2s\,[\textstyle\sum_n \nu_n e^{-s\nu_n}]/[\sum_n e^{-s\nu_n}]. \tag{14}$$

The exponentials in Eq. (14) contain the product $s\nu_n$. The equivalent definition $d_s = -2\, d\ln P/d\ln s$ avoids numerical differentiation when the spectrum is known. For a finite connected graph, d_s tends to zero both as $s \to 0$ and as $s \to \infty$. The physically informative feature is therefore a stable intermediate scaling window, not a single limiting or accidental value of d_s.

For a cycle of 256 vertices, the calculation gives $d_s(10) \approx 1.01318$; for the Cartesian product of two cycles of 32 vertices each, $d_s(10) \approx 2.02636$. In these examples the geometric class of the graph is chosen in advance. They validate the diagnostic but do not demonstrate dynamical selection of dimensionality. Agreement among spectral, volume-growth, and topological dimensions requires additional conditions. A spatial $d_s \approx 3$ also does not establish either Lorentzian signature or a 3+1-dimensional spacetime.

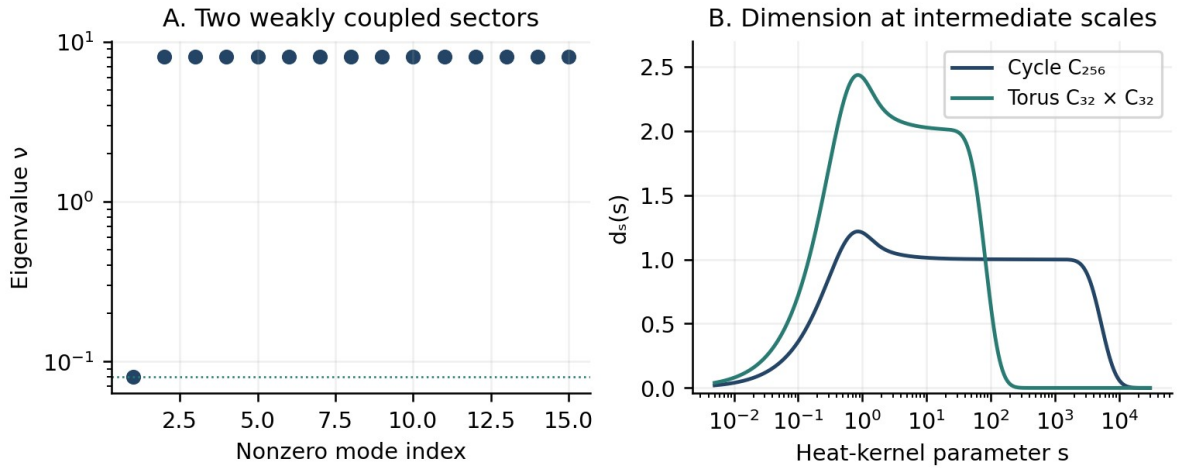


Figure 1. *Spectral separation and scale-dependent diagnostics. A: nonzero eigenvalues of two weakly coupled complete graphs with n = 8 and η = 0.04. B: spectral dimension for a preselected cycle and a two-dimensional torus. For a finite number of vertices, the intermediate plateaus do not persist in the large-s limit.*

5. Why a Common Network Rearrangement Does Not Equalize Speeds

For two independent fields on the same fixed background, consider

$$\ddot{\varphi}_1 + \alpha_1 L(W)\varphi_1 = 0, \quad \ddot{\varphi}_2 + \alpha_2 L(W)\varphi_2 = 0, \quad \alpha_1, \alpha_2 > 0. \tag{15}$$

Proposition 3. For every nonzero eigenmode of L, the ratio of squared frequencies is α_1/α_2. If there exists a common long-wavelength limit $L \to -\beta h^{ij}\partial_i\partial_j$ with $\beta > 0$, then $c_1^2/c_2^2 = \alpha_1/\alpha_2$. Changing W while keeping distinct α_1 and α_2 fixed cannot make the two speeds coincide.

Proof. Substituting $Lz_n = \nu_n z_n$ into Eq. (15) gives $\omega_{a,n}^2 = \alpha_a \nu_n$. The common factor ν_n cancels in the ratio. In a shared continuum limit $c_a^2 = \alpha_a\beta$, and β cancels as well. □

The statement concerns the spectrum of a frozen background and the local leading coefficients when such a continuum limit exists. For rapidly varying W, constant mode frequencies may not be defined at all. Nonadiabatic mode mixing does not invalidate the algebraic result, but it requires a different operational notion of propagation.

A penalty for unequal squared speeds makes the point particularly transparent. With a common β it takes the form

$$V_{lo<k} = \Gamma(c_1^2 - c_2^2)^2 = \Gamma(\alpha_1 - \alpha_2)^2\beta^2, \quad \Gamma > 0. \tag{16}$$

If the remaining terms in the functional do not prevent $\beta \to 0$, the penalty vanishes simply by stopping propagation. This does not produce a finite universal speed. If β is constrained by $\beta \geq \beta_{min} > 0$ and $\alpha_1 \neq \alpha_2$, the penalty remains positive. For $\beta > 0$ the normalized relative difference is entirely independent of the network:

$$\Delta = 2|c_1 - c_2|/(c_1 + c_2) = 2|\sqrt{\alpha_1} - \sqrt{\alpha_2}|/(\sqrt{\alpha_1} + \sqrt{\alpha_2}). \tag{17}$$

To alter this conclusion one must change the model class itself: allow interactions that renormalize the field kinetic forms, employ different spatial operators with a justified relation among them, or invoke a symmetry or renormalization-group mechanism. Independent rescaling of the clock for each field does not solve the problem either; sectors to be compared must share one operational temporal calibration.

6. Coupled Components and a Collective Massless Branch

6.1. Exact Solution of the Two-Component Model

On a fixed network defined by Eq. (3), choose

$$A = \mathrm{diag}(a_1,a_2), \quad B = \mathrm{diag}(b_1,b_2), \quad U = \eta[[1,-1],[-1,1]], \tag{18}$$

$$a_1,a_2,b_1,b_2,\eta > 0. \tag{19}$$

The potential $\eta(\varphi_i - \chi_i)^2/2$ couples the components at each vertex. It preserves the common constant shift $\varphi_i \to \varphi_i + c_0$, $\chi_i \to \chi_i + c_0$ while breaking the independence of the two shifts. Consequently, at $\nu = 0$ there is one free collective coordinate, whereas the relative coordinate is gapped. Positivity of all coefficients guarantees nonnegative squared frequencies for every nonnegative eigenvalue of L.

For an eigenmode of L with eigenvalue ν, the problem reduces to a two-dimensional generalized eigenvalue equation:

$$\det(U + \nu B - \omega^2 A) = 0. \tag{20}$$

Introduce $T(\nu)$ and $D(\nu)$:

$$T(\nu) = (b_1/a_1 + b_2/a_2)\nu + \eta(1/a_1 + 1/a_2), \tag{21}$$

$$D(\nu) = [(b_1/a_1 - b_2/a_2)\nu + \eta(1/a_1 - 1/a_2)]^2 + 4\eta^2/(a_1a_2). \tag{22}$$

The exact branches are

$$\omega_+^2(\nu) = [T(\nu) + \sqrt{D(\nu)}]/2, \quad \omega_-^2(\nu) = [T(\nu) - \sqrt{D(\nu)}]/2. \tag{23}$$

For small ν, the second expression is numerically better evaluated from the product of the roots:

$$\omega_-^2(\nu) = [b_1b_2\nu^2 + \eta(b_1 + b_2)\nu]/[a_1a_2\omega_+^2(\nu)]. \tag{24}$$

At $\nu = 0$, $\omega_-^2 = 0$ and $\omega_+^2 = g_0^2$, where

$$g_0^2 = \eta(a_1 + a_2)/(a_1a_2). \tag{25}$$

Expanding the lower branch at fixed $\eta > 0$ gives

$$\omega_-^2(\nu) = v_0^2\nu - K\nu^2 + O(\nu^3), \tag{26}$$

$$v_0^2 = (b_1 + b_2)/(a_1 + a_2), \quad K = (a_2b_1 - a_1b_2)^2/[\eta(a_1 + a_2)^3] \geq 0. \tag{27}$$

The leading squared speed is a weighted mean of the microscopic squared speeds because

$$v_0^2 = [a_1(b_1/a_1) + a_2(b_2/a_2)]/(a_1 + a_2). \quad (28)$$

It therefore lies between b_1/a_1 and b_2/a_2. If these ratios coincide, $K = 0$ and the collective and relative coordinates separate exactly. If they differ, a negative correction of order ν^2 appears. The sign of this correction does not imply an instability of the complete model: the exact eigenvalues of Eq. (20) are nonnegative. The truncated expansion in Eq. (26) must not be extrapolated beyond its domain of validity.

6.2. What Has Actually Become Universal

At frequencies well below g_0, relative motion is suppressed and both microscopic components project onto the same collective mode. Its long-wavelength speed is finite for any positive a_1, a_2, b_1, b_2 and fixed $\eta > 0$; equality of the original microscopic speeds is not required. However, only one massless branch remains in the system. This is a robust reduction of the number of low-energy degrees of freedom, not an equalization of the speeds of two independent massless fields.

Even the upper gapped branch generally has a different low-momentum slope: the coefficient of ν in ω^2_+ is $b_1/a_1 + b_2/a_2 - v_0^2$. Matching the effective kinematics of the massive and massless sectors would therefore require additional conditions, as in analogue models [6]. A mass term by itself does not change the principal symbol of a local equation, but the kinetic coefficients of the sectors must still be matched.

In the full two-field continuum equation the high-frequency principal symbol is determined by A and B; the intravertex matrix U is of lower differential order. The coupling in Eq. (18) therefore does not turn two distinct ultraviolet cones into a single cone. The common cone discussed here belongs to the leading theory of the single low-energy mode that remains after the gapped relative mode is eliminated. This distinction is especially important when discussing signal-front velocity.

6.3. Gap Closing and Noncommuting Limits

As $\eta \to 0$ the gap in Eq. (25) closes. At $\eta = 0$ there are two independent massless branches with speeds $\sqrt{(b_1/a_1)}$ and $\sqrt{(b_2/a_2)}$. The infrared region in which the relative mode can be integrated out shrinks together with g_0. For unequal microscopic speeds,

$$\lim(\eta\to0^+)\,\lim(\nu\to0^+)\,\omega^2_-(\nu,\eta)/\nu = (b_1 + b_2)/(a_1 + a_2), \quad (29)$$

$$\lim(\nu\to0^+)\,\lim(\eta\to0^+)\,\omega^2_-(\nu,\eta)/\nu = \min(b_1/a_1, b_2/a_2). \quad (30)$$

Thus the appearance of an additional free sector as the coupling is removed does not guarantee universality of its cone.

A hypothesis in which free null modes emerge collectively must be tested together with changes in the dimension of the massless subspace. It should not be identified with the appearance of an individual photon or with recombination in standard cosmology.

6.4. Long-Wavelength Example on a Cycle

Take $a_1 = 1$, $a_2 = 2$, $b_1 = 1$, $b_2 = 8$, and $\eta = 3$. Then $v_0^2 = 3$, $g_0^2 = 4.5$, and $K = 4/9$. For a unit-spacing cycle, $\nu(k) = 4\sin^2(k/2)$, hence

$$\omega^2_-(k) = 3k^2 - (25/36)k^4 + O(k^6). \quad (31)$$

The coefficient 25/36 combines the lattice discreteness of the cycle with the elimination of the relative mode. When a physical lattice spacing and time calibration are restored, all dimensional factors must be reinstated consistently. The numerical parameters in this example are dimensionless model parameters, not estimates of physical constants.

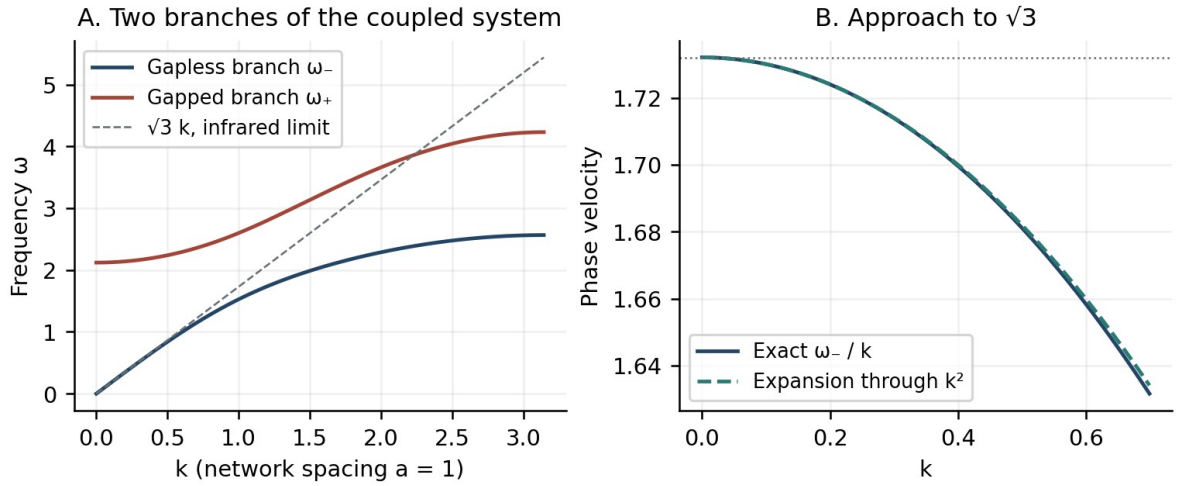


Figure 2. *Exact spectrum of the model in Eq. (18) for $a_1 = 1$, $a_2 = 2$, $b_1 = 1$, $b_2 = 8$, and $\eta = 3$ on a unit cycle. A: gapless and gapped branches; the line $\sqrt{3}\,k$ is shown as an infrared reference. B: phase velocity of the lower branch and its expansion $\sqrt{3}[1 - 25k^2/216]$. Agreement as $k \to 0$ does not imply the absence of dispersion at finite k.*

7. Criterion for a Common Cone with Several Massless Modes

7.1. Normalized Spectral Problem

For the general matrices A, B, and U of Section 3 define

$$T = A^{-1/2}UA^{-1/2}, \quad D = A^{-1/2}BA^{-1/2}. \tag{32}$$

After the change of variables $x = A^{1/2}\Phi$, the squared frequency on a mode with spectral parameter ν is an eigenvalue of $T + \nu D$. Here T and D denote matrices; the scalar functions of the same names introduced in Section 6 are not used below. Let $\dim \ker T = r \geq 1$. On its orthogonal complement assume a gap $g^2 > 0$. If $r = m$, the complement is absent and the compression described below is exact, with no mode elimination required.

Choose a matrix Z whose columns form an orthonormal basis of $\ker T$, and an orthonormal complement Y. Define the blocks

$$D_{00} = Z^{T}DZ, \quad D_{01} = Z^{T}DY, \quad D_{11} = Y^{T}DY, \quad T_{11} = Y^{T}TY \geq g^2 I. \tag{33}$$

The matrix D_{00} is positive definite. Its eigenvalues $v_1^2,\ldots,v_r^2$ determine the leading low-energy slopes:

$$\omega_j^2(\nu) = v_j^2\nu + O(\nu^2), \quad j = 1,\ldots,r. \tag{34}$$

It is the dimension r, not the number m of microscopic components, that determines the number of massless branches. On a finite graph this means r branches emanate from zero when ν is treated as a spectral parameter. A physical limit $\nu \to 0$ requires a family of graphs or another controlled limiting procedure, because a single finite connected graph has a finite gap between the zero mode and the next spectral level.

7.2. Elimination of Gapped Modes and Error Control

Write $x = Z\xi + Y\zeta$ and $z = \omega^2$. The second block equation gives

$$\zeta = -\nu(T_{11} + \nu D_{11} - zI)^{-1}D_{10}\xi. \tag{35}$$

Substitution into the first block yields the exact Schur-complement equation whenever the indicated inverse exists:

$$[\nu D_{00} - zI - \nu^2 D_{01}(T_{11} + \nu D_{11} - zI)^{-1}D_{10}]\xi = 0. \tag{36}$$

For $0 \leq z \leq g^2/2$ and $\nu \geq 0$, the norm of the inverse is at most $2/g^2$. Hence the norm of the correction to $\nu D_{00} - zI$ is bounded by $2\nu^2\|D\|^2/g^2$. The condition $\nu\|D\| \ll g^2$ controls the elimination of the gapped modes. Expansion gives

$$\nu D_{00} - \nu^2 D_{01}T_{11}^{-1}D_{10} + O(\nu^3). \tag{37}$$

Equation (37) refers to the low branch $z = O(\nu)$; constants in the remainder depend on fixed D and g. If D_{00} is degenerate, the second-order correction must be diagonalized within the degenerate subspace. The matrix

$D_{01}T_{11}^{-1}D_{10}$ is nonnegative, and the minus sign in Eq. (37) explains the sign of K in the two-component example.

7.3. Conditional Lorentzian Interpretation

Now impose a separate assumption: a continuum limit exists. Suppose a family of graphs and a map of observables admit, on the scales of interest, a common local elliptic spatial operator with positive form h^{ij} and spectrum

$$\nu(k) = \beta h^{ij}k_i k_j + O(|k|^4), \quad \beta > 0. \tag{38}$$

Condition (38) is not merely a statement about the list of eigenvalues. Suitable eigenfunctions, locality of the limiting operator, and a consistent map of observables are also required. Defining $k_n = \sqrt{\nu_n}$ without these data would merely rename the spectrum. The spatial tensor h and the number of its directions belong to the reconstructed limit, not to the fundamental data W.

In the leading two-derivative theory of the massless modes, the principal symbol after kinetic normalization is

$$P(\omega,k) = -\omega^2 I_r + \beta h^{ij}k_i k_j D_{00}. \tag{39}$$

Theorem 4. Within the class defined by Eqs. (32)–(39), all r massless branches have the same leading characteristic cone if and only if

$$D_{00} = v^2 I_r \quad \text{for some } v^2 > 0. \tag{40}$$

In that case the common symbol has the zero set

$$-\omega^2 + c^2 h^{ij}k_i k_j = 0, \quad c^2 = \beta v^2. \tag{41}$$

Proof. D_{00} is a real symmetric positive matrix and can be orthogonally diagonalized. The determinant of Eq. (39) is the product of factors $-\omega^2 + \beta v_j^2 h^{ij}k_i k_j$. For a positive spatial form, the cones coincide exactly when all v_j^2 are equal. A symmetric matrix with only one eigenvalue is that eigenvalue times the identity. □

The condition can be written without selecting a normalized basis. If the columns of Q form any basis of ker U, it is equivalent to

$$Q^T B Q = v^2 Q^T A Q. \tag{42}$$

For r = 1 the condition is automatic: one branch necessarily has one leading speed. For $r \geq 2$ it is a genuine constraint on the relative kinetic forms. If the graph continuum limit is anisotropic, the existence of a common spatial form preserves the meaning of Eq. (41); a common isotropic scalar c is obtained after choosing a local orthonormal spatial frame. In a more general class with different matrix coefficients B^{ij}, the projections of all such matrices would have to be matched, not only a single B.

The theorem concerns principal symbols of the leading low-energy theory. It does not assert equality of the exact dispersion relations at all k, full Lorentz symmetry of interactions, or a common ultraviolet signal-front velocity. For r > 1, the second-order matrix in Eq. (37) may split the branches at order v^2 even when Eq. (40) holds.

7.4. Symmetry Protection and Its Limits

Corollary 4.1. Suppose an orthogonal group acts on ker T, its representation on the entire r-dimensional massless subspace is irreducible and unbroken by the background, and the full dynamics preserves the symmetry. Then any symmetric operator D_{00} commuting with the group action is scalar.

Indeed, each eigenspace of D_{00} is invariant under the group. Irreducibility therefore permits only the entire space. This argument applies to symmetric real operators and does not require any assumption about the type of the full real commutant.

An explicit example is obtained by replacing $A \to A \otimes I_r$, $B \to B \otimes I_r$, and $U \to U \otimes I_r$ in the two-component model. This produces r collective massless modes, and O(r) forbids their splitting in a symmetric state. In this particular quadratic construction even the complete dispersion relations of the copies are identical. Their identical microscopic structure and the internal symmetry, however, are inserted assumptions. Such a construction cannot be regarded as a derivation of universality for all particles.

If the massless subspace decomposes into distinct irreducible representations, the symmetry allows different speeds between them. Multiplicities of equivalent representations also require analysis of the permitted mixing. For realistic scalar, fermionic, gauge, and gravitational excitations, a common irreducible representation has not been constructed here. Extending the theorem to those sectors requires their physical principal symbols after constraints and gauge redundancy have been removed.

When interactions are added, an exact internal symmetry continues to constrain the allowed matrix structures only if it is preserved by the regulator, the quantum theory, and the background. This is not a calculation of the renormalization-group flow. For mechanisms of the type discussed in Refs. [4, 5], one must instead show that sector-dependent differences are irrelevant directions of a specific flow; a postulated damping equation for the differences is not a substitute for such a derivation.

8. Quasilocal Balance on a Changing Network

The Hamiltonian in Eqs. (2)–(4) makes it possible to state precisely what is conserved when the relational structure changes. Define the field energy assigned to vertex i by dividing every intervertex energy equally between its endpoints:

$$e_i = \tfrac{1}{2} p_i^{T} A^{-1} p_i + \tfrac{1}{2} \Phi_i^{T} U \Phi_i + \tfrac{1}{4} \sum_j w_{ij} (\Delta\Phi_{ij})^{T} B \Delta\Phi_{ij}. \tag{43}$$

Let $v_i = A^{-1} p_i$. Differentiation using Eq. (5) gives

$$\dot{e}_i = -\sum_j J_{ij} + S_i, \tag{44}$$

$$J_{ij} = \tfrac{1}{2} w_{ij} (v_i + v_j)^{T} B (\Phi_i - \Phi_j), \qquad J_{ij} = -J_{ji}, \tag{45}$$

$$S_i = \tfrac{1}{4} \sum_j \dot{w}_{ij} (\Delta\Phi_{ij})^{T} B \Delta\Phi_{ij}. \tag{46}$$

For a set Q of vertices, the internal fluxes cancel pairwise:

$$d\left(\sum_{i \in Q} e_i\right)/d\tau = -\sum_{i \in Q, j \notin Q} J_{ij} + \sum_{i \in Q} S_i. \tag{47}$$

This is an exact balance law with respect to a boundary defined by edges that cross the partition; no geometric surface is required. For a fixed network, $S_i = 0$. If the weights change, the field sector gains or loses energy to the relational degrees of freedom themselves:

$$\dot{H}_p = \tfrac{1}{2} \sum_{\langle ij \rangle} \dot{w}_{ij} (\Delta\Phi_{ij})^{T} B \Delta\Phi_{ij} = -\dot{H}_r, \qquad \dot{H} = 0. \tag{48}$$

The symmetric assignment of each edge energy to its endpoints is convenient but not unique. Other local conventions change the detailed form of e_i and J_{ij} while preserving the total balance. We do not introduce two independent resources, a bulk energy and a separate boundary flux; the flux describes the change of the same conserved quantity.

In particular, a gradient law $\dot{W} = -\nabla V$ by itself describes dissipation of V because $\dot{V} = -\|\nabla V\|^2 \le 0$. It may be used as an optimization algorithm or as an effective law for an open subsystem. A closed description must include the degrees of freedom that receive the energy or explicitly adopt a different type of fundamental dynamics. Equations (5)–(6) provide one such closed realization.

Equation (48) gives concrete content to the original idea that redistribution and geometric restructuring are interdependent, but only at the level of changing relations. To call y or W geometric variables, one must exhibit a map to an effective metric. To identify H with gravitational energy, one must additionally match the relevant constraints and boundary conditions. Results such as Refs. [10, 11] may enter at that subsequent stage, but they do not follow from Eq. (48).

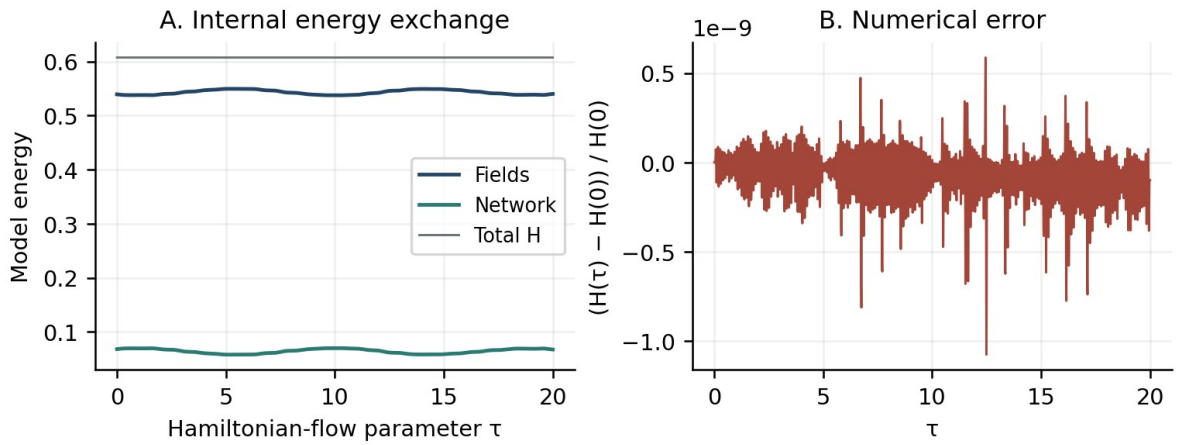


Figure 3. *Closed dynamical network of five vertices and ten relations. A: field energy, relational energy, and their conserved sum. B: relative total-energy error for DOP853 integration on τ ∈ [0,20]. The graph is not interpreted as physical space; the computation tests internal exchange and closure of the equations.*

9. Correlations, Response, and Causality

9.1. Analytic Tails on a Finite Network

Consider the scalar equation $\ddot{q} + \alpha L q = 0$ with $\alpha > 0$. The response at vertex i to a unit initial velocity at vertex j is the entire matrix function

$$R_{ij}(\tau) = [\sum_{n=0}^{\infty} (-\alpha L)^n \tau^{2n+1}/(2n+1)!]_{ij}. \quad (49)$$

It may equivalently be written as $\sin(\tau\sqrt{(\alpha L)})/\sqrt{(\alpha L)}$, with the value on the zero mode understood as τ. Let i and j be separated by r edges along a shortest path, and assume all edge weights are positive.

Proposition 5. For $r \geq 1$, the first r terms of the series in Eq. (49), namely those with $n < r$, do not contribute. The first nonzero term is positive:

$$R_{ij}(\tau) = \alpha^r W_r(i,j) \tau^{2r+1}/(2r+1)! + O(\tau^{2r+3}), \quad (50)$$

where $W_r(i,j) > 0$ is the sum of products of edge weights over all shortest r-edge paths from j to i.

Proof. With fewer than r matrix multiplications, L^n cannot connect two vertices at graph distance r; diagonal entries do not reduce the number of off-diagonal transitions required. For $n = r$ every nonzero contribution corresponds to a shortest path and contains r off-diagonal factors $-w$. The sign $(-1)^r$ from L^r cancels the factor $(-1)^r$ in the series. All products of positive weights are positive. □

Thus, for every connected pair of vertices the response is nonzero for sufficiently small positive τ. At fixed graph distance this excludes an exact law $R_{ij}(\tau) = 0$ for all $0 < \tau < r/v$ with finite v. The model may possess a sharply defined region of rapid propagation and extremely small tails outside it, but that is not a strict light cone.

For example, on a nine-vertex path with unit weights and $\alpha = 1$, the response at distance $r = 4$ and $\tau = 0.1$ is approximately 2.75347×10^{-15}; the leading term in Eq. (50) gives 2.75573×10^{-15}. Smallness does not make the response identically zero. In the corresponding quantum harmonic model, the relevant coordinate commutator is proportional to the same response function.

Lieb–Robinson-type estimates [8] can bound such tails for suitable interactions and observables. They must not be replaced by an equality to zero. Strict causality may emerge in a continuum limit as a property of the limiting equation and its solutions, but that requires a precise limiting procedure, normalization of the couplings, and control of the convergence of response functions. No such theorem for dynamically emerging graphs is proved in the present work.

9.2. Correlation Structure and Operational Signalling

For quantum subsystems A and B, the mutual information

$$I(A:B) = S(\rho A) + S(\rho B) - S(\rho AB) \quad (51)$$

measures the failure of the joint state to factorize, but it does not specify a direction of controllable influence. Symmetric correlation data do not distinguish retarded from advanced response.

Moreover, the choice of subalgebras or a tensor-product factorization must be defined before mutual information is computed; using mutual information to derive the factorization itself therefore requires an additional self-consistent construction.

In standard quantum mechanics, a local trace-preserving operation ΛA with Kraus operators K^{α} satisfying $\sum^{\alpha} K^{\alpha}\dagger K^{\alpha} = I$ leaves the remote reduced state unchanged when averaged over the local outcome:

$$\mathrm{Tr}A[(\Lambda A \otimes \mathrm{id}B)(\rho AB)] = \mathrm{Tr}A\ \rho AB. \tag{52}$$

The equality follows from cyclicity of the partial trace over A and trace preservation. It is independent of the amount of entanglement. A conditional state selected by a particular outcome may differ, but use of that selection by the other party requires information about the outcome.

This kinematic no-signalling statement concerns a local operation with respect to an already specified decomposition. It does not prove the absence of later influence through an interaction Hamiltonian. If geometrically remote subsystems are directly connected by a weakly suppressed term in H, the dynamical response must be bounded separately. Equation (52) therefore neither eliminates the tails in Eq. (50) nor completes the reconstruction of causality.

The expression $d_{ij} = -\ln(W_{ij}/W_0)$ is likewise only a possible diagnostic. Even for positive normalized weights it can violate the triangle inequality: with $W_{12} = W_{23} = \exp(-1)$ and $W_{13} = \exp(-3)$, one obtains $d_{13} = 3 > d_{12} + d_{23} = 2$. Constructing a shortest-path metric repairs this defect when edge lengths are positive, but it adds a chosen rule of geometrization. Its connection to physical propagation still has to be demonstrated.

In this paper C is not introduced as an independent classical field. In a possible quantum extension, correlations should be computed from the state, observables, and dynamics. The choice of state remains part of the model. For example, the free zero mode of a finite massless system has no normalizable oscillator ground state unless the collective coordinate is fixed or regulated. Even the calculation of “vacuum” mutual information therefore requires an explicitly stated prescription.

10. Geometry, Conformal Class, and Domain of Applicability

When the conditions of Theorem 4 are satisfied, the leading system has a hyperbolic symbol with one temporal and a positive spatial part. In a local description one may choose a representative of the conformal class whose inverse form is $G = \mathrm{diag}(-1, c^2 h^{ij})$. The sign of the temporal part follows here from the assumed wave Hamiltonian and positivity of A and B. This is a conditional derivation of hyperbolicity for the effective equation, not an origin of time or Lorentzian signature from the positive matrix W alone.

A symmetric W does not select a time orientation. Retarded response is introduced only after an initial-value problem and a direction of the flow parameter have been chosen. A reversible Hamiltonian system does not by itself explain a thermodynamic arrow of time. Causal reconstruction requires additional data and assumptions; for example, known theorems for conformal reconstruction already require suitable spacetimes of dimension greater than two with past and future distinguishability [1, 2].

On a slowly varying background, a Lorentzian interpretation of characteristics is meaningful only when scales separate: the wavelength must be short relative to the background-variation scale but long relative to the network scale, and the frequency must be small relative to the gap of the eliminated modes. Formally one requires an intermediate regime

$$L_{\mathrm{bg}}^{-1} \ll |k| \ll a_{\mathrm{net}}^{-1}, \quad \omega^2 \ll g^2. \tag{53}$$

Here L_{bg} and a_{net} are scales defined within the already adopted effective description. The coefficients must also vary slowly over a wave period. If no such interval exists, geometric optics, the continuum approximation, and elimination of gapped modes cannot be applied simultaneously.

A conformal transformation $g \to \Omega^2 g$ with $\Omega > 0$ preserves null directions. However, equality of cones among sectors, a conformal change of one metric, and constancy of a measured c in time are different statements. If only the spectral prefactor $\beta(W)$ changes while A is fixed, then $c^2 = \beta v^2$ changes relative to the

chosen τ. Constancy of an operationally measured c requires tracing how internal clocks and spatial calibration change as well. The present model does not yet establish the required compensation.

Likewise, the number of vertices is not automatically a physical volume. Mapping counting measure to volume depends on the limiting procedure and on the density with which degrees of freedom represent space. A causal order can determine conformal information only under suitable conditions, whereas an overall scale requires additional data. This preprint does not construct a unified spacetime measure, a global topology, or dynamics at cosmological scales.

Obtaining a common cone is also weaker than obtaining full special relativity. One must study nonlinear interactions, transformations of observables, states, conserved currents, and the possibility of a preferred frame. A gravitational limit further requires a dynamical metric, constraints, coupling to an energy–momentum tensor, and recovery of tested general-relativistic limits. These requirements cannot be replaced by a successful linear dispersion relation.

11. Reproducible Computations and Phenomenological Benchmarks

11.1. Protocol for Finite-System Checks

All numerical results reported here were computed directly for the models defined in this work. The script reproduce.py uses Python 3.12.13, NumPy 2.3.5, SciPy 1.17.0, and Matplotlib 3.10.8. The random seed is 20260905. Parameters, initial states, numerical outputs, and the trajectory of the dynamical network are stored in the accompanying computational supplement. No external observational data were used in these calculations.

For forty random connected weighted graphs with 24 vertices, the frequency ratio implied by Eq. (15) was checked for $\alpha_1 = 1$ and $\alpha_2 = 4$. It equals 2 for every nonzero mode to machine precision. This check validates the implementation of the algebraic identity; it is not an independent proof of the general no-go statement.

For the spectrum in Eq. (13), the maximum discrepancy between numerical diagonalization and the exact value was 1.42×10^{-14}. With an additional inhomogeneous interblock perturbation, $\varepsilon = 0.104673$ and $\|P - P_0\| = 0.000442474$ were obtained, below the bound $\varepsilon/(\gamma - \varepsilon) = 0.0132576$. This tests subspace stability for one selected perturbation; the general estimate follows from the proof, not from the example.

Equation (23) was checked for one hundred independent choices of positive a_1, a_2, b_1, b_2, η and four values of ν. The maximum normalized error relative to generalized diagonalization of Eq. (20) was 8.43×10^{-16}, with the error divided by $1 + |\omega^2|$. Results for the lower branch on the cycle C_{256} are listed in Table 2.

Mode n	k = 2πn/256	ω₋/k	(ω₋/k)/√3 − 1
1	0.0245437	1.7319300	−0.00006972
2	0.0490874	1.7315677	−0.00027894
4	0.0981748	1.7301171	−0.00111641
8	0.1963495	1.7242984	−0.00447584
16	0.3926991	1.7007871	−0.01805013

Table 2. *Exact two-component spectrum on the cycle. The limit √3 is compared with the phase velocity; the table does not determine group velocities or signal-front velocity.*

An additional check uses $T = \mathrm{diag}(0,0,2,3,4,5)$ and a positive D with $D_{00} = 3I_2$ and nonzero D_{01}. The exact two low branches share the leading slope 3 but differ at order ν^2. The script compares them with the correction matrix in Eq. (37). This independently confirms that a common leading cone does not imply identical complete dispersion relations.

For the dynamical network we choose five vertices, all ten possible edges, the matrices in Eq. (18) with the parameters of Section 6.4, $I_e = 4$, and $V_r = \sum_e (y_e + 0.8)^2$. DOP853 integration is performed on $\tau \in [0,20]$ with $\mathrm{rtol} = 10^{-10}$, $\mathrm{atol} = 10^{-12}$, and 2001 output points. The initial total energy is 0.60739633; the maximum relative energy drift is 1.08×10^{-9}. The field energy varies approximately from 0.53744 to 0.54953. The residual of the local analytic balance in Eq. (44), evaluated in two equivalent ways at the initial state, is 1.04×10^{-17}. The calculation tests closure of these particular equations, not a gravitational conservation law.

11.2. What Is Required for Comparison with Experiment

The parameters of the demonstration calculations have not yet been mapped to physical energies and lengths. Consequently, the coefficient −25/36 in Eq. (31) cannot be converted into a bound on a quantum-gravity scale in electronvolts. One must first identify which observables correspond to the field branches, how clocks are defined, what dimensional relation connects k and ν, and which operators arise after quantum corrections.

Experimental constraints depend on the sector, operator dimension, anisotropy, dispersion, and conventions used. The 2026 update of the Kostelecký–Russell data tables [13] systematizes limits for matter, photons, neutrinos, and gravity. It would therefore be incorrect to summarize all such cases by a single number called “the precision of Lorentz invariance.”

A useful benchmark is provided by the joint observation of GW170817 and GRB 170817A [14]. Under the source-emission-delay assumptions of that analysis, the relative difference between gravitational-wave and electromagnetic-wave speeds is constrained approximately to the interval -3×10^{-15} to $+7 \times 10^{-16}$. This constrains those two propagation channels; it is not a demonstration that all particles have identical limiting speeds. In our scalar demonstration model neither a photon sector nor a gravitational sector has yet been identified, so the interval is not applied directly.

A physical implementation would require calculation of dimensionless intersector differences and dispersive coefficients in a common calibration, a test of their radiative stability, and only then selection of the relevant observations. Reference [12] explains why microscopically small Lorentz-violating effects need not remain equally small in effective kinetic coefficients. The finite quadratic model presented here does not calculate such quantum corrections.

12. Results Achieved and the Open Part of the Program

The first technical stage separates several problems that are often conflated. Rearrangement of a common network, emergence of a collective massless branch, coincidence of the cones of several branches, and strict microcausality are mathematically distinct requirements. They cannot be tested by a single plot or by one number c.

Problem	Result of this work	What remains necessary
Relational redistribution	Closed H and local balance, Eqs. (44)–(48)	Physical interpretation of the conserved quantity
Quasilocal distinguishability	Spectral criterion and bound, Eq. (12)	Dynamical selection of a block/local regime
Common speed of independent fields	No-go result for the class in Eq. (15)	Interaction, symmetry, or a calculated RG flow
Collective massless mode	Exact model, Eqs. (18)–(31)	Connection to observable physical sectors
Several common infrared cones	Criterion, Eq. (40), and conditional symmetry protection	Origin of the symmetry and intersector matching
Strict causality	Finite-network tails, Eq. (50)	Controlled response limit or a different microscopic law
Dimension, time, and gravity	Explicitly stated conditions	Independent dynamical derivations

Table 3. *Summary of results obtained in the present work and the principal open requirements.*

The next stage should preserve computability while reducing the number of independent assumptions. This requires a concrete permutation-invariant potential, or a quantum law for the relations, that produces a local geometric regime from a broad class of initial data. Several diagnostics should be tested simultaneously: spectral gaps, growth of accessible volume, stability of partitions, and locality of the effective operator. Manually preparing a cycle or lattice is useful as a control but does not complete this step.

Once such a regime is obtained, the field matrix must be studied on the entire massless subspace. Criterion (42) tests for a common cone without assigning a common speed in advance. If the condition is reached dynamically, an attraction mechanism and the suppression rate of deviations must be demonstrated. If it is

protected by symmetry, the origin of that symmetry and its preservation under background evolution must be explained. Closing of a gap requires the low-energy modes to be recounted.

A separate line of work concerns the origin of oriented causality and internal clocks. The reparametrization in Eq. (7) does not solve that problem. Information-theoretic, thermodynamic, and algebraic mechanisms should be compared with concrete observer operations and with the limiting behavior of the dynamical response. Within the original process-based picture, an observer must itself be a stable subsystem rather than an external absolute measuring device.

Finally, a gravitational stage requires deriving dynamics for the effective metric and matching them to field constraints, quasilocal charges, cosmological expansion, and tested limits of general relativity. Hypotheses of boundarylessness, simple connectivity, technical compactness, or a single dimensionless invariant remain possible directions, but none is used in the proofs of this preprint. The absence of an external reservoir does not by itself select topology, dimensionality, or signature.

13. Conclusion

The program of a universal causal structure has been given a concrete spectral core. For a common network with distinct fixed coefficients of independent fields, equalization of propagation speeds is impossible. Introducing a coupling changes the low-energy content of the system and produces a collective massless branch, but this does not suffice for universality across several independent physical sectors. Within the quadratic class considered here, universality of their leading cones is equivalent to scalarity of the normalized stiffness operator on the massless subspace.

The closed dynamics constructed here ties energy exchange to restructuring of relations, while a spectral gap provides a quantitative criterion for stable quasilocal distinguishability. At the same time, analytic response tails on a finite network establish a limit to the causal interpretation: a common low-energy wave kinematics is not the same as strict microcausality of the underlying system. These results narrow the space of admissible models and provide testable conditions for further derivations of geometry, time, and observable universality.

Part II. Response limits and coarse-graining

1. The question after the first spectral criterion

The first version of this programme [15] made several necessary distinctions precise. Rearranging one common graph Laplacian cannot equalise independent fields with unequal fixed kinetic ratios. Coupling two components can leave one collective massless mode without making two independent massless species universal. For a positive quadratic model with kinetic matrix A, spatial matrix B and nonnegative coupling U, the leading cones of all surviving massless modes coincide precisely when the normalised spatial operator is scalar on the massless subspace. Finally, a finite continuous-time network generally has analytic propagation tails, even when its long-wavelength dispersion resembles a wave equation.

The next question is therefore not the numerical value of c. It is how a relation, an effective response, a propagation cone and an oriented causal order can be connected by controlled changes of description. Each transition requires both an identification of observables and a topology of convergence. Eigenvalues alone do not supply either locality or an operational meaning for distance.

This version concentrates on three tasks. First, we establish a concrete response limit in which microscopic tails disappear outside an enlarged limiting cone. Second, we ask whether the set of common-cone configurations is invariant or attracting under a specified dynamics. Third, we compare explicit coarse-graining maps and identify the assumptions under which their limiting predictions agree. Internal observers, gauge sectors and gravity remain a short research programme, rather than additional purported derivations.

The positive results use periodic chains whose continuum geometry is prescribed. This is a control problem: it establishes what a successful response limit must contain and gives quantitative estimates that can later be tested on dynamically selected networks. It does not establish that a generic relational Hamiltonian

selects those chains. The contribution lies in the stated estimates, their compatibility conditions and the obstructions separating the stages. The Fourier, projection and recurrence methods themselves are standard mathematical tools.

Work on convergence of spectral structures [16] provides a general framework for comparing operators on varying Hilbert spaces. Lieb–Robinson estimates [8] distinguish controlled propagation from exact vanishing at finite lattice spacing. Interaction-driven approaches to Lorentz universality [4, 5] concern scale evolution in particular theories. These three subjects address different parts of the present problem; none is used as an automatic implication from an undirected graph to an oriented spacetime.

2. Structures and three topologies of convergence

Let H_n be the Hilbert space of microscopic linear perturbations, with its physically specified measure, and let H be a proposed effective space. An extraction requires maps $J_n : H_n \to H$ and $Q_n : H \to H_n$, as well as an operator and a class of states. In the examples below J_n is an isometry and $Q_n = J_n^*$, so $Q_nJ_n = I$ while J_nQ_n is a projection converging strongly to I.

For a nonnegative self-adjoint operator L_n define the velocity response

$$S_n(t) = \sin(t\sqrt{L_n})/\sqrt{L_n}, \quad S_n(t)|\ker L_n = tI. \quad (1)$$

It solves $\ddot{q} + L_nq = 0$ with $q(0) = 0$ and $\dot{q}(0) = f_n$. The retarded response is zero for $t < 0$ and equals $S_n(t)$ for $t \geq 0$. This use of "retarded" already makes a choice of temporal orientation and preparation; the undirected operator does not select that choice.

We distinguish three levels. Spectral convergence concerns eigenvalues or spectral projections after specified identifications. Operator and locality convergence concerns actions on identified states, resolvents, quadratic forms and localisation operators. Response convergence concerns the dynamical functions of those operators. Our principal response topology is

$$\sup(0 \leq t \leq T) \, \|J_nS_n(t)Q_nf - S(t)f\|H \to 0 \quad \text{for every } f \in H \text{ and } T < \infty. \quad (2)$$

Equation (2) is uniform strong convergence on compact time intervals. It is not operator-norm convergence over all unit vectors, and it is not pointwise convergence of a singular Green kernel. A separate kernel estimate will be proved for a specific model. The distinction matters because microscopic states with wavelength tending to zero need not be covered uniformly by convergence for a fixed f.

Locality also requires a comparison of spatial operations. If $M\chi$ multiplies an effective field by a bounded local test function χ, compatible microscopic localisation operators should satisfy, on the relevant states,

$$J_nM\chi_{,n}Q_nf \to M\chi f. \quad (3)$$

Conjugating an operator by an arbitrary nonlocal unitary preserves its spectrum while changing these localisation operations. Thus matching eigenvalues without matching observables cannot establish local geometry. In Section 4 we give two explicit reconstructions and compare their responses directly.

The logical hierarchy is relation, response, propagation cone, then causal order. A wave symbol defines a pair of cone components. Time orientation selects one component as future. A global order additionally depends on the manifold, admissible causal curves and global causal conditions. A local Lorentzian cone field does not require a separate Frobenius integrability condition to define causal curves, but it does not by itself exclude closed causal curves.

3. A controlled response limit and disappearance of tails

3.1. The finite systems and their measure

Fix a circle of circumference ℓ. Take odd N, spacing $a = \ell/N$ and sites $x_j = ja$ modulo ℓ. The microscopic inner product is

$$\langle q,u\rangle_a = a\sum_j \bar{q}_ju_j, \quad \|q\|_a^2 = a\sum_j|q_j|^2. \quad (4)$$

For a fixed finite hopping range M, choose $\kappa_m \geq 0$, $\kappa_1 > 0$ and

$$c^2 = \sum_m \kappa_m m^2 > 0, \quad \mu_4 = \sum_m \kappa_m m^4, \quad 1 \leq m \leq M. \quad (5)$$

Define the operator and canonical Hamiltonian

$$(L_a q)_j = a^{-2}\sum_m \kappa_m(2q_j - q_{j+m} - q_{j-m}), \quad (6)$$

$$H_a = \sum_j \pi_j^2/(2a) + (2a)^{-1}\sum_{j,m} \kappa_m(q_{j+m} - q_j)^2. \quad (7)$$

Here $\pi_j = a\dot{q}_j$ and $\{q_i,\pi_j\} = \delta_{ij}$. Hence $\ddot{q} + L_a q = 0$ is a finite, autonomous, closed Hamiltonian system. Positivity excludes unstable linear modes. Its constant mode is free; compactness of an energy surface is not automatic unless that mode is fixed or reduced.

Let $K(N) = \{2\pi n/\ell : |n| \le (N-1)/2\}$. The map J_a sends grid values to the unique trigonometric polynomial with these modes and these values at the sites. Discrete Fourier orthogonality makes J_a an isometry into $L^2(S^1,dx)$. Set $Q_a = J_a^*$ and $P_a = J_a Q_a$. Thus $Q_a f$ consists of samples of $P_a f$, not generally samples of an arbitrary unsmoothed f. This distinction avoids an unmentioned aliasing assumption.

On a mode $k \in K(N)$, the discrete eigenvalue is

$$\lambda_a(k) = (4/a^2)\sum_m \kappa_m \sin^2(mak/2). \quad (8)$$

The candidate limit is $L = -c^2\partial_x^2$ with periodic domain. Both the circle and its measure have been introduced explicitly. They have not been inferred from a spectrum.

3.2. Uniform strong convergence and a microscopic universality class

Theorem 1. For the family (4)–(8), every $f \in H^3(S^1)$ and $T < \infty$ satisfy

$$E_{a,T}(f) = \sup(0 \le t \le T)\ \|J_a S_a(t) Q_a f - S(t) f\|_2, \quad (9)$$

$$E_{a,T}(f) \le [T^2\mu_4 a^2/(24c) + T(a/\pi)^3]\ \|\partial_x^3 f\|_2. \quad (10)$$

For every $f \in L^2(S^1)$, the response difference (9) tends to zero uniformly on [0,T], without an asserted rate. Different fixed finite-range coefficient sets with the same c^2 therefore have the same identified continuum response.

Proof. Taylor's theorem and positivity give

$$0 \le c^2k^2 - \lambda_a(k) \le \mu_4 a^2 k^4/12, \quad (11)$$

$$0 \le c|k| - \sqrt{\lambda_a(k)} \le \mu_4 a^2 |k|^3/(12c). \quad (12)$$

The scalar response can be written as $f_t(\omega) = \int_0^t \cos(s\omega)ds$, including $\omega = 0$. Therefore $|f_t(\omega_1) - f_t(\omega_2)| \le t^2|\omega_1 - \omega_2|/2$. Apply this estimate mode by mode and use Parseval's identity. The omitted Fourier modes have $|k| > \pi/a$, so their L^2 norm is at most $(a/\pi)^3\|\partial_x^3 f\|_2$; multiplying by $\|S(t)\| \le T$ gives the second term. Finally, H^3 is dense in L^2 and both response operators have norm at most T. Approximation by a fixed smooth f proves uniform strong convergence. □

The theorem controls eigenfunctions through J_a and controls the entire time interval. It is stronger than matching a small-k slope. It is still relative to a chosen identification and fixed effective states.

Three distinct examples at $c = 1$ are $\kappa = (1)$, $\kappa = (1/2,1/8)$, and $\kappa = (1/2,1/16,1/36)$. Their fourth moments are 1, 5/2 and 15/4. They therefore have different finite-spacing dispersions but the same limit. This is a concrete universality class of prescribed linear chains, not a demonstrated universality class of arbitrary nonlinear relational Hamiltonians.

3.3. Why the Green kernel must be normalised

Write $R^a_{ij}(t) = [S_a(t)]_{ij}$ as a matrix in site coordinates. Then

$$q_i(t) = \sum_j R^a_{ij}(t) f_j = a\sum_j G^a_{ij}(t) f_j, \quad G^a_{ij} = R^a_{ij}/a. \quad (13)$$

The continuum Green density is represented by G^a, not by R^a. A unit vector at a site has integrated weight a, whereas a unit-integral impulse has value 1/a at that site. With nonuniform site measures μ_j, the corresponding density is R_{ij}/μ_j.

Consequently, vanishing unnormalised matrix elements can reflect dilution of an impulse. Such vanishing does not by itself distinguish the inside from the outside of a continuum cone. The measure and the normalisation of the intervention must be fixed before a tail criterion is interpreted physically.

3.4. An explicit bound outside every enlarged cone

Restrict now to nearest-neighbour chains, $\kappa_1 = c^2$. Let $d_a(i,j) = a\min(|i-j|, N-|i-j|)$ be the physical circle distance. For $T > 0$ and $\delta > 0$ define

$$\theta = \operatorname{arcosh}(1 + \delta/(cT)), \quad (14)$$

$$b(T,\delta) = 2[\delta\theta - cT(\sinh\theta - \theta)] > 0. \quad (15)$$

Theorem 2. The normalised finite-cycle Green kernel satisfies

$$\varepsilon_a(T,\delta) = \sup(0 \le t \le T) \sup(d_a(i,j) > ct + \delta) |G^a_{ij}(t)|, \quad (16)$$

$$\varepsilon_a(T,\delta) \le [2T/\{a(1 - \exp(-2\theta\ell/a))\}] \exp(-b(T,\delta)/a) \to 0. \quad (17)$$

An empty spatial supremum is taken to be zero. No claim of uniform convergence on the cone boundary $\delta = 0$ is made.

Proof. On the infinite chain, the cosine propagator has Fourier coefficients of the entire periodic function $\cos[(2ct/a)\sin(z/2)]$. Shift its Fourier contour by imaginary height 2θ with the sign appropriate to the site separation n. The coefficient has absolute value at most $\exp(-2|n|\theta + 2ct \sinh\theta/a)$. Integrating the cosine propagator from 0 to t bounds the velocity response by t times this expression.

Periodising over $n + mN$ and summing the resulting geometric series gives

$$C_a(t,\theta) = 2t/\{a(1 - \exp(-2\theta N))\}, \quad (18)$$

$$|G^a_{ij}(t)| \le C_a(t,\theta) \exp[-2\theta d_a(i,j)/a + 2ct \sinh\theta/a]. \quad (19)$$

If $d_a > ct + \delta$, replace t by T in the positive term $c\,t(\sinh\theta - \theta)$. The choice (14) maximises $\delta\theta - cT(\sinh\theta - \theta)$, yielding (17). This maximum is positive because its value is zero at $\theta = 0$ and its derivative there is $\delta > 0$. □

Every fixed connected chain still has a nonzero analytic response at sufficiently small positive time between any pair of sites. The theorem establishes uniform disappearance of the physically normalised tails in a singular refinement limit. It does not assign an exact finite-speed cone to any finite N.

The limiting response has the periodic d'Alembert representation

$$[S(t)f](x) = (2c)^{-1} \int f(y)dy, \quad x - ct \le y \le x + ct, \quad (20)$$

where f is periodically extended. Equivalently, its retarded Green density is the periodisation of $(2c)^{-1}1\{|x| < ct\}$ for $t > 0$, with the boundary understood distributionally. It is the retarded fundamental solution of $\partial_t^2 - c^2\partial_x^2$.

Theorem 1 gives convergence of responses tested against fixed spatial L^2 observables, uniformly in time, and therefore convergence against smooth compactly supported spacetime tests. The local reconstruction below gives the same tested limit. Theorem 2 additionally proves a pointwise exterior bound for the discrete density. These are complementary statements; one must not promote the strong-operator result alone to a pointwise kernel theorem.

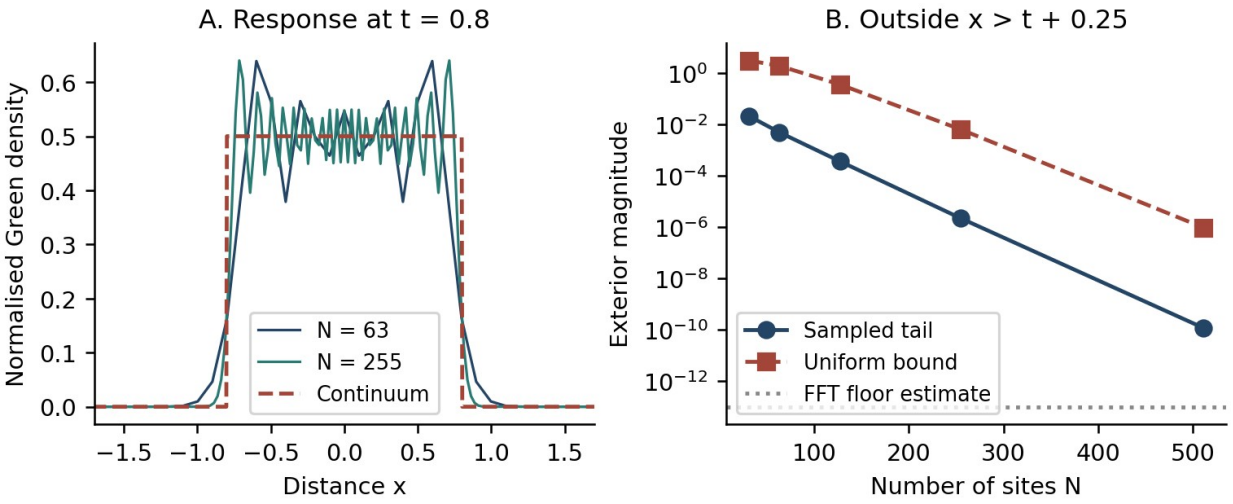


Figure 1. A: normalised finite-chain Green kernels at t = 0.8 on the circle ℓ = 2π, c = 1, compared with the continuum response. Oscillations near the fronts are compatible with convergence away from them. B: sampled exterior maxima for t ∈ [0,1], d > t + 0.25, and the bound proved for the whole time interval. Numerical sampling does not prove a supremum.

4. Coarse-graining, locality and the order of limits

4.1. Extraction is more than a spectral projection

A useful infrared extraction contains three ingredients: a selected subspace, an identification with effective variables, and a measure. For a graph cutoff $P_{n,\Lambda}$, a massless-component basis Z and kinetic normalisation $A^{1/2}$, a possible linear map is

$$\Pi_{n,\Lambda}\Phi = J_n P_{n,\Lambda}(I \otimes Z^T A^{1/2})\Phi. \qquad (21)$$

The tensor notation distinguishes the graph sector from the component sector. Formula (21) is a specification, not a universal definition for every relational system. The cutoff, maps, norm and states must be reported together. Under a change of effective coordinates the limit may be compared by an explicitly stated intertwining map, rather than by requiring identical coordinate entries.

For block-structured graphs, the earlier spectral estimate [15] can stabilise the selected subspace: if $L = L_0 + E$, $E \geq 0$, $\|E\| = \varepsilon < \gamma/2$ and L_0 has r zero modes separated by γ, then the low projection differs from the block-constant projection by at most $\varepsilon/(\gamma - \varepsilon)$. This controls selection under perturbations. It does not construct J_n, a local manifold or a causal response.

4.2. A comparison of Fourier and local reconstructions

Let $E_a : H_a \to L^2(S^1)$ assign q_j to the cell of width a centred on x_j. Its adjoint $F_a = E_a^*$ is cell averaging, $(F_a f)_j = a^{-1}\int_{\text{cell } j} f$. Both maps are local in position, and E_a is an isometry. They differ from trigonometric interpolation, which is spatially nonlocal at finite N.

Theorem 3. For the operators of Theorem 1 and $f \in H^1(S^1)$,

$$\mathscr{E}_{a,T}(f) = \sup_{0 \leq t \leq T} \|E_a S_a(t) F_a f - J_a S_a(t) Q_a f\|_2, \qquad (22)$$

$$\mathscr{E}_{a,T}(f) \leq aT(1 + 1/\pi)\|\partial_x f\|_2. \qquad (23)$$

Both reconstructions therefore have the same uniform strong response limit on every fixed L^2 state.

Proof. For a trigonometric polynomial $u = J_a q$, the fundamental theorem of calculus and Cauchy–Schwarz on each centred cell give

$$\|E_a q - J_a q\|_2 \leq (a/2)\|\partial_x J_a q\|_2. \qquad (24)$$

Apply this to $u = P_a f$. Since omitted modes have $|k| > \pi/a$ and cell averaging is contractive,

$$\|F_a f - Q_a f\|_a \leq a(1/\pi + 1/2)\|\partial_x f\|_2. \qquad (25)$$

Split the difference in (22) into $E_a S_a(F_a - Q_a)f$ and $(E_a - J_a)S_a Q_a f$. The first term is bounded using $\|S_a(t)\| \le T$. For the second, (24) applies and the Fourier response commutes with ∂_x, so $\|\partial_x J_a S_a Q_a f\|_2 \le T\|\partial_x f\|_2$. Adding the bounds proves (23). Density and the uniform operator bounds extend convergence to L^2. □

This is independence under two specified, physically interpretable reconstructions. It does not imply independence under arbitrary maps. In particular, a coarse-graining that mixes distant cells without an appropriate limiting locality condition can change the meaning of an intervention even when it preserves the spectrum.

4.3. A fixed infrared cutoff cannot have an exact cone

Proposition 4. On the circle, retain a fixed finite set of Fourier modes containing the zero mode and apply it to the continuum wave response. For any $t > 0$ the resulting kernel is a nonzero trigonometric polynomial. It cannot vanish on a nonempty open spatial interval.

Proof. A finite Fourier sum is analytic. Vanishing on an open interval would force it to vanish identically, whereas its zero-mode coefficient equals t/ℓ. □

On the line, compact frequency support gives the corresponding analyticity obstruction. Thus a fixed band-limited response is incompatible with exact spatial support, even if its unfiltered wave operator has finite propagation speed. One cannot demonstrate strict effective causality merely by applying a fixed Π_{IR} and reading the leading symbol.

For an exact local wave response, a limiting reconstruction must recover arbitrarily high effective Fourier modes, or another physically justified local completion must be supplied. This does not mean that a finite-order infrared expansion predicts microscopic fronts. It means that a strict cone belongs to a specified limiting local theory, while finite-band observations only resolve an approximation to it.

4.4. A sufficient large-gap completion for collective modes

The same issue appears when massive relative modes are eliminated. In kinetically normalised coordinates consider, on a spatial eigenvalue $\nu \ge 0$,

$$\ddot{x} + (T_g + \nu D)x = 0, \quad T_g = \mathrm{diag}(0, T_{11}),$$
$$T_{11} \ge g^2 I, \quad D > 0. \qquad (26)$$

Here the first block has dimension r; D_{00}, D_{01} and D_{11} are its component blocks. The matrices are constant in time for this statement. Prepare $x(0) = 0$, $\dot{x}(0) = (f,0)$, and write $x = (\xi,\zeta)$. Let ξ_0 solve $\ddot{\xi}_0 + \nu D_{00}\xi_0 = 0$ with the same collective initial data.

Theorem 5. For $0 \le t \le T$,

$$\|\zeta(t)\| \le (2\nu\|D\|t/g^2)\|f\|, \quad (27)$$
$$\|\xi(t) - \xi_0(t)\| \le [\nu^2\|D\|^2 t^3/(3g^2)]\|f\|. \qquad (28)$$

Proof. Positivity of the full operator implies $\|x(t)\| \le t\|f\|$ and $\|\dot{x}(t)\| \le \|f\|$ by spectral calculus. Put $H_{11} = T_{11} + \nu D_{11} \ge g^2 I$. The forced equation for ζ gives an integral against $\sin((t-s)\sqrt{H_{11}})/\sqrt{H_{11}}$. Integration by parts yields

$$\zeta(t) = -\nu H_{11}^{-1}[D_{10}\xi(t)$$
$$- \int_0^t \cos((t-s)\sqrt{H_{11}})D_{10}\dot{\xi}(s)ds]. \qquad (29)$$

The term containing $\xi(0)$ vanishes. The two remaining terms each have norm at most $\|D\|t\|f\|$, giving (27). The difference $\xi - \xi_0$ is forced by $-\nu D_{01}\zeta$, and its homogeneous velocity response has norm at most $t-s$. Integrating $2\nu^2\|D\|^2(t-s)s/g^2$ gives (28). □

Now take the unit-speed graph operator from Theorem 1 and assume $D_{00} = \nu^2 I_r$. Let $a \to 0$ and $g \to \infty$ with fixed bounded D. For each fixed finite Fourier band, (28) controls collective-mode elimination and Theorem 1 controls the spatial limit. Uniform response bounds and density extend the resulting convergence to every fixed L^2 collective initial velocity. The limit is the r-component local wave response with speed v.

This statement makes the completion explicit. At fixed finite g, the projected exact response generally retains dispersive and temporal-memory corrections. A strictly local common-cone limit is obtained here by a

singular large-gap family, not by interpreting the finite-g expansion as an exact microscopic law. The preparation of heavy modes is part of the theorem.

If one uses an increasing intermediate cutoff $|k| \le \Lambda$, sufficient conditions from the stated bounds are $a^2\Lambda^3 \to 0$ and $\Lambda^4/g^2 \to 0$, with $\Lambda \to \infty$. These conditions are convenient, not optimal. For example, $g = a^{-1}$ and $\Lambda = a^{-1/3}$ satisfy both. The large-gap limit is a variation of microscopic models; it is not a temporal attraction mechanism for D_{00}.

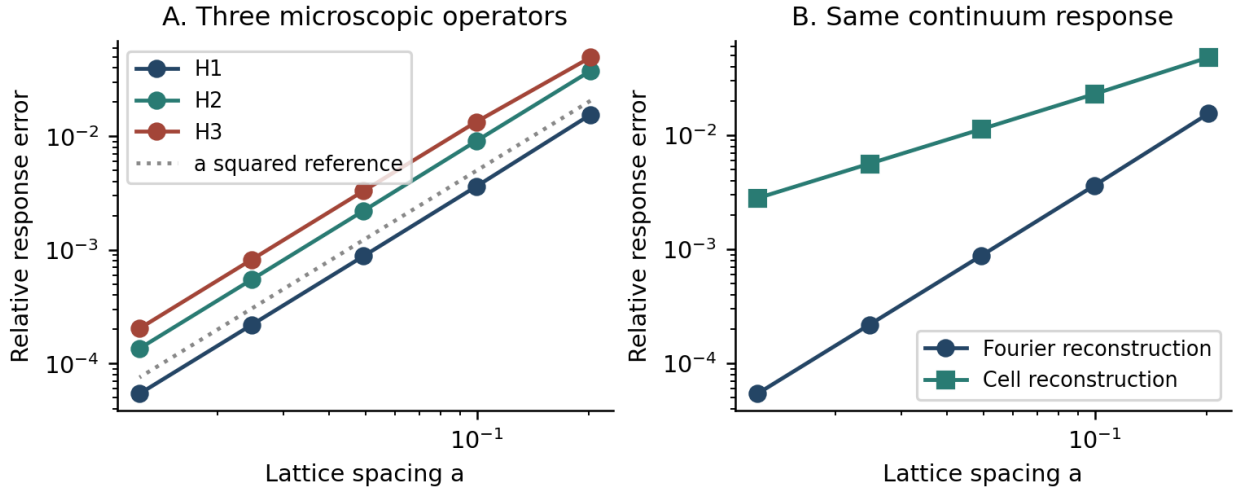


Figure 2. Relative L^2 response errors on $\ell = 2\pi$ for $f(x) = \cos x + 0.35 \sin 3x + 0.2 \cos 7x$, sampled over $t \in [0,1]$. A: three different finite-range operators with the same second moment approach the same continuum response. B: Fourier interpolation and local cell reconstruction converge at different rates to that response.

5. Invariance, protection and attraction of universality

5.1. The common-cone set on a moving massless space

For state-dependent positive matrices A(X), B(X) and nonnegative U(X), define

$$T(X) = A^{-1/2}UA^{-1/2}, \quad D(X) = A^{-1/2}BA^{-1/2}. \quad (30)$$

Work on a smooth region of state space where dim ker T = r is constant and the positive spectrum is uniformly separated from zero. Let P(X) be the orthogonal projection onto ker T. Define the basis-independent defect

$$v^2(X) = \mathrm{tr}(PD)/r, \quad F(X) = PDP - v^2P, \quad \Delta_D(X) = \|F(X)\|. \quad (31)$$

The set M_{cone} is defined by F = 0. In an orthonormal basis Z of ker T, its condition is $D_{00} = Z^TDZ = v^2I_r$. Under the common elliptic spatial-symbol assumption, the physical leading determinant factors into $-\omega^2 + v_j^2h^{ij}k_ik_j$. Positive symmetry of D_{00} shows that all these cones coincide if and only if F = 0. Equivalently, for any basis Q of ker U, $Q^TBQ = v^2Q^TAQ$ [15]. At r = 1 this condition is automatic and says nothing about universality between independent species.

Calling M_{cone} a manifold requires an additional regularity condition. On a fixed-rank stratum it is locally a smooth submanifold when the independent trace-free constraints have constant rank. We use “universality set” when this is not established. Nor is Δ_D automatically equal to distance in the full microscopic state space; local comparison requires a regular parametrisation and a specified norm.

5.2. A tangency test that includes motion of the projection

Let X(t) follow a C^1 autonomous vector field. Choose a horizontally transported orthonormal basis Z(t), meaning $Z^T\dot{Z} = 0$. Differentiating PZ = Z gives $\dot{Z} = \dot{P}Z$ in this gauge. Hence

$$dD_{00}/dt = Z^T(\dot{D} + \dot{P}D + D\dot{P})Z. \quad (32)$$

For an r-by-r matrix K write $\mathrm{tf}(K) = K - \mathrm{tr}(K)I_r/r$.

Proposition 6. On a regular part of M_{cone}, the condition for the vector field to be tangent to the universality constraints is

$$\mathrm{tf}[Z^{T}(\dot{D} + \dot{P}D + D\dot{P})Z] = 0 \quad \text{whenever } F = 0. \qquad (33)$$

If this holds throughout that submanifold, local uniqueness of the flow implies local invariance for as long as the trajectory stays in the regular, gapped region. If it fails at a point, a trajectory through that point immediately acquires a nonzero first-order trace-free defect.

The formula follows directly from (32) after subtracting the trace. It is covariant under orthogonal changes of basis on the massless subspace. In an arbitrary moving basis, a commutator with the basis connection is added; on $D_{00} = v^2 I$ it vanishes. Thus (33) tests a physical condition rather than a choice of eigenvector gauge.

If T has an exact fixed-rank kernel, differentiating $TP = 0$ also gives the useful estimate $\|\dot{P}\| \le \|\dot{T}\|/g^2$, where T is at least g^2 on the complement. This follows by solving the off-diagonal block with the restricted inverse of T; the derivative of a projection has only off-diagonal blocks. A closing gap can therefore invalidate both the smooth bundle and an adiabatic interpretation.

For the original adaptive-weight model [15], A, B and U are fixed while the edge weights move. Consequently T, P and D_{00} are constant. That model preserves an already universal D_{00} and preserves an already nonuniversal one. It contains no temporal mechanism driving Δ_D to zero. Introducing state dependence in A, B or U changes the model and requires the test above together with the full equations of motion.

5.3. Zero anisotropy is not automatically invariant

Take a canonical pair (q,p) with $H_{ani} = (p^2 + q^2)/2$, and a probe-sector matrix $D(q) = 3I_2 + q\,\mathrm{diag}(1,-1)$, $U = 0$. Along the invariant zero-field background, $q(t) = 0.2 \sin t$ and $p(t) = 0.2 \cos t$ are allowed by the closed Hamiltonian dynamics. The matrix stays positive and $\Delta_D = |q|$.

At $t = 0$ the cone condition holds, but $p \neq 0$ gives $\dot{q} \neq 0$, so it immediately fails. Constraining only q while ignoring its conjugate momentum does not define an invariant set. A nonzero field background can add further forces through its coupling to q. This elementary example shows why an equation for Δ_D alone is insufficient and why the full state, constraints and backreaction matter.

5.4. The Hamiltonian obstruction to a genuine attractor

Proposition 7. Let an autonomous Hamiltonian flow have an invariant region carrying a finite invariant measure μ. Suppose M_{cone} is closed and has μ-measure zero in that region. Then the set of initial states satisfying $\mathrm{dist}(X(t),M_{cone}) \to 0$ has μ-measure zero.

Proof. For a finite measure-preserving flow, almost every state is recurrent. A recurrent state outside a closed M_{cone} returns arbitrarily close to itself and thus infinitely often remains a positive distance from M_{cone}. It cannot converge in distance to that set. The remaining initial states lie in a null set or in M_{cone} itself. □

This is the relevant consequence of volume preservation and recurrence [17]. The finite-measure assumption is essential. It may hold on a compact regular energy surface with its invariant measure, or on a finite-volume invariant region. It is not automatic for the free collective coordinate of a harmonic network, and the statement is not applied to that coordinate without reduction.

The obstruction concerns attraction in full microscopic phase space under physical Hamiltonian time. It does not exclude attraction of reduced observables, energy transfer to unresolved degrees of freedom, an infinite-volume limit, or a renormalisation-group flow in scale. Each of these alternatives changes the mathematical question. A dissipative law $\dot{\Delta}D = -\gamma\Delta_D$ cannot simply be inserted as a fundamental closed Hamiltonian equation without specifying the additional structure.

5.5. What symmetry does establish

If an unbroken orthogonal symmetry acts irreducibly on $\ker T$ and D_{00} commutes with it, D_{00} is scalar: every eigenspace of this real symmetric operator is invariant, so irreducibility permits only one eigenvalue. If the

full flow preserves the symmetry and the background belongs to its invariant symmetric sector, this protects a subset of M_{cone}.

Protection is distinct from attraction from nonsymmetric initial states. Different irreducible sectors can have different speeds, and repeated representations permit additional mixing. Interaction-driven convergence such as [4, 5] requires a calculated scale flow, rather than the tangency condition alone. The present version derives tests and obstructions; it does not present a new interacting renormalisation-group calculation or a generic attracting common-cone phase.

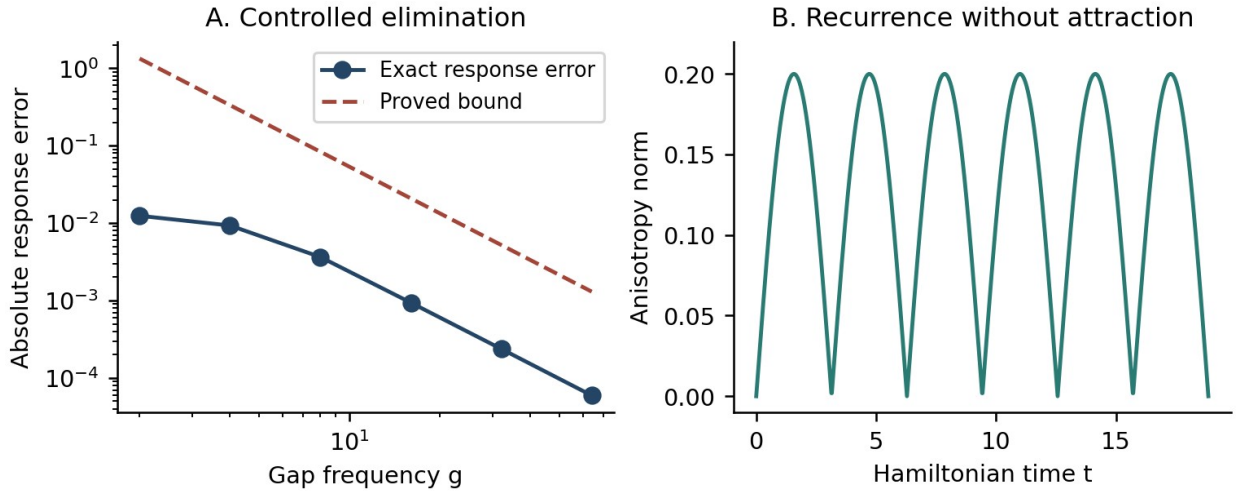


Figure 3. A: collective-response error for $D = [[3,-\sqrt{2}],[-\sqrt{2},2]]$, $T_g = diag(0,g^2)$, $v = 1$ and collective initial velocity, compared with the bound of Theorem 5 on $t \in [0,1]$. B: $\Delta_D = 0.2|\sin t|$ in the conservative anisotropy example. Increasing a model's gap and evolving a Hamiltonian state in time are different operations.

6. Orientation, global order and metric scale

6.1. What a symmetric microscopic law cannot select

For an even-in-momentum Hamiltonian, time reversal Θ changes the signs of momenta and leaves coordinates and undirected weights unchanged. It obeys $\Theta\Phi_t = \Phi_{-t}\Theta$. A rule that selects an oriented future must transform oppositely under Θ; a rule depending only on the Θ-even W cannot do so.

Proposition 8. No deterministic assignment $\varepsilon(W) \in \{+1,-1\}$ can both depend only on a time-reversal-invariant W and satisfy $\varepsilon(\Theta W) = -\varepsilon(W)$.

The proof is immediate: $\Theta W = W$ would require $\varepsilon(W) = -\varepsilon(W)$. More generally, time-reversal-invariant laws and invariant preparation data do not distinguish a retarded experiment from its reversed advanced experiment without an additional choice. A momentum-sensitive state, boundary condition, statistical condition or internal clock may supply such a choice. The existence and stability of that structure must be shown separately.

This separates an unoriented pair of cone components from a choice of future, and separates that choice from a thermodynamic arrow. The chain limit already uses a time parameter and an initial-value preparation; it is not a derivation of temporal orientation from W.

6.2. A cone and an order are different results

On the prescribed cylinder $\mathbb{R} \times S^1$, the wave symbol gives the Lorentzian metric class represented by $ds^2 = -c^2dt^2 + dx^2$. With increasing t chosen as future, define

$$(t,x) \preccurlyeq (s,y) \quad \text{iff} \quad s - t \geq dS^1(x,y)/c. \qquad (34)$$

The circle triangle inequality proves transitivity. Nonnegativity of the time difference gives antisymmetry, because two reverse relations force $s = t$ and $dS^1(x,y) = 0$. Thus (34) is a partial order and t is a global time function. This conclusion uses the product topology and the nonperiodic time coordinate.

If time is instead periodically identified, the same local cone field descends to a spacetime with closed timelike curves. A trajectory at fixed x can wind around time, destroying antisymmetry of causal reachability. Hence matching local cones cannot establish a global causal order. In general spacetimes the required causality assumptions must be stated before invoking causal reconstruction results [1, 2].

6.3. Conformal data and measure data

A common quadratic Lorentzian cone determines a conformal class, not its scale. This conclusion assumes that the cone is the null cone of a Lorentzian quadratic form; a general common dispersion surface need not be Lorentzian. Once a smooth D-dimensional manifold and a representative g are available, an additional positive volume density μ fixes the conformal factor by

$$\tilde{g} = \Omega^2 g, \quad \mathrm{dvol}(\tilde{g}) = \Omega^D \mathrm{dvol}(g), \quad (35)$$

$$\Omega = [d\mu/\mathrm{dvol}(g)]^{1/D}. \quad (36)$$

For Lorentzian metrics, dvol denotes the positive density $\sqrt{|\det g|}$. Equation (36) is conditional on identified compatible measure data. Counting graph vertices without proving their measure limit does not supply those data. It also does not explain why the observed dimension should be 3+1.

7. Reproducible checks and their scope

The accompanying reproduce_v2.py performs deterministic Fourier calculations and finite matrix diagonalisation. It uses Python 3.12.13, NumPy 2.3.5, SciPy 1.17.0 and Matplotlib 3.10.8. It requires no external datasets. All values below are dimensionless benchmarks on a prescribed circle $\ell = 2\pi$.

For nearest-neighbour propagation, c = 1, T = 1 and δ = 0.25. The response density is computed as the inverse discrete Fourier transform of $t\,\mathrm{sinc}(t\sqrt{\lambda}/\pi)$, divided by a. The exterior maximum is sampled at 801 uniformly spaced times. Theorem 2, rather than the sampling, controls the continuous interval.

For convergence, the initial velocity is $f(x) = \cos x + 0.35 \sin 3x + 0.2 \cos 7x$, with zero initial displacement. Errors are L^2 distances divided by $\|f\|_2$ and maximised over 1001 sampled times in [0,1]. Fourier errors are evaluated by Parseval's identity. For the cell reconstruction, the initial data are exact cell averages and the L^2 comparison with the continuum is integrated analytically, so no additional spatial plotting grid enters the reported error.

N	Exterior kernel	Fourier error	Cell error
31	2.13713×10^{-2}	1.54863×10^{-2}	4.80821×10^{-2}
63	4.98650×10^{-3}	3.62274×10^{-3}	2.29583×10^{-2}
127	3.69388×10^{-4}	8.82617×10^{-4}	1.13206×10^{-2}
255	2.19631×10^{-6}	2.18370×10^{-4}	5.63044×10^{-3}
511	1.18654×10^{-10}	5.43442×10^{-5}	2.80879×10^{-3}

Table 1. Sampled tail magnitudes and response errors for the nearest-neighbour benchmark. The kernel column is physically normalised by 1/a; the two error columns compare fixed L^2 data. These are different diagnostics, not competing estimates of one quantity.

The three hopping families of Section 3 approach the same response, with differences visible in their finite-spacing errors. Local cell reconstruction has a leading discretisation error of order a in this example, while the smooth Fourier reconstruction has order a^2. The shared limiting response is therefore compatible with different convergence rates.

For the gap test, use $D = [[3,-\sqrt{2}],[-\sqrt{2},2]]$, $T_g = \mathrm{diag}(0,g^2)$, ν = 1 and collective initial velocity (1,0). Comparing the exact projected response with $\sin(\sqrt{3}t)/\sqrt{3}$ at 2001 times in [0,1] gives maximum errors 9.36955×10^{-4} at g = 16, 2.36968×10^{-4} at g = 32 and 5.93913×10^{-5} at g = 64. They lie below (28). These calculations test a family of increasingly stiff relative modes; they do not show that a physical system evolves toward that family.

Numerical tails eventually approach floating-point roundoff, and sampled maxima cannot establish exact zeros. None of these data is a fit to a photon, graviton or laboratory propagation experiment. The mathematical

proofs establish the stated model properties; physical identification of sectors and calibration of their scales remain necessary before observational constraints can be used.

8. Counterexamples and the remaining research programme

The non-equivalence of successive stages can be documented by explicit controls.

Available structure	Missing implication	Counterexample or obstruction
Connected relation	Several separated slow sectors	Complete graph: only the constant mode is slow; the other eigenvalues coincide
Separated sectors	A selected temporal orientation	Two weakly coupled blocks with a symmetric operator and reversal-invariant data
Propagation	A common cone	Independent fields with unequal fixed kinetic ratios on one Laplacian
Common leading cone	Exact finite-network microcausality	A harmonic chain or the collective model with analytic tails
Exact local wave operator	Locality after a fixed IR cutoff	Proposition 4: a nonzero finite Fourier kernel cannot vanish on an open interval
Common local cones	A global causal partial order	Periodic identification of time creates closed timelike curves
Zero anisotropy at one instant	An invariant universality set	q = 0 with nonzero conjugate momentum in Section 5.3

Table 2. Previous-stage data do not determine the next stage. These examples constrain the proposed transitions; they do not rule out transitions under additional hypotheses.

The central uncompleted step is selection of geometry. A genuinely pregeometric continuation needs a specified permutation-invariant law and state ensemble that produce a local regime from a sufficiently broad class of initial relations. Graphity-type models [3] give useful starting points, but the required claim must include control of eigenfunctions, localisation operations and response, not only a spectral dimension. The prescribed chains of this paper provide a benchmark against which such a result can be compared.

An internal observer can be specified operationally by an accessible algebra A^{O}, a clock observable T^{O}, a set of admissible controls and a preparation procedure. A relation $A \rightsquigarrow B$ then means that an allowed intervention at A changes later accessible statistics at B in that clock convention. The kernels used here describe ideal coordinate impulses and readouts. Whether those interventions can be implemented by an internal subsystem is an additional model question, not a consequence of a nonzero mathematical matrix entry.

For gauge fields and gravity, universality must be formulated on the physical degrees of freedom after constraints and gauge redundancy are treated. The desired condition is equality of physical characteristic sets, $\mathrm{Char}_{\mathrm{phys}}(P_a) = N^*$, across sectors. A determinant with gauge-zero directions is not a physical cone test. The scalar compression D_{00} supplies a controlled model, while extensions to constrained principal symbols require separate analysis; analogous multicomponent media already show the importance of these distinctions [6].

Three further distinctions remain essential. A scale flow is not Hamiltonian time evolution. A stable invariant set need not attract nearby states. A common leading cone need not be stable under quantum corrections. Testing an interacting physical model requires its actual symmetries, regulator, effective operators and observable sectors, as emphasised by the radiative-stability problem [12]. No such interacting calculation is claimed here.

The remaining topics—vacuum selection, a dynamical relational description, topological restrictions, compactness and the gravity interface—should be attached to a specific missing transition rather than treated as independent explanations. Pregeometry is then a level of description on which the operations later interpreted as geometric and causal are not yet defined. Its content lies in invariants, admissible maps and conditions for transitions between levels, rather than in naming a final underlying substance.

9. Conclusion

The first spectral criterion can be extended into a controlled response programme. On prescribed finite chains, physically normalised microscopic tails disappear uniformly outside every enlarged continuum cone, and several different hopping Hamiltonians approach the same local wave response. Fourier and local cell reconstructions agree in the limit. A fixed infrared cutoff does not possess exact spatial support, and a large-gap estimate identifies one sufficient completion of collective-mode elimination.

Cone universality has a separate dynamical problem. Its basis-independent defect must be transported with the massless projection; a tangency condition tests invariance, while Hamiltonian recurrence obstructs a positive-measure attracting set under the stated finite-measure assumptions. Symmetry protection, attraction of reduced observables and scale-flow universality must therefore be investigated separately.

The limit supplies a propagation structure under explicit geometric and temporal identifications. It does not choose a future from reversal-invariant data, establish a global order without causal hypotheses, or fix a metric scale without measure data. These distinctions turn the next stage of the relational programme into three testable problems with stated mathematical boundaries.

Part III. Universal null calibration

1. From a dimensional number to a structural condition

The proposed invariant is not the number $299\,792\,458$. In the SI, that numerical value of the vacuum speed of light is exact because it enters the definition of the metre, together with the definition of the second [18]. The experimental content of relativistic propagation is not removed by that metrological choice. One still has to establish which physical sectors share a characteristic structure and whether their agreement survives changes of scale, state and description. The debate about dimensionful constants [19] usefully exposes the dependence on conventions; our results require no particular position in that broader debate.

The preceding papers of this programme established a spectral common-cone criterion [15] and controlled response limits on prescribed chains [20]. The present article uses those results to answer a different question: what exactly is preserved when spatial and temporal measurements are calibrated against the same null structure? The earlier light papers [21–23] supplied compact geometric examples; the DRD framework [24] supplied the distinction between a carrier and structures preserved by specified morphisms. Neither body of work is taken as an independent proof of universality. Corrections to statements relevant here are given in Section 9 and the accompanying corpus audit.

Hypothesis of the programme. Spatial and temporal measures may be effective calibrations compatible with a more primitive common response structure. This is a hypothesis about reconstruction. It is not an identification of a three-dimensional vector space with a one-dimensional time axis. Our precise, weaker object is a compatibility relation between positive length and time scales after an observer splitting and physical characteristics have been reconstructed.

There are four main technical results. Theorem 1 separates common-cone physics from unit normalisation. Theorem 2 supplies a sufficient criterion for independence under specified reconstructions. Conditional Proposition 3 gives spectral length–time bounds. Theorem 4 lifts the harmonic control model to unitary quantum dynamics and a microcausal limit. Proposition 5 corrects a previous recurrence claim. None is presented as a new derivation of Lorentz symmetry for the Standard Model. The matrix, spectral-calculus and CCR methods are standard; the contribution is their explicit assembly, quantitative control and the identification of conditions under which the proposed invariant is meaningful.

1.1. The neutral level and the controlled level

At the proposed microscopic level, let R denote a neutral relational structure, for example a symmetric weighted adjacency matrix considered up to vertex permutations. It is not called a causal relation. Before a metric reconstruction has been defined, $d(A, B)$ is undefined, rather than zero. Before a physical clock has

been constructed, the same applies to $\Delta t(A,B)$. We assume neither a Planck grain nor S^3, compactness, a photon field or a future-directed order at this level.

The theorems below begin at a later, explicitly controlled level. Positive quadratic dynamics, a parameter for evolution, Hilbert-space measures and candidate reconstruction maps are given. In the chain example, the circle and its refinement are prescribed. An evolution parameter used to prove a limit is not thereby a derived physical clock. Thus the results test an interface which a pregeometric model must eventually produce; they do not supply the missing transition from an arbitrary R to that interface.

1.2. Physical characteristics and the meaning of "massless"

Let M be a smooth effective manifold. For each retained physical sector a, let $P_a^{phys}(x,k)$ denote its principal symbol after constraints and gauge redundancy have been treated. We write

$$Char_{phys}(P_a)=\{(x,k)\in T^{*}M : k\neq 0, ker_{phys}P_a(x,k)\neq\{0\}\}. \quad (1)$$

A gauge kernel present for every k is excluded from this definition. In a system with mixing, the sectors are the physical propagation branches, not the original field names. Our finite-dimensional proofs concern real symmetric positive kinetic and spatial forms; strong hyperbolicity is then automatic after diagonalisation. Equality of determinants in an arbitrary matrix system would not establish well-posedness.

"Massless" denotes branches whose frequency tends to zero as the spatial spectral parameter tends to zero. A mass term in an already local relativistic equation is lower order and does not change its principal characteristics. Consequently, universality among massless sectors is a useful test but does not by itself establish universality of all matter. A single remaining massless branch is especially insufficient to test intersector agreement.

Definition 1 (universal null calibration). A specified family of physical sectors admits universal null calibration on a region if their characteristic sets coincide with one smooth Lorentzian null cotangent cone N^{*}, and if local observer standards can be calibrated against its dual tangent cone using the same positive length–time conversion for every sector. No future component or conformal representative is included in this definition. The Lorentzian qualification is substantive: a common higher-degree characteristic variety need not be the cone of a metric [33].

Here a calibration means a local frame and unit convention, not a globally integrable system of coordinates. This distinction is essential when coefficients vary. The standard formula $d\ell = c\,d\tau$ uses the time increment measured in an observer's decomposition of a null displacement; $d\tau$ is not the proper time of the null curve itself.

2. The common calibration theorem and a dimensionless defect

2.1. Model class and allowed transformations

Fix one positive spatial form h^{ij}. On a retained r-dimensional massless space, let $A_0 = A_0^T > 0$ and $B_0 = B_0^T > 0$. The physical principal symbol is

$$P(\omega,k) = -\omega^2 A_0 + h^{ij}k_i k_j B_0,\quad D_0 = A_0^{-1/2} B_0 A_0^{-1/2}. \quad (2)$$

The eigenvalues of D_0 are $v_j^2 > 0$. A common change of numerical coordinates $x' = \alpha x$, $t' = \beta t$, with $\alpha,\beta > 0$, sends all speeds to $v'_j = \rho v_j$, where $\rho = \alpha/\beta$. Equivalently, changing the units of length and time multiplies every numerical speed by one common positive factor. A sector-specific clock change is not an allowed common recalibration.

Theorem 1 (common calibration). In the class (2), the following are equivalent: all physical branch cones coincide; $B_0 = v^2 A_0$ for some $v > 0$; $D_0 = v^2 I_r$; and one common positive recalibration makes every branch speed equal to one. The necessary and sufficient recalibration has $\rho = 1/v$. Before that choice, $D_0 = v^2 I_r$ is the physical universality condition; afterwards, $D'_0 = I_r$ is its unit-normalised representation.

For a smooth field of such systems the assertion is pointwise and yields smooth local calibrations wherever the forms remain positive and the rank is fixed. A single constant pair (α, β) on a whole region works only if the shared $v(x)$ is constant in the initially chosen conventions.

Proof. Kinetic normalisation and the spectral theorem give

$$\det\left(A_0^{-1/2} P A_0^{-1/2}\right) = \prod_{j=1}^{r} \left(-\omega^2 + v_j^2 h^{ij} k_i k_j\right). \quad (3)$$

Choose any nonzero spatial covector. Positivity of h implies that two factors have the same zero set exactly when their positive speeds coincide. A real symmetric matrix with only one eigenvalue is scalar. Finally $D'_0 = \rho^2 D_0$, so $D'_0 = I_r$ exactly when $D_0 = \rho^{-2} I_r$. The pointwise statement follows by the same argument, and the assertion about a constant recalibration follows from $\rho\, v(x) = 1$. ▫

The common spatial-form assumption cannot be omitted from the claim about scalar unit changes. For example $-\omega^2 + k_x^2 + 4k_y^2$ may already be the common symbol of every sector, yet isotropic unit rescaling does not make it $-\omega'^2 + |k'|^2$. One additionally needs a common spatial frame adapted to h. This is a frame change, not a physical equalisation mechanism.

More generally, at a point in spacetime dimension at least three, two Lorentzian quadratic forms of the same signature have the same null set if and only if they are positively proportional. To see this, put one form in Minkowski form and evaluate the other on $(1, \pm e_i)$: mixed time–space coefficients vanish and the spatial diagonal entries are equal to minus the time coefficient. Evaluation on $\left(1, (e_i + e_j)/\sqrt{2}\right)$ removes the remaining cross terms. The signature fixes the positive factor. A common Lorentzian frame therefore normalises all such cones locally. In dimension two the same argument requires a common choice of which complementary directions are timelike; that datum is already fixed by the positive wave class (2). These algebraic statements are distinct from global causal-reconstruction theorems [1, 25–29].

2.2. Defect, optimal normalisation and basis independence

Definition 2 (relative calibration defect). For the positive pencil (B_0, A_0) define

$$\Delta_c = \max_{i,j} \left| \log \frac{v_i}{v_j} \right| = \frac{1}{2} \log \frac{\lambda_{max}(D_0)}{\lambda_{min}(D_0)}. \quad (4)$$

It vanishes exactly at the common-cone condition. It is invariant under a common unit rescaling, orthogonal changes of normalised basis, and arbitrary changes of the original basis when both forms are transformed by the same congruence. In the latter case these are generalised eigenvalues of $B_0 z = \lambda A_0 z$, rather than ordinary eigenvalues of the coordinate matrix B_0. Under $r = 1$, $\Delta_c = 0$ is automatic and cannot certify universality beyond that branch.

The best common calibration in a worst-case logarithmic sense is explicit:

$$\inf_{\rho > 0} \max_j \left| \log(\rho v_j) \right| = \frac{\Delta_c}{2}, \quad \rho_¿ = \left(v_{min} v_{max}\right)^{-1/2}. \quad (5)$$

Indeed adding $\log \rho$ translates the interval $[\log v_{min}, \log v_{max}]$; its smallest centred radius is half its length. The irreducible residual is positive whenever two cones differ. Thus the convention $c = 1$ cannot hide a relative-speed defect.

The link to [15, 20] is direct. With $T = A^{-1/2} U A^{-1/2}$, $D = A^{-1/2} B A^{-1/2}$ and Z an orthonormal basis of $\ker T$, the relevant matrix is $D_{00} = Z^T D Z$. Formula (4) applied to D_{00} removes the scale dependence of the earlier norm defect. If $\acute{v}^2 = \mathrm{tr}\, D_{00}/r$ and $\varepsilon = \| D_{00} - \acute{v}^2 I \| / \acute{v}^2 < 1$, then

$$\Delta_c \le \frac{1}{2} \log \frac{1+\varepsilon}{1-\varepsilon}. \quad (6)$$

This follows by bounding the extremal eigenvalues. It is a continuity estimate, not a relaxation law. On a moving gapped massless space, the invariance test remains the trace-free part of $Z^T\left(\dot{D}+\dot{P}\,D+D\,\dot{P}\right)Z$, with $P=Z\,Z^T$ and horizontal transport of Z [20]. Internal symmetry may protect a scalar compression on a symmetric background. It need not unify inequivalent representations or attract generic states. The finite-invariant-measure Hamiltonian recurrence obstruction of [20] continues to apply under its stated hypotheses.

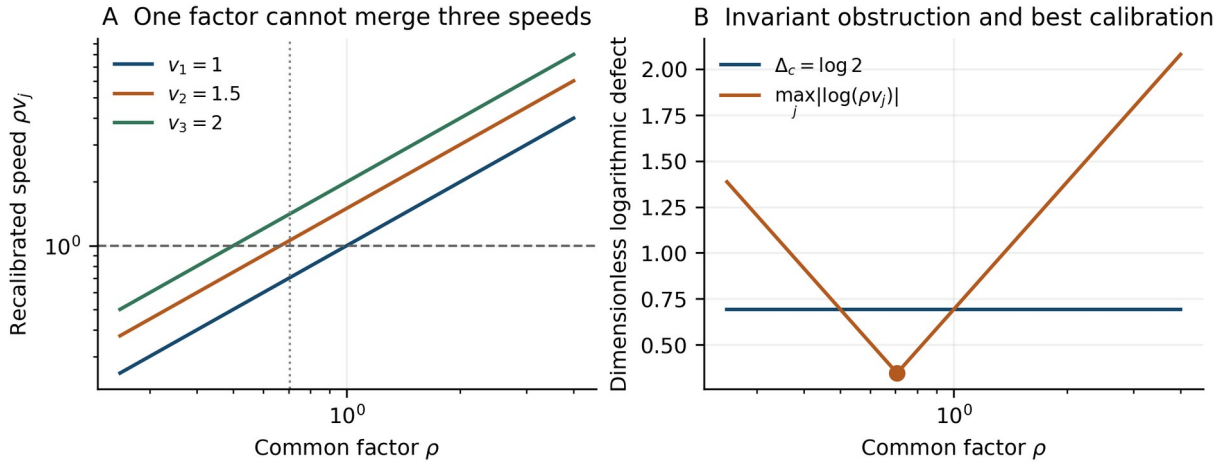


Figure 1. Common unit changes multiply the three speeds together. The nonzero calibration defect remains fixed, while the smallest worst-case logarithmic residual equals half that defect. The curves illustrate Theorem 1 and equation (5); they are not a fit to physical sectors.

2.3. What kind of invariant is obtained?

The cone is a geometric subset of the cotangent bundle $T^{\dot{\iota}}M$ with its zero section removed, transforming covariantly under diffeomorphisms. Equality of cones is invariant under a common diffeomorphism, invertible physical field redefinitions and positive conformal changes. The numbers v_i/v_j and Δ_c have the stated invariant meaning in the common spatial-form and common observer splitting of (2); they are not general Lorentz-boost-invariant scalars for arbitrary nonuniversal cones. Under a change of observer the nonzero value can change, even though failure of cone equality cannot disappear. In anisotropic or mixed-order systems, the primary object is the family of physical characteristic sets. A numerical distance between such sets needs an additional normalisation or background norm.

Likewise, a positive conformal rescaling preserves the vanishing of the universal-calibration defect, but generally changes spectra and response amplitudes. The several invariances must therefore be named separately. The strongest carrier-independent claim proved here will be invariance within the reconstruction class of Theorem 2, rather than under every imaginable coarse-graining.

3. Admissible reconstruction and a sufficient invariance theorem

3.1. State, measure, localisation and operator data

For clarity this section uses time-independent coefficients on a prescribed effective spatial manifold Σ. Let H_n be real Hilbert spaces, $K_n \geq 0$ self-adjoint, and $H=L^2\left(\Sigma, d\mu; R^r\right)$. Complexification is understood when spectral calculus requires it. A reconstruction contains $J_n: H_n \to H$, its adjoint $Q_n = J_n^{\dot{\iota}}$, a physical measure, and localisation operators. We use isometric J_n, with $J_n Q_n \to I$ strongly.

Definition 3 (admissibility in the static control class). The reconstruction is admissible if the preceding state and measure conditions hold, microscopic multiplication operations converge to the declared effective multiplication algebra, and the identified resolvents converge strongly:

$$\begin{aligned} J_n M_{\chi,n} Q_n f &\to M_\chi f, \\ J_n (K_n + z)^{-1} Q_n f &\to (K+z)^{-1} f \ (z>0). \end{aligned} \tag{7}$$

Here χ is a bounded smooth local test function and K is a nonnegative self-adjoint realisation of a local second-order spatial differential operator. Its physical wave system must have the positive hyperbolic form under investigation. Neither a common cone nor $\Delta_c = 0$ is assumed in admissibility. For wave limits outside this static positive class, the topology and well-posedness assumptions must be replaced explicitly.

Resolvent convergence in (7) gives, for every fixed f,

$$\begin{aligned} S_n(t) &\to \frac{\sin\left(t\sqrt{K_n}\right)}{\sqrt{K_n}}, S_n(t) \to_{\ker K_n} t\, I, \\ \sup_{|t|\le T} \| J_n S_n(t) Q_n f - S(t) f \| &\to 0. \end{aligned} \tag{8}$$

For completeness, $s \mapsto \sin\left(t\sqrt{s}\right)/\sqrt{s}$ is continuous on σ, vanishes at infinity and has value t at zero. Strong resolvent convergence therefore implies convergence of this function of the operators, by uniform approximation with resolvent functions. Also $\| S_n(t) - S_n(s) \| \le |t-s|$. A finite time mesh upgrades pointwise convergence to (8). This is an application of the spectral-convergence framework [16], not a claim of operator-norm convergence on all unit vectors.

Localisation is independent input. An arbitrary nonlocal unitary conjugation preserves spectra but changes the association between states and places. Resolvent or response convergence without that association cannot by itself reconstruct physical distance.

3.2. Comparison without presupposing the limiting cone

Consider two isometric reconstructions $J_n^{(i)} : H_n \to H^{(i)}$, $i = 1, 2$, of the same microscopic family. Let $V : H^{(1)} \to H^{(2)}$ be unitary, induced by one smooth spatial diffeomorphism and a smooth orthogonal physical-bundle map, with the density factor required by the measures. Thus it intertwines the local multiplication algebras. The time convention is initially the same in both descriptions.

Theorem 2 (reconstruction invariance from bounded-energy agreement). Suppose reconstruction 1 has limit (8). On a dense subspace $D \subset H^{(1)}$, assume

$$\| Q_n^{(2)} V f - Q_n^{(1)} f \| \to 0, C_f := \sup_n \| K_n^{1/2} Q_n^{(1)} f \| < \infty. \tag{9}$$

For every finite E, suppose also

$$\eta_n(E) := \left\| \left(J_n^{(2)} - V J_n^{(1)} \right) 1_{[0,E]}(K_n) \right\| \to 0. \tag{10}$$

Then reconstruction 2 has the response limit $V S(t) V^*$, uniformly on compact time intervals in the strong topology. If its effective local operator is denoted $K^{(2)}$, it equals $V K^{(1)} V^*$. Its physical characteristic sets are the common cotangent transform of those in reconstruction 1. Consequently universal null calibration is preserved. No equality of limiting cones is assumed in (9)–(10).

Proof. Put $A_n = J_n^{(2)} - V J_n^{(1)}$, so $\| A_n \| \le 2$, and split the difference of the reconstructed responses into an input mismatch and $A_n S_n(t) Q_n^{(1)} f$. Write $R_n^{(i)}(t) = J_n^{(i)} S_n(t) Q_n^{(i)}$. Since S_n commutes with the spectral projections, for $f \in D$ and $E > 0$,

$$\begin{aligned} e_n(T, f) &:= \sup_{|t| \le T} \| R_n^{(2)}(t) V f - V R_n^{(1)}(t) f \|, \\ e_n(T, f) &\le T \| Q_n^{(2)} V f - Q_n^{(1)} f \| + T\, \eta_n(E) \| f \| + \frac{2 T C_f}{\sqrt{E}}. \end{aligned} \tag{11}$$

The last term follows from the energy bound on the input and $\| S_n(t) \| \le T$. First take $n \to \infty$ at fixed E, then $E \to \infty$. Uniform boundedness and density extend convergence to every f. For $s > 0$, the Laplace transform of the response is $\left(K + s^2\right)^{-1}$. The bound $\| S(t) \| \le t$ permits dominated integration, so equality

of the responses identifies the resolvents and hence the self-adjoint operators. A smooth local unitary induced by a diffeomorphism transforms a differential principal symbol by its cotangent map and conjugation in the physical fibre; invertible conjugation cannot create or remove characteristic covectors. ▫

The theorem is a sufficient condition, not a classification of every admissible reconstruction. Condition (10) is deliberately imposed only on bounded-energy subspaces; requiring global operator-norm agreement would unnecessarily exclude natural discretisations. It does not imply that Δ_c is invariant under discarding a physical sector. The retained sector content is part of the object being compared.

3.3. Fourier and cell maps satisfy the criterion

Take a circle of circumference L, odd N, spacing $a = L/N$ and microscopic norm $\| q \|_a^2 = a \sum_j |q_j|^2$.

Trigonometric interpolation J_a uses modes $k = 2\pi n/L$, $|n| \le (N-1)/2$; $Q_a = J_a^{\dagger}$ samples the Fourier projection of the input. The local map E_a assigns q_j to the centred cell of width a, with adjoint F_a given by cell averages. Both are isometries into $L^2(S^1, dx)$.

For the nearest-neighbour unit-speed operator L_a, its eigenvalue and elementary bounds are

$$v_a(k) = \frac{4}{a^2}\sin^2\frac{ak}{2}, \quad \frac{4}{\pi^2}k^2 \le v_a(k) \le k^2 \quad (|ak| \le \pi). \tag{12}$$

The cell estimate $\| (E_a - J_a) q \|_2 \le (a/2) \| \partial_x J_a q \|_2$ from [20] therefore yields

$$\| (E_a - J_a) 1_{(0,E]}(L_a) \| \le \frac{\pi a \sqrt{E}}{4}. \tag{13}$$

For $f \in H^1$, $\| F_a f - Q_a f \|_a \le a(1/\pi + 1/2) \| f' \|_2$ and $\| L_a^{1/2} Q_a f \|_a \le \| f' \|_2$. Thus (9)–(10) hold with $V = I$. For $K_a = L_a \otimes D_0$ one uses the positive lower eigenvalue of D_0 in the energy bound. Multiplication by sampled smooth χ has the same local limit for both maps: for cells this follows from uniform continuity; for Fourier interpolation it follows first on fixed trigonometric polynomials and then by density and uniform boundedness. The resolvent limit for Fourier maps follows directly from (12), and for cells from the same bounded-energy comparison applied to resolvents. Hence both are admissible.

This also states precisely what is reconstruction independent: the common limiting response and its local characteristics. Finite-spacing response amplitudes and approximation rates need not coincide. A fixed finite spectral cutoff is excluded from a strict local completion: its nonzero trigonometric response kernel cannot vanish on an open interval [20]. Removing the cutoff, with a controlled limit, is part of recovering strict support.

A restricted change of microscopic carrier is also controlled. Replace nearest-neighbour weights by fixed finite-range weights $w_s \ge 0$, with $w_1 > 0$ and $\sum_s w_s s^2 = 1$. The spatial eigenvalue becomes $v_a^w(k) = 4a^{-2}\sum_s w_s \sin^2(sak/2)$. On the retained Fourier band it obeys $4w_1 k^2/\pi^2 \le v_a^w(k) \le k^2$ and $0 \le k^2 - v_a^w(k) \le (a^2 k^4/12)\sum_s w_s s^4$. Taylor's bound and the preceding spectral-calculus argument therefore give the same local response for fixed positive D_0 and either reconstruction. These weights form a concrete universality class with the same second moment and different finite-spacing corrections. This does not establish universality under arbitrary graph rewiring or a change of physical sector content.

3.4. A limited categorical formulation

Consider the groupoid whose objects are the effective local response systems just specified, including their measures, local observable algebras and physical sector content. Arrows are smooth local unitaries V intertwining the responses; a common positive time-unit change may be included as $K^{(2)} = \beta^{-2} V K^{(1)} V^{\dagger}$. Identities, inverses and composition preserve these conditions. Assignment of the physical characteristic

family is a functor to the groupoid of cone families with their induced common cotangent maps. Equality of all sector cones is preserved by this functor.

Two admissible reconstructions are equivalent when their limiting systems are isomorphic in this groupoid. Theorem 2 gives a checkable sufficient condition for such equivalence. This is the precise DRD interface used here: carrier, morphism and preserved structure are specified separately [24]. It neither identifies a DRD profile with the null cone nor imports a suspension chain, Planck cutoff or S^3 topology. General noninvertible coarse-grainings are not arrows of this groupoid; a larger category would require explicit rules for loss of sectors and localisation data.

4. Spectral length–time correspondence with error control

4.1. Two spatial scales and a non-circular comparison

Let $L\phi_n=\lambda_n\phi_n$ be a spatial spectral problem with a specified measure and localisation map. For $\lambda_n>0$ define $\ell_n^{spec}=\lambda_n^{-1/2}$; for a positive angular frequency define $\tau_{j,n}^{dyn}=\omega_{j,n}^{-1}$. These are reduced spectral scales, not a geodesic distance and a full oscillation period. Choosing full wavelengths and periods inserts the same factor 2π into both. Zero modes have no finite inverse-frequency scales and are excluded from these definitions, but not from the dynamics.

If one defines $\lambda_n:=\omega_n^2/c^2$ from the outset, $\ell_n/\tau_n=c$ is an identity. The following result instead starts with an independently specified spatial operator, separate kinetic/stiffness data and an explicit approximation regime.

Conditional Proposition 3 (spectral correspondence). Let

$$T_g=\begin{pmatrix}0 & 0\\ 0 & T_{11}\end{pmatrix}, T_{11}\geq g^2 I, D=\begin{pmatrix}D_{00} & D_{01}\\ D_{10} & D_{11}\end{pmatrix}>0. \tag{14}$$

Write $d_1\leq\cdots\leq d_r$ for the eigenvalues of D_{00}, $v_j=\sqrt{d_j}$, $b=\|D_{01}\|$, and $d_¿=\|D\|$. If $v>0$ and $v\,d_¿\leq g^2/2$, the lowest r eigenvalues $z_j(v)=\omega_j^2(v)$ of T_g+vD obey

$$0\leq v\,d_j-z_j(v)\leq\frac{2v^2b^2}{g^2}, 0\leq v_j-\frac{\omega_j(v)}{\sqrt{v}}\leq\frac{2vb^2}{g^2v_j}. \tag{15}$$

On the prescribed nearest-neighbour chain, use $v=v_a(k)$ from (12), with $k\neq0$. The spectral scale is $v_a(k)^{-1/2}$; the independently reconstructed Fourier scale is $|k)^{-1}$. They satisfy

$$0\leq v_j-\frac{\omega_j(v_a(k))}{|k)}\leq\frac{v_ja^2k^2}{12}+\frac{2k^2b^2}{g^2v_j}. \tag{16}$$

In particular, if $D_{00}=v^2I_r$, every retained branch has the same limiting length–time ratio v, and the same common recalibration makes those ratios tend to one. The convergence is uniform on $0<|k)\leq\Lambda$ whenever $a\Lambda\to0$, $\Lambda/g\to0$, and D remains fixed. These spectral conditions are weaker than bounds needed for some response norms; they do not replace a response-convergence theorem.

Proof. Restricting the Rayleigh quotient to the first block gives $z_j\leq v\,d_j$ by the min–max principle. For x in that block and y in its complement, Young's inequality gives

$$2v|\langle x,D_{01}y\rangle|\leq\frac{2v^2b^2}{g^2}\|x\|^2+\frac{g^2}{2}\|y\|^2. \tag{17}$$

Thus T_g+vD is bounded below in quadratic-form order by the block diagonal matrix with blocks $vD_{00}-2v^2b^2g^{-2}I$ and $(g^2/2)I$. All eigenvalues of its first block are at most $g^2/2$, by the assumed

smallness condition. Min–max gives the lower bound in (15); positivity of $T_g + \nu D$ ensures positive frequencies. Dividing the difference of squares by $\nu_j + \omega_j / \sqrt{\nu} \geq \nu_j$ gives the frequency estimate.

Taylor's inequality gives $0 \leq k^2 - \nu_a(k) \leq a^2 k^4/12$, and hence $0 \leq 1 - \sqrt{\nu_a(k)}/|k| \leq a^2 k^2/12$. Split $\nu_j - \omega_j/|k|$ into the lattice contribution and the gap contribution, and use $\nu_a(k) \leq k^2$. This proves (16) and its uniform consequence. ▫

At fixed finite g on a fixed circle, refinement alone need not eliminate the second term. Either the gap must increase or a genuine long-wavelength family must make $|k|/g$ small. A fixed circle has a smallest nonzero Fourier mode; saying "$k \to 0$" without changing the family is insufficient. If $D_{01} = 0$, the gap correction vanishes identically. Different positive finite-range chains with the same second moment have the same limit, while their fourth moments modify the lattice term [20].

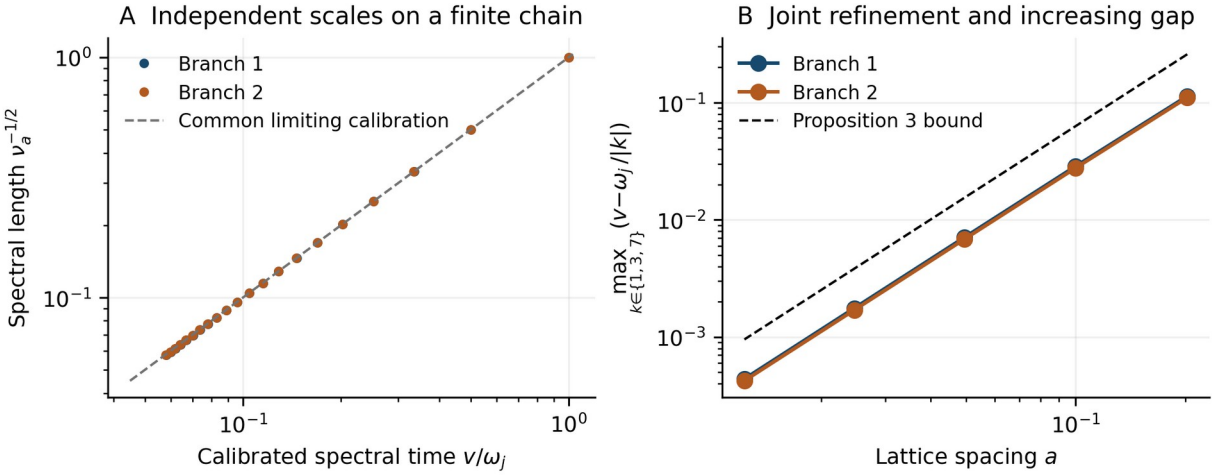


Figure 2. A: independently defined spectral lengths and calibrated dynamical times for two low branches at finite spacing. B: refinement with $g = 4/a$ makes both Fourier length–time ratios approach $\nu = 1.3$, within the bound of Proposition 3. The two branches have a common leading cone and different finite-scale dispersions.

4.2. What survives reconstruction and what is operational?

Changing J_a to E_a does not change the finite-dimensional frequencies or spatial eigenvalues. It changes their representation as effective fields. Under the general response isomorphisms of Section 3, the dynamical operators are unitarily equivalent after a common time recalibration. A separate spatial operator L and its spectral scales are preserved only if that operator is also part of the intertwined data. Cone equality alone does not imply isospectrality; arbitrary conformal changes are a counterexample.

For a local wave equation on a specified ultrastatic geometry, signal travel and an observer's clock can operationally calibrate length. On the circle, the earliest causal travel time between two points is $d_{S^1}(x, y)/\nu$; a local round-trip experiment measures distance as half the round-trip time times ν. This uses the reconstructed metric, an orientation and controllable emission and detection. Proposition 3 alone establishes a spectral correspondence. It does not turn $\lambda_n^{-1/2}$ into the distance between two arbitrary events or construct a measuring observer from R.

5. Quantum dynamics and emergent microcausality

5.1. Canonical normalisation, including the zero mode

Consider the nearest-neighbour chain on S^1 with speed $c > 0$, $K_a = c^2 L_a$, and canonical Hamiltonian

$$H_a = \sum_j \frac{\pi_j^2}{2a} + \frac{c^2}{2a} \sum_j (q_{j+1} - q_j)^2, [q_i, \pi_j] = i\hbar \delta_{ij} I. \quad (18)$$

Thus $\dot{q}_j = \pi_j / a$. On $L^2(R^N, dq)$, with the usual Schrödinger representation, first use the canonical rescaling $y_j = \sqrt{a}\, q_j$, $p_j = \pi_j / \sqrt{a}$. A real orthogonal normal-mode transformation then reduces H_a to one unit-mass free particle and $N-1$ positive-frequency oscillators. The resulting nonnegative self-adjoint Hamiltonian generates a strongly continuous unitary group. Essential self-adjointness can also be read directly on the Schwartz core in these normal coordinates. No oscillator vacuum is assigned to the free mode.

Let $e_0 = L^{-1/2}$ and $e_{n,\sigma}$ be the real orthonormal sine/cosine basis, with $n \geq 1$ and σ labelling sine and cosine. Define $Q_0 = \langle e_0, q \rangle_a$ and $P_0 = \langle e_0, \dot{q} \rangle_a$. Then $[Q_0, P_0) = i\hbar$ and $Q_0(t) = Q_0 + t\, P_0$. Nonzero frequencies are

$$\omega_{a,n} = \frac{2c}{a} \sin \frac{a k_n}{2}, k_n = \frac{2\pi n}{L}, 1 \leq n \leq \frac{N-1}{2}; \omega_n = c\, k_n. \backslash(19\backslash)$$

One convenient common Hilbert space is

$$H = L^2(R, dQ_0) \otimes F_s\left(\ell^2\{(n,\sigma) : n \geq 1\}\right). \backslash(20\backslash)$$

Each finite oscillator space embeds by placing the omitted modes in their vacuum. The identification uses the creation operators of the corresponding finite frequencies; it is specified, not inferred from an arbitrary abstract Hilbert-space isomorphism. On (20) extend the normal-ordered finite generator by leaving omitted modes dynamically idle:

$$\begin{aligned} \overline{H}_a &\;\text{¿}\; \frac{P_0^2}{2} + \sum_{n \leq (N-1)/2, \sigma} \hbar \omega_{a,n} N_{n,\sigma}, \\ H &\;\text{¿}\; \frac{P_0^2}{2} + \sum_{n \geq 1, \sigma} \hbar \omega_n N_{n,\sigma}. \end{aligned} \qquad \backslash(21\backslash)$$

On the embedded finite system, $\overline{H}_a$ differs from H_a only by its oscillator zero-point energy. Subtracting this scalar fixes an otherwise divergent overall phase; it does not affect Heisenberg fields or commutators. The series defining H is the nonnegative second-quantised generator on its natural domain.

5.2. Statement and proof

For a real spatial test function f, let

$$\Phi_a^J(t,f) = a \sum_j (Q_a f)_j q_j(t), \Phi_a^E(t,f) = a \sum_j (F_a f)_j q_j(t). \backslash(22\backslash)$$

For a real $F \in C_c^\infty(R \times S^1)$, spacetime smearing means $\Phi_a^J(F) = \int \Phi_a^J(t, F_t)\, dt$, and similarly for E. The continuum field is the usual massless mode expansion, with its full free zero-mode term retained. Write

$$E(F,G) = \int\int \langle F_t, S(t-s) G_s \rangle_{L^2} dt\, ds, S(t) = \frac{\sin\left(t c \sqrt{-\partial_x^2}\right)}{c\sqrt{-\partial_x^2}}. \backslash(23\backslash)$$

Theorem 4 (unitary quantum continuum control). For the model (18)–(22):

1. Each finite evolution is unitary. After the explicit embeddings and energy subtraction in (21), $e^{-it\overline{H}_a/\hbar} \to e^{-itH/\hbar}$ strongly, uniformly for t in compact intervals.

2. The field commutator is exactly the classical response with the physical measure normalisation:

$$\begin{aligned} [q_i(t), q_j(s)) &\;\text{¿}\; -\frac{i\hbar}{a} [S_a(t-s))_{ij} I, \\ [\Phi_a^J(t,f), \Phi_a^J(s,g)) &\;\text{¿}\; -i\hbar \langle Q_a f, S_a(t-s) Q_a g \rangle_a I. \end{aligned} \qquad \backslash(24\backslash)$$

The analogous identity holds with F_a. Both spacetime-smeared commutator distributions converge to $-i\hbar E(F,G) I$.

3. On (20), the corresponding spacetime-smeared Weyl operators $W_a(F)=\exp\left(i\Phi_a(F)\right)$ converge strongly to $W(F)$ for either reconstruction. If the compact supports of F and G are spacelike separated in the limiting cylinder, then

$$\left\| \left[W_a(F), W_a(G)\right] \right\| = 2\left|\sin\frac{\hbar E_a(F,G)}{2}\right| \to 0, \left[W(F), W(G)\right]=0. \quad (25)$$

The same assertion holds for spatially smeared fields at two fixed spacelike separated time slices. The limiting free-field Weyl net therefore satisfies microcausality. This is a free scalar control theorem, not a derivation of quantum mechanics, interacting QFT or gravity.

Proof of part 1. Finite-mode diagonalisation gives the self-adjoint finite Hamiltonians. Vectors consisting of an arbitrary zero-mode wavefunction and a finite linear combination of finite-occupation, finitely supported oscillator states form a dense subspace of (20). For each such vector, only finitely many $\omega_{a,n}$ enter; they converge to ω_n, and the free zero-mode unitary is identical for all a. The finite phase factors converge uniformly on compact time intervals. Density and unitary norm one give the asserted strong convergence on all of H. This does not assert operator-norm convergence of the unitary groups or of their generators.

Proof of part 2. The Heisenberg solution is $q(t)=C_a(t)q(0)+S_a(t)\pi(0)/a$, where $C_a=\cos\left(t\sqrt{K_a}\right)$. The canonical commutators and the sine subtraction identity give (24), first on the common invariant core; the right-hand side is a bounded central operator. Its coefficient after spatial smearing is the identified classical bilinear response. Equation (8), and the Fourier–cell comparison of Theorem 2, give convergence uniformly on compact time differences for fixed spatial tests. The uniform bound $\left|\langle f, S_a(t)g\rangle_a\right| \le |t| \left\| f \right\|_a \left\| g \right\|_a$ then gives (23) for spacetime tests by approximation with finite sums of separated smooth functions and dominated integration. These bounds also make the kernels continuous distributions. Finite a retains the analytic tails of the classical response.

Proof of part 3. In the real Fourier basis, the nonzero-mode part of $\Phi_a^J(t,f)$ has creation and annihilation coefficients proportional to

$$\sqrt{\frac{\hbar}{2\omega_{a,n}}}\, f_{n,\sigma} e^{\pm i\omega_{a,n}t}, f_{n,\sigma}=\langle e_{n,\sigma}, f\rangle. \quad (26)$$

For retained modes, $\omega_{a,n} \ge (2c/\pi)k_n$. Thus inverse square-root frequencies are uniformly bounded and the Fourier coefficients of a fixed smooth test provide a summable majorant. Pointwise frequency convergence and dominated convergence give convergence of the coefficient vectors in ℓ^2, uniformly on compact times. For cells, the input mismatch $\left\| F_a f - Q_a f \right\|_a \to 0$ gives the same conclusion, since the least nonzero frequency has a positive uniform lower bound. The zero-mode averages agree exactly for both maps, so its coefficients converge without an infrared prescription. Time integration preserves these conclusions. Strong continuity of the Fock and Schrödinger Weyl representations proves the asserted strong convergence of the bounded operators.

Finally the Weyl relations and the central commutator give the norm identity (25). The limiting d'Alembert kernel has support in $d_{S^1}(x,y) \le c|t-s|$. Hence $E(F,G)=0$ for spacelike separated supports. Finite products and linear combinations generate commuting local algebras, giving microcausality in the usual algebraic sense [30, 31]. ▫

These constructions also give convergence of products of the indicated Weyl observables in any fixed vector state on (20). For example, the tensor product of a normalised zero-mode wavepacket and the nonzero-mode Fock vacuum is permitted. There is no normalisable stationary ground state of the full massless scalar on this compact space, because its zero-mode Hamiltonian is a free particle. A limit of vacuum states is therefore not silently assumed.

The difference between strong and norm statements matters: the individual Weyl operators converge strongly in the chosen representation, while the spacelike commutator norms tend to zero because their scalar symplectic coefficients do. We do not claim norm convergence of the individual Weyl operators.

5.3. Uniform exterior control and the essential zero mode

For the site-density kernel $G_{a,ij}(t)=\left[S_a(t)\right)_{ij}/a$, the exterior estimate proved in [20] is

$$\inf_{0\le t\le T} d_a(i,j)>ct+\delta \Rightarrow \left|G_{a,ij}(t)\right)\le \frac{2T}{a\left(1-e^{-2\theta L/a}\right)}e^{-b(T,\delta)/a}, \tag{27}$$

$$\theta \qquad \text{¿}\operatorname{arcosh}\left(1+\frac{\delta}{cT}\right), \tag{28}$$

$$b(T,\delta) \quad \text{¿}\, 2\{\delta\theta - cT(\sinh\theta-\theta)\}>0.$$

Equation (24) transfers this bound directly to the coefficient of the coordinate commutator, with factor $\hbar$. It is a bound on a central coefficient, not an operator-norm bound on an unbounded field. It holds for every fixed $T,\delta>0$; uniform boundary convergence at $\delta=0$ is not claimed. The exact finite system has no sharp causal support. Distributional convergence and (27) are compatible with oscillations near the limiting front.

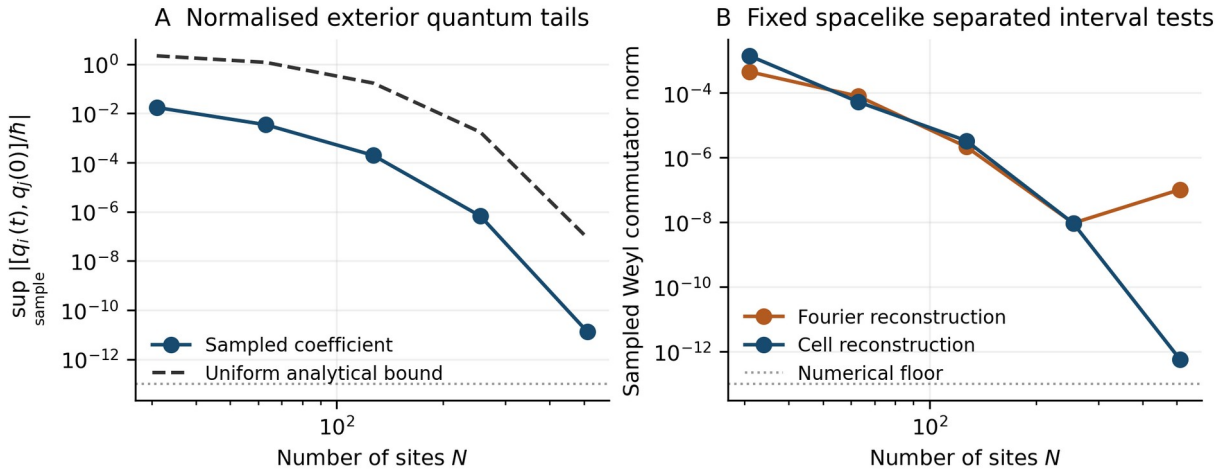


Figure 3. A: sampled exterior coordinate-commutator coefficients for $c=\hbar=1$, $T=0.8$, $\delta=0.25$, compared with the bound over the full time interval. B: commutator norms of two fixed interval-smeared Weyl observables, using Fourier or cell reconstruction. Their continuum commutator vanishes. Fourier errors need not decrease monotonically. Values near the dotted numerical floor do not establish exact zeros.

Deleting the scalar zero mode is not an innocuous way of obtaining a vacuum while keeping the same local field. If P_0 projects onto constants, then the altered response is

$$S^{\perp}(t)=S(t)-tP_0, G^{\perp}(t;x,y)=G(t;x,y)-\frac{t}{L}. \tag{29}$$

Outside the wave cone, the full kernel is zero but the second kernel is generally $-t/L$. The reduced equal-time bracket is likewise $\delta(x-y)-1/L$. Removing a canonical degree of freedom changes the observable algebra. Mean-zero test observables, derivative fields or a mass regulator can be treated consistently as different constructions; one cannot remove the mode and still claim locality of the original scalar field for arbitrary local tests.

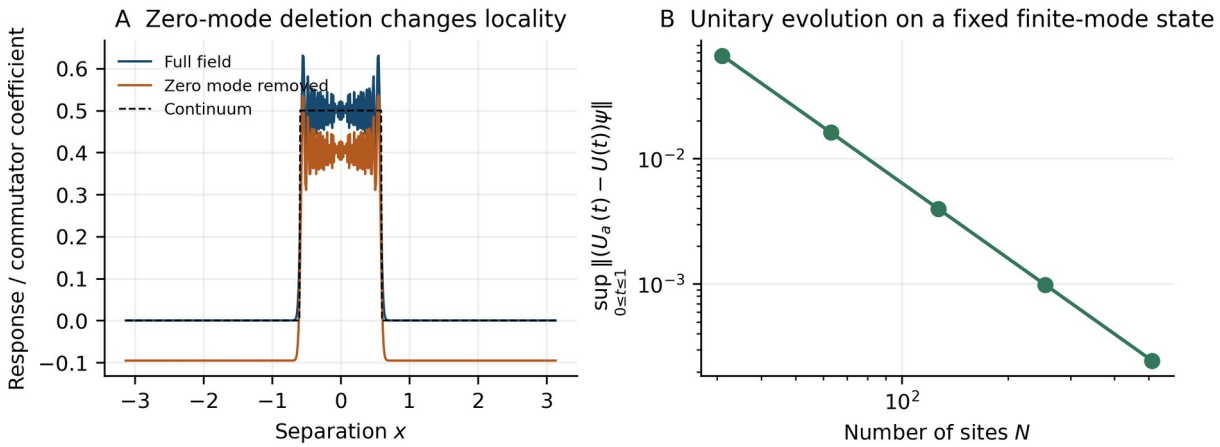


Figure 4. A: removing the zero mode shifts the kernel by $-t/L$, producing a nonlocal exterior contribution even after refinement. B: strong unitary convergence illustrated on a fixed superposition of the vacuum and a finite-occupation state. The free zero-mode evolution is identical in the two descriptions and cancels from this norm comparison.

The existence of rigorous lattice-to-field constructions is established prior work. In particular, Morinelli, Morsella, Stottmeister and Tanimoto build local free-field nets through wavelet scaling maps, with results on states, dynamics and locality [32]. Harmonic Lieb–Robinson bounds [8] control finite-lattice propagation. Our theorem is a simpler massless finite-volume control with an explicit zero-mode representation, calibrated wave speed and comparison of two reconstruction maps. It is not a claim that compatibility of microscopic unitarity with continuum locality is newly discovered.

6. Orientation, clocks and the conformal scale

6.1. Four distinct structures

An unoriented Lorentzian cone, agreement of sector calibrations, a time orientation and a thermodynamic arrow are four different data. A deterministic orientation selector $\epsilon(R) \in \{+1, -1\}$ cannot depend only on time-reversal-even data and obey $\epsilon(\Theta R) = -\epsilon(R)$ when $\Theta R = R$: those requirements contradict each other. This elementary obstruction from [20] survives quantisation; unitarity does not select a future.

Given a smooth Lorentzian conformal cone field, selecting a continuous component of the timelike cones is precisely the additional time-orientation datum. Such a selection exists globally only when the spacetime is time orientable; equivalently, it admits a continuous nowhere-zero timelike vector field. A globally monotone clock is stronger. For example, a temporal function with timelike gradient supplies a global orientation and excludes closed causal curves, whereas a time-oriented spacetime may still contain such curves. Standard causal-hierarchy and low-regularity qualifications are discussed in [29]. Our control cylinder has an externally specified nonperiodic time parameter; making that time periodic would preserve the local cone while introducing closed timelike curves.

A retarded response further includes an initial-value or preparation convention. A thermodynamic arrow would require additional statistical and state information. Neither is a consequence of setting $c = 1$.

6.2. Relative calibration leaves an arbitrary common scale

Choose a representative g of the reconstructed conformal class and a local unit timelike observer u, using length units so that $g(u,u) = -1$. For a tangent displacement X, set

$$\tau_u(X) = -g(u,X), \quad X_\perp = X - \tau_u(X)u, \quad \ell_u(X) = \sqrt{g(X_\perp, X_\perp)}. \tag{30}$$

On the null cone, $\ell_u(X) = |\tau_u(X)|$. Restoring a time unit converts this relation to $d\ell = c\, d\tau$. Replacing g by $\Omega^2 g$ and u by $\Omega^{-1}u$ multiplies both τ_u and ℓ_u by Ω. Their null equality remains, with no choice of Ω. Thus $c = 1$ fixes a relative calibration and does not fix the common conformal scale.

If a smooth positive spacetime density $d\mu$ is additionally supplied in dimension D, a representative is selected by

$$\tilde{g}=\Omega^2 g,\Omega=\left(\frac{d\mu}{d\,vol_g}\right)^{1/D},d\,vol_g=\sqrt{|det\,g|}\,d^D x.\ \backslash(31\backslash)$$

This is a conditional reconstruction from compatible scale data. Microscopic vertex counting alone does not provide a physical spacetime density. Clock and rod standards are another possible route but must be physically realised and compared, rather than stipulated to prove their own compatibility.

The causal-reconstruction literature is relevant with its hypotheses intact. Hawking–King–McCarthy relate a path topology to differential and conformal structure in their causal setting [25]; Malament strengthens the relation between timelike curves, topology and causal distinguishability [1]. These are not theorems that an arbitrary relation is a spacetime. The EPS programme uses light trajectories, free fall, compatibility conditions and further clock/scale assumptions to reach progressively richer geometry [26–28]. The additional assumptions are precisely why one cannot replace the full metric by a bare null cone. None of these sources derives gravitational field equations from the calibration convention.

7. Interactions, gauge sectors and the limits of universality

A common free symbol need not be stable under interactions. Anber and Donoghue calculate convergence of limiting speeds under specific renormalisation-group flows, with slow logarithmic convergence in some regimes [4]. Strongly coupled holographic examples can instead suppress deviations by powers of an infrared-to-ultraviolet scale ratio [5]. These mechanisms concern specified interacting theories and already defined field content; they are not implications of a symmetric adjacency matrix. The radiative-stability problem of [12], and continuing QED analyses such as the 2026 revision of [34], make sector-dependent corrections an explicit part of the question. We import no observational bound or gauge-symmetry claim from the latter preprint.

Effective metrics in coupled media likewise require matching conditions [6]. A recent Euclidean scalar–tensor construction obtains a Lorentzian massless tensor dispersion while other degrees of freedom generally remain elliptic [35]. This is a useful counterpoint: obtaining one Lorentzian branch is not obtaining one universal physical cone. These examples support a conditional research programme while preventing an inference of automatic universality.

Gauge reduction must precede a characteristic comparison. For vacuum Maxwell theory on a given Lorentzian background, the potential symbol has the schematic form $k^2\delta^\mu{}_\nu - k^\mu k_\nu$ and annihilates a pure-gauge amplitude proportional to k for every covector. A raw determinant is therefore useless. After the gauge quotient and physical constraints, the propagating polarisations have $k^2=0$. In more general diffeomorphism-invariant theories the reduced characteristic polynomial can contain distinct quadratic and quartic factors [33]. Accordingly, the scalar compression criterion is a control model, not a replacement for physical characteristic analysis in gauge fields or gravity.

Next stability problem. On a fixed-rank physical bundle, determine whether interactions and admissible elimination preserve $\Delta_c=0$, and whether its transverse deviations are irrelevant under a calculated scale flow. This requires a specified action, regulator, state class and symmetry constraints. No interacting beta function is derived in this article. The theorem about calibration cannot answer that separate dynamical question.

8. Reproducible calculations

All new calculations are produced by reproduce_v3.py; the filename records the internal research step, not the article title. The script writes results_v3.json, four PNG figures at 300 dpi and their vector PDF counterparts. It uses Python 3.12.13, NumPy 2.3.5, SciPy 1.17.0 and Matplotlib 3.10.8. The seed is 20260906 and is used only for a reproducible basis rotation. Parameters and raw sampled maxima are recorded without rounding in the JSON. No external dataset or observational fit enters the calculations.

The pencil test uses speeds $(1, 1.5, 2)$ and an invertible nonorthogonal change of basis applied to both kinetic and spatial forms. Its generalised eigenvalues reproduce $(1, 2.25, 4)$, and $\Delta_c = \log 2$ is unchanged over common factors from $1/4$ to 4.

For the two low and two heavy modes, $L = 2\pi$, $N \in \{31, 63, 127, 255, 511\}$, $g = 4/a$, $v = 1.3$, and

$$D = \begin{pmatrix} 1.69 & 0 & 0.35 & 0.10 \\ 0 & 1.69 & -0.20 & 0.40 \\ 0.35 & -0.20 & 2.10 & 0 \\ 0.10 & 0.40 & 0 & 2.70 \end{pmatrix}, T_g = diag(0, 0, g^2, g^2). \backslash(32\backslash)$$

Every tabulated mode $k \in \{1, 3, 7\}$ satisfies the smallness hypothesis of Proposition 3. At $N = 511$, $k = 7$, the two Fourier ratios are approximately 1.29956116 and 1.29957722, with a common target 1.3; the absolute-error bound is 9.53148×10^{-4}. The finite-mode interbranch defect is 1.23598×10^{-5} despite the exact leading condition $D_{00} = 1.69\, I_2$.

For quantum tails, $c = \hbar = 1$, $T = 0.8$, $\delta = 0.25$, and 801 time samples are used. For the Weyl tests, f and g are unit-L^2 periodic indicator functions of intervals with half-width 0.15 and centres 0 and 1.2. Their supports have distance 0.9, while the tested time difference is in $[0, 0.7)$ with 701 samples. The continuum bilinear response is therefore exactly zero. These L^2 time-slice tests are allowed by the bounded-response and Weyl-coefficient estimates in Theorem 4; spacetime test distributions in the theorem itself use smooth functions. Fourier coefficients and cell averages of the intervals are evaluated analytically, not by a spatial quadrature.

N	Exterior coefficient	Weyl norm: Fourier	Weyl norm: cells
31	1.74271×10^{-2}	4.45368×10^{-4}	1.38756×10^{-3}
63	3.47458×10^{-3}	7.77902×10^{-5}	5.19459×10^{-5}
127	1.96427×10^{-4}	2.15010×10^{-6}	3.29905×10^{-6}
255	6.69764×10^{-7}	9.38007×10^{-9}	9.26398×10^{-9}
511	1.34687×10^{-11}	1.02148×10^{-7}	5.73035×10^{-13}

Table 1. Sampled maxima of three different quantities. The coordinate coefficient uses the measure normalisation $1/a$; the Weyl norms concern fixed extended observables. The Fourier sequence is nonmonotone, as permitted by strong convergence. The last cell value is close to the conservative 10^{-13} numerical floor.

The unitary comparison uses ¿, tensored with any fixed normalised zero-mode wavepacket. The supremum of the vector-norm difference on $[0, 1)$ is calculated analytically from the two frequency sums: it decreases from approximately 6.62429×10^{-2} at $N = 31$ to 2.44989×10^{-4} at $N = 511$. This example illustrates convergence on a fixed state, not on all states uniformly. Proofs, rather than plots or sampled maxima, establish the limiting assertions.

9. Earlier compact-sphere models: retained controls and explicit corrections

9.1. Geometry determines travel times only after calibration

As an optional effective control, take the ultrastatic spacetime $(R \times S^3_R, -c^2 dt^2 + g_R)$. Its round spatial diameter is πR, and causal access from a fixed point to the whole spatial slice becomes possible at $T = \pi R/c$. This follows from the geodesic distance bound and the null condition, not from a selection equation for c. The diameter is a metric quantity, not a topological invariant.

For a fixed source point, homogeneity makes the accessible *volume fraction* independent of that point. With $u = \min(c\,T/R, \pi)$,

$$\Gamma(T) = \frac{u - 1/2 \sin(2u)}{\pi}. \backslash(33\backslash)$$

The geodesic ball volume used here is $4\pi R^3 \int_0^u \sin^2\chi \, d\chi$. This restores a missing factor of π in the displayed proof in [21]; the final normalised fraction there was correct. The phrase "some (hence all) source points" should not be applied to membership in a fixed ball: homogeneity equates the fractions, not the sets. Complete accessibility means that an appropriate signal can be directed to each destination, not that a single null trajectory visits every point.

Compactness alone does not guarantee a finite access time in an evolving geometry. For $-c^2 dt^2 + R(t)^2 d\Omega_3^2$, coverage by time T requires $\int_{t_0}^{T} c\,dt/R(t) \geq \pi$. For example $R(t) = R_0 e^{H(t-t_0)}$ has finite total future angular reach $c/(H R_0)$, which may be below π. An unwrapped null angle is not bounded by π; the reached-ball fraction saturates there. These qualifications prevent the static result of [21] from being transferred unchanged to the evolving setting of [23].

The closure map $\Phi(c) = \pi R/(\pi R/c) = c$ has every positive c as a fixed point. Likewise choosing $M(c) = c^2 R \varphi^2/(2G)$ in the ansatz of [22] and substituting back into its speed equation returns the input for every $c > 0$. Neither operation selects a unique speed or establishes agreement of independent sectors. The remaining language of "selection" in [21, 22] is replaced here by an algebraic identity. A Fibonacci hierarchy can remain a declared historical ansatz; it enters none of the present proofs.

9.2. A corrected recurrence statement

Let $L = -\Delta_{S^3_R} \geq 0$ and $K_c(t) = e^{ict\sqrt{L}}$. This convention includes c explicitly. The wave equation is $\partial_t^2 \psi + c^2 L \psi = 0$. If $k_n = \sqrt{n(n+2)}/R$, the dispersion $\omega_n = c\,k_n$ has ratio $\omega_n/k_n = c$ exactly. Differentiating with respect to a different variable, such as the mode number divided by R, does not compute $d\omega/dk_n$; the group-velocity calculation in [21] mixed these variables. The half-wave operator is also not the retarded wave response and does not have its exact support property.

Proposition 5 (strong recurrence is not norm recurrence). For this K_c, every orbit $t \mapsto K_c(t)\psi$ is almost periodic in the Hilbert-space norm, and the full operator has no nonzero exact period. Nevertheless,

$$\| K_c(t) - I \| \geq \sqrt{2} \text{ for every } t \neq 0. \backslash(34\backslash)$$

It is therefore not almost periodic in operator norm. Moreover the operator $e^{ict\sqrt{L+R^{-2}}}$ has the same wave principal cone and has the exact period $2\pi R/c$.

Proof. Finite spectral sums have almost-periodic orbits, and approximate any fixed orbit uniformly in time by unitarity. A nonzero exact operator period would require both $ct\sqrt{3}/R$ and $ct\sqrt{8}/R$ to be integer multiples of 2π, contradicting the irrationality of $\sqrt{3/8}$.

For the norm bound put $\vartheta = ct/R$. Since $\sqrt{n(n+2)} = n + 1 + o(1)$, if $\vartheta \notin 2\pi Z$, the limiting phase sequence has either a dense circle closure or a nontrivial finite cyclic closure. Its supremum distance from 1 is at least $\sqrt{3}$, hence so is the operator norm. If $\vartheta = 2\pi m$, $m \neq 0$, choose $n = |m)$ and write

$$\alpha = \frac{2\pi |m)}{|m) + 1 + \sqrt{(|m) + 1)^2 - 1}}, \quad \frac{\pi}{2} < \alpha < \pi. \backslash(35\backslash)$$

The corresponding phase has distance $2\sin(\alpha/2) > \sqrt{2}$ from 1. Thus no nonzero time is a norm almost-period for tolerance below $\sqrt{2}$. Finally the shifted frequencies are $c(n+1)/R$, which proves their stated exact period. ▫

This corrects Theorem IV(ii) of [21], whose proof established strong orbit recurrence but whose statement asserted operator-norm recurrence. Conservation of $\| K_c(t)\psi \|$ is unitarity, not a theorem that a cosmological geometry cannot evolve or that no individual state can be periodic. The shifted example also shows that the common null cone does not determine detailed spectral recurrence.

9.3. The DRD connection has a precise boundary

The carrier–morphism distinction in [24] is retained. A homological profile, a spectral profile and a physical null structure have different preserving maps; calling each an invariant does not make them identical. The new reconstruction groupoid of Section 3 is defined independently of the more extensive DRD package.

There is one further distinction relevant to importing causal claims. For a homogeneous shell-complete spectral projector, its constant diagonal indeed gives $Tr(1_B \Pi 1_B)/Tr\,\Pi = Vol(B)/Vol(S^3)$. This is an exact *spectral-weight fraction*. It does not by itself count the number of independently accessible logical modes or establish a communication protocol. For a finite scalar band on the round sphere, analyticity makes restriction to any nonempty open set injective in the ideal noiseless mathematical sense, even when that trace fraction is small. Operational reconstruction requires accuracy, noise, energy and intervention constraints. Accordingly the stronger operational interpretation of the trace ratio in [24] remains conditional, and is not used to prove causal calibration here.

10. What is proved, what is assumed, what remains open

Proved within the stated classes. One common scalar unit normalisation exists exactly at scalarity of the positive compressed stiffness; relative-speed ratios obstruct it otherwise. The bounded-energy comparison implies equality of reconstructed response limits and preservation of their physical cone family. The independent spectral length–time ratios satisfy (15)–(16). Finite harmonic quantisation and its specified embeddings give strong unitary convergence, convergence of commutator distributions and a microcausal limiting Weyl net. Proposition 5 gives the stated recurrence correction. These proofs are conditional on their mathematical inputs, not on the philosophical hypothesis in Section 1.

Assumed. Positive quadratic wave dynamics, a time parameter for the control systems, specified physical sectors and measures, prescribed chain geometry, suitable operator domains and local reconstructions are inputs. Scalarity of D_{00} is tested by Theorem 1 and explicitly imposed in the common-cone examples; it is not generated dynamically by those examples. Quantum CCR and the representation choice are also inputs. Neither S^3 nor a Planck cutoff is an input to the central four results.

No-go results and non-implications. Distinct sector speeds cannot be equalised by a common unit change, by (4). A fixed-point identity cannot select a unique fixed point. Positive conformal rescaling leaves the cone while changing metric scale. Time-reversal-even data cannot supply a reversal-odd orientation selector. Spectral eigenvalues alone do not identify localisation operations. The convention $c = 1$ does not establish universality, and one massless branch does not test agreement with an independent branch. Finally, the definitions $\ell_P = \sqrt{\hbar G/c^3}$ and $t_P = \ell_P/c$ assign scales; a lower bound on measurable intervals is an additional physical postulate, not a mathematical consequence of those definitions.

The transition structure is therefore:

Transition	Status in this article
Neutral relation to stable local degrees of freedom	Open geometry-selection problem
Specified operators and maps to local response	Controlled chain result; sufficient general comparison
Several physical branches to one Lorentzian cone	Exact conditional matrix criterion; no generic attraction

Transition	Status in this article
Common cone to common unit calibration	Theorem 1, locally
Spatial spectral scales to dynamical times	Proposition 3 with explicit error bounds
Finite unitary network to microcausal quantum field	Theorem 4 on the prescribed cylinder
Unoriented cone to future and thermodynamic arrow	Separate orientation and state data required
Conformal class to a metric and its dynamics	Scale data required; gravitational dynamics open

Table 2. Status of the successive transitions. A controlled implication does not remove the hypotheses entering its left-hand side.

The next substantial theorem should start with a specified permutation-invariant relational dynamics or state ensemble and prove that a nontrivial class of its histories admits local reconstructions with a stable physical massless bundle. It should then establish that the compressed kinetic/spatial forms approach a common scalar class, with quantitative response estimates and controlled sector content. A prescribed chain is a benchmark for that theorem, not its conclusion. Internal clock sectors and accessible control algebras must eventually replace the external preparation parameter. Gauge fields and gravity require their own constraint and characteristic analysis before universality can be claimed across them.

The sphere, hierarchy, density-tracking and gravitational-selector ideas of earlier papers remain deferred hypotheses to the extent that they are logically compatible with these requirements. The identity-based speed selection and operator-norm recurrence claims do not survive. No Hubble-tension ansatz or numerical Planck calibration is used as evidence for the present results.

11. Conclusion

The defensible invariant content of the c programme is the existence, for specified physical sectors and admissible descriptions, of one Lorentzian characteristic structure. In the positive common-spatial-form class, this is exactly scalarity of the massless compression. A common natural calibration represents that condition by $c=1$; it does not explain why the condition holds.

The controlled results show that this distinction can be carried through spectral comparison, reconstruction changes and canonical quantisation. In particular, microscopic unitary systems with analytic tails can converge to a unitary local free field with exact microcausality, provided the full field algebra, including its zero mode, is treated consistently. What remains open is the origin and stability of the required geometric and intersector structures from neutral relational data. The programme is thereby a problem about existence and preservation of transitions between descriptions, rather than a computation of a dimensionful number.

Part IV. Functional consistency of light

1. From universal calibration to functional consistency

The preceding manuscript, *Universal Null Calibration*, is the base result of this paper [36]. It established a distinction between a common physical cone and its numerical representation by $c=1$. The next question is whether propagation, phase, energy, momentum, pressure and quantum response are independent primitives, or whether several of them are constrained by one effective variational structure.

We answer this question in a specified Maxwell sector. This choice is a physical anchor, not a derivation of electromagnetism from a neutral network. The principal symbol, metric derivative of the action, symmetry generators and boundary momentum balance are different mathematical constructions. Agreement among their predictions is consequently a useful test of a proposed effective description. Logical dependence within a theory must be distinguished from independence of experimental readout channels.

The main additions are the null-sector closure statement in Section 10, the operational pressure loop in Section 11, and the reconstruction conditions and counterexample in Section 18. Standard Maxwell, relativistic and quantum identities are derived as control results and are not claimed as new laws. The new programme-level content is the explicit dependency structure, including the extra data required to transport

it between descriptions. Sections 3–7 set the scope of the relational language; they do not supply assumptions hidden inside the physical proofs.

2. Results retained from the preceding manuscript

For a positive multicomponent wave pencil with one positive spatial form, the kinetically normalised symbol on the retained massless space is

$$P(\omega,k)=-\omega^2 I_r+h^{ij}k_i k_j D_{00}, D_{00}=D_{00}^T>0. \tag{2.1}$$

The common-cone theorem of [36] states

$$\text{one common branch cone} \Leftrightarrow D_{00}=v^2 I_r, D'_{00}=I_r \text{ after one common calibration}. \tag{2.2}$$

Here D_{00} is the compression of the normalised spatial form to the massless bundle; its rank and physical sector content are fixed. The dimensionless defect and optimal common rescaling are

$$\Delta_c=\frac{1}{2}\log\frac{\lambda_{max}(D_{00})}{\lambda_{min}(D_{00})}, \inf_{\rho>0}\max_j\left|\log(\rho v_j)\right|=\frac{\Delta_c}{2}. \tag{2.3}$$

Both statements concern the common spatial form and observer splitting in that model class. The nonzero numerical defect is not asserted to be a Lorentz scalar for arbitrary nonuniversal cones. Cone inequality itself cannot be removed by one common change of units. For one retained branch, $\Delta_c=0$ is automatic and supplies no intersector test.

Admissible reconstruction in [36] includes Hilbert-space state identification, measure, localisation and resolvent/response convergence. For isometric maps $J_n^{(i)}$, adjoints $Q_n^{(i)}$, microscopic $K_n\geq 0$, and a smooth local unitary V, sufficient comparison data are

$$\begin{aligned}\left\|Q_n^{(2)}Vf-Q_n^{(1)}f\right\|\to 0, \text{¿}n\left\|K_n^{1/2}Q_n^{(1)}f\right\|<\infty,\\ \left\|\left(J_n^{(2)}-VJ_n^{(1)}\right)1_{[0,E]}(K_n)\right\|\to 0\,(E<\infty),\end{aligned} \tag{2.4}$$

on a dense set of inputs. They give the same limiting sine response, up to V, strongly and uniformly on finite time intervals. Equality of limiting cones is a conclusion when the limiting operators are local; it is not inserted into these assumptions. Agreement of spectra without localisation does not suffice.

The controlled spectral correspondence compares independently specified $\lambda_n^{-1/2}$ and ω_n^{-1}, with lattice and eliminated-mode errors. It is not a microscopic theorem about the distance between arbitrary events. The chain quantum theorem retains finite-system unitarity and analytic response tails, while the continuum free scalar field has exactly microcausal local Weyl algebras. It gives strong, finite-time-uniform unitary convergence after the stated embeddings and energy subtraction. It does not give operator-norm convergence of the evolution groups.

The compact-space massless scalar has a free zero mode, with no normalisable ground state of that mode. Keeping the unreduced field requires keeping this mode: its response is $t\,P_0$, and deleting it spoils the cancellation needed for local scalar commutators. These results are retained, not reinterpreted as a photon construction. The earlier recurrence correction and compactness limitations are recalled in Section 20.

The exact SI value $299\,792\,458\,m\,s^{-1}$ belongs to the definition of the metre [18]. Neither an identity $c\mapsto c$ nor compactness selects a unique physical speed. A common cone, time orientation, conformal scale and global causal order remain distinct. Gauge reduction precedes characteristic comparison. The transition from a neutral relation to a stable local physical sector remains open.

3. Levels of description and interpretive scope

Definition 3.1 (Four different objects). A *relation* specifies permitted relational data; a *carrier* supports specified distinctions; a *reconstruction* maps controlled microscopic data to an effective description; an *observable* is a quantity in that description together with its readout convention. No identification of these four objects is assumed.

For example, a symmetric adjacency matrix considered up to vertex permutation is not a retarded response, a propagation cone or an oriented causal order. A spectral projection is not yet a spatial localisation map. A field amplitude is not a detector count before a detector model and a state have been supplied. This is the carrier/morphism interface used by the DRD programme [24]; no additional DRD theorem is needed here.

Interpretation 3.2 (Relational distinction). Phenomenological presence is not identified with independent substantial self-being. A distinction may belong to a relation rather than to either isolated pole. This motivates studying preserved distinctions and non-factorisation, but is not a consequence of topology or field theory.

The mathematical tasks are to specify objects, morphisms and preserved structures. We use *quasilocal manifestation*, *effective reconstruction* and *operational accessibility* for the relevant interfaces. No theorem in this paper describes physical reality arising from non-reality.

4. The two-point carrier and the Hopf interface

Let $S^0=\{-1,+1\}$, equipped with its discrete topology. Define a distinction on a target carrier Y by δ_Y¿ when $y_-\neq y_+$¿¿ and 0 otherwise.

Lemma 4.1 (Preservation of a two-point distinction). A map $f:S^0\to Y$ preserves the two-point distinction exactly when its restriction to the two poles is injective. Removing either pole leaves no pair on which this particular distinction can be instantiated.

Proof. The preserved predicate is $\delta_Y\big(f(-1),f(+1)\big)=1$, which is precisely injectivity on a two-element domain. A singleton has no two distinct elements. □

This elementary statement supplies a minimal carrier and a class of distinction-preserving maps; noninjective maps can erase the distinction. It does not attach a physical size or a microscopic constituent to S^0.

The suspension identity $\Sigma S^n\cong S^{n+1}$ gives a mathematical carrier hierarchy. With specified equatorial inclusions it can be written

$$S^0\to S^1\to S^2\to S^3. \tag{4.1}$$

The optional labels distinction, phase, direction and spinor total space refer to possible uses of these carriers; the arrows are not an evolution law. A rigorous non-factorisation example is the Hopf bundle

$$S^1\to S^3\xrightarrow{\pi}S^2,\pi(z_1,z_2)=\Big(2\Re\big(z_1\acute{z}_2\big),2\Im\big(z_1\acute{z}_2\big),\big|z_1\big|^2-\big|z_2\big|^2\Big), \tag{4.2}$$

where $\big|z_1\big|^2+\big|z_2\big|^2=1$ and the fibre acts by simultaneous phase multiplication [37]. It is not a global product: $\pi_1\big(S^3\big)=0$ whereas $\pi_1\big(S^2\times S^1\big)=Z$. A global section of this principal bundle would trivialise it and is therefore impossible.

An optical use of this construction needs care. Normalised two-component Jones vectors at a fixed propagation direction form an S^3, and quotient by overall optical phase gives the Stokes sphere. That sphere parametrises polarisation, not automatically spatial propagation direction. Its optical phase quotient is also not, without further construction, the spacetime gauge transformation $A\mapsto A+d\chi$. Mathematical non-factorisation, physical interaction and philosophical interdependence are three separate claims. The first does not prove either of the other two.

5. Predimensional data and four meanings of vacuum

Before a physical dimension functional has been constructed, the statement $dim_{phys}X$ is undefined can be appropriate. It must not be replaced by $dim_{phys}X=0$. The topological dimension of the discrete carrier S^0 is indeed zero; that is a different assertion. Likewise, unreconstructed distance $d(A,B)$ and clock separation $\Delta t(A,B)$ may be undefined rather than zero.

Topological dimension refers to a topological space; spectral dimension typically uses the scaling of a specified heat trace or return probability; operational spatial dimension concerns reconstructed independent spatial measurements; spacetime dimension includes the time direction; field-theory scaling dimensions concern fields and operators. These need not agree, and several are unavailable before their defining structures exist. The combinations $\ell_P = \sqrt{\hbar G/c^3}$ and $t_P = \sqrt{\hbar G/c^5}$ do not by themselves prove minimal discrete intervals.

Meaning of vacuum	Data and limitation
No detected photons	A detector, exposure and readout give no counts; this does not uniquely specify a quantum state.
QFT vacuum state	A distinguished state of a specified field algebra, usually selected using spacetime symmetry or another prescription.
Hamiltonian ground state	A lowest-energy state for a specified Hamiltonian, if one exists. It need not be unique or be the vacuum chosen in another description.
Relational ground sector	A hypothetical sector of microscopic relational dynamics; no photon algebra, metric or Hamiltonian ground-state property is inferred from the name.

Vacuum correlations will be examined only after the effective field algebra and hyperbolic dynamics have been specified. The compact scalar zero mode in Section 2 is a concrete warning against silently assuming a ground state.

6. Domain, possible accessibility and signal history

Let B be a declared domain of a physical description, with whatever boundary conditions are needed for its dynamics. Predicates of this model are defined on B; the model need not assign a truth value outside B. For an observer world tube O up to clock reading τ, the causal past $J_B^-(O_{\le\tau})$ bounds possible incoming access. Actual accessible information can occupy a smaller set because sensors and controls are limited. Let $H_{sig} \subset B \times B$ record actual emission–reception pairs. Then

$$\text{domain } B \neq \text{current accessibility} \neq \text{actual signal history } H_{sig}. \tag{6.1}$$

Two regions can share the same local dynamics and cone without having exchanged a photon in a given time window. Causal structure describes possible influences subject to the dynamics, not a census of photons.

A particle horizon depends on the available past history; an event horizon depends on an appropriate future boundary or asymptotic accessibility; a Hubble radius uses an expansion rate and is not generally a causal horizon. None is automatically B. A local cone field defines directed causal curves only after orientation, and a global causal relation can have pathologies unless extra conditions hold [29]. Global hyperbolicity is explicitly assumed when used below.

7. Present-state data and an internal observer

On a chosen Cauchy surface, complete classical data evolve as $X(t) = \Phi_{t-t_0}(X(t_0))$ when the flow exists and is reversible. In a specified unitary quantum representation, $|\psi(t)\rangle = U(t,t_0)|\psi(t_0)\rangle$. Neither equation grants an internal subsystem access to the full state.

Definition 7.1 (Operational observer data). An observer is an assumed stable quasilocal subsystem equipped with an accessible algebra A_O, a clock readout T_O over a stated working interval, admissible controls, record degrees of freedom and readout maps. No consciousness variable, perfect global clock or tensor factorisation of every local algebra is assumed.

Global state data, restrictions to A_O, stored records used to infer past states and conditional predictions of future readings are different objects. In particular, a clock readout need not be a self-adjoint operator canonically conjugate to a semibounded Hamiltonian on its full spectrum.

Lemma 7.2 (Finite-access nonuniqueness). If a unitary V commutes with every $A \in A_O$, the states ω and $\omega_V(A) = \omega(V^{¿} A V)$ agree on A_O even if they differ on the global algebra.

Proof. $V^{¿} A V = A$ on the accessible algebra. No inversion of the global unitary evolution can distinguish these restrictions from accessible data alone. □

Local commutativity supplies physical examples with controls in a spacelike separated region. Classically, finite domain of dependence gives the analogous indistinguishability of initial data that differ outside the relevant causal past. A time orientation chooses a cone component; retarded preparation specifies a response convention; records require a state and recording interaction; a thermodynamic arrow requires statistical and boundary information. Universal calibration selects none of these arrows.

8. One action and several different operations

Assumption 8.1 (Maxwell control sector). The effective domain is a smooth oriented, time-oriented, four-dimensional Lorentzian manifold with signature ¿. In the region studied the field is linear vacuum Maxwell theory with a positive constant action normalisation, with no dispersive material or nonlinear constitutive correction. The theory is considered locally in a trivial gauge patch; global topological flux sectors require separate data. Global hyperbolicity is additionally imposed for global retarded Green operators.

Unless SI factors are displayed, $c = 1$ and rationalised field units are used; $\hbar$ is retained. Start with

$$S_\gamma[A, g] = -\frac{\gamma}{4}\int_B F_{\mu\nu} F^{\mu\nu} \sqrt{-g}\, d^4x, F = dA, \gamma > 0. \tag{8.1}$$

The requested standard action is $\gamma = 1$. Keeping γ visible exposes a normalisation which the cone cannot determine. With compactly supported variations,

$$\delta_A S_\gamma = \gamma \int_B \left(\nabla_\mu F^{\mu\nu}\right) \delta A_\nu \, dvol_g, \nabla_\mu F^{\mu\nu} = 0, dF = 0, \tag{8.2}$$

$$T_{\mu\nu} := -\frac{2}{\sqrt{-g}} \frac{\delta S_\gamma}{\delta g^{\mu\nu}} = \gamma \left(F_{\mu\alpha} F_\nu{}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right). \tag{8.3}$$

The metric variation holds the covariant components of A fixed. Using $\delta\sqrt{-g} = -1/2 \sqrt{-g}\, g_{\mu\nu} \delta g^{\mu\nu}$ and $\delta(F^2) = 2 F_{\mu\alpha} F_\nu{}^{\alpha} \delta g^{\mu\nu}$ gives the second identity. Thus equations of motion and stress are derivatives with respect to different arguments of one functional.

The field equations imply $\nabla_\mu T^{\mu\nu} = 0$ in a source-free region. For a Killing field ξ, the current $J^\mu_\xi = T^{\mu\nu} \xi_\nu$ obeys

$$\nabla_\mu J^\mu_\xi = T^{\mu\nu} \nabla_{(\mu} \xi_{\nu)} = 0. \tag{8.4}$$

A global charge additionally needs a suitable integration surface and control of boundary flux. Generic curved spacetimes need not have a conserved global energy. In Minkowski space, translations provide the energy and momentum charges; gauge symmetry supplies constraints and identities rather than an additional freely propagating polarisation. These standard variational and conservation constructions are the anchors for the following calculations [38, 39].

Proposition 8.2 (Normalisation is invisible to the cone). At fixed g and fixed classical field amplitude, all positive constants γ give the same source-free field equations and physical characteristics, while $T_\gamma = \gamma T_1$.

Proof. Divide the first equation of (8.2) by γ and inspect (8.3). □

This comparison preserves homogeneous evolution and its cone; it does not preserve response amplitudes to a source whose coupling is held fixed. The sourced inverse operator and canonical brackets scale inversely with γ. Hence it is not a counterexample to a theorem assuming full source and observable normalisation.

9. Physical characteristics after gauge reduction

Up to a nonzero overall factor the principal symbol of the potential equation is

$$P(k)a=-k^2a+k(k\cdot a),k^2=g^{\mu\nu}k_\mu k_\nu. \tag{9.1}$$

Every covector has the gauge kernel $a \propto k$. Counting this kernel as a physical characteristic would declare every covector characteristic.

Proposition 9.1 (Maxwell physical cone and screen space). For nonzero k, a nontrivial physical polarisation exists precisely when $k^2=0$. At such a covector the physical polarisation space is $k^\perp/span\{k\}$ and has dimension two over the field of amplitudes.

Proof. If $k^2\neq 0$, the equation forces $a=k(k\cdot a)/k^2$, a pure gauge amplitude. If $k^2=0$, it forces $k\cdot a=0$. The three-dimensional orthogonal space contains the one-dimensional gauge direction, leaving a two-dimensional quotient. □

Consequently

$$Char_{phys}(P)=\{(x,k):k\neq 0,g^{\mu\nu}k_\mu k_\nu=0\}. \tag{9.2}$$

The determinant on unreduced potentials is not used. Lorenz gauge gives a normally hyperbolic potential equation with a lower-order curvature term, while the gauge-invariant field strength and physical quotient carry the observable content. A massless scalar, Maxwell field and linearised gravitational field can share this quadratic cone while having different physical polarisations [40].

10. Null energy–momentum and its exact domain

10.1. Geometric optics and normalisation

Consider a smooth geometric-optics family away from caustics and boundaries,

$$A_\mu^{(\varepsilon)}=\Re\{(a_\mu+O(\varepsilon))e^{i\theta/\varepsilon}\},q_\mu=\nabla_\mu\theta,\kappa_\mu=q_\mu/\varepsilon. \tag{10.1}$$

Here κ is the physical phase covector; q is its scaled counterpart. The wavelength is small compared with the variation scales of the background and envelope. The leading equations and next transport order give

$$q^2=0,q\cdot a=0,q^\nu\nabla_\nu q^\mu=0,\nabla_\mu(a^2q^\mu)=0,q^\nu\nabla_\nu e^\mu=0\,\text{modulo}\,q^\mu, \tag{10.2}$$

where $a=a\,e$ is a screen-polarisation decomposition with $e\cdot\acute{e}=1$. The geodesic identity follows directly from $q=d\theta$ and $\nabla(q^2)=0$. The transport statements require the next order of the field equation, not just its characteristic set. They are the standard leading Maxwell geometric-optics equations [40].

Average over rapid phase while keeping the slowly varying amplitude fixed. Since the average of two real oscillatory factors supplies $1/2$, contraction of $q_\mu a_\nu-q_\nu a_\mu$ in (8.3) gives

$$\varepsilon^2\langle T^{\mu\nu}[A^{(\varepsilon)}]\rangle=\frac{\gamma}{2}(a\cdot\acute{a})q^\mu q^\nu+O(\varepsilon). \tag{10.3}$$

Equivalently the leading unscaled tensor is $\frac{\gamma}{2}(a\cdot\acute{a})\kappa^\mu\kappa^\nu$. The error statement concerns smooth asymptotic families on compact regions where the expansion is valid, not an estimate uniform through caustics. Gauge shifts by multiples of q leave the leading screen norm and tensor unchanged. Co-propagating mixed polarisations may be averaged within the same null ray sector; different propagation directions generally cannot.

The notation $T^{\mu\nu}=\rho\,q^\mu q^\nu$ has a scaling redundancy: $q\mapsto b\,q$, $\rho\mapsto\rho/b^2$. The null direction alone therefore does not set energy density. Given a unit future observer U in $c=1$ units, normalise $\ell=q/(-U\cdot q)$ and write $\ell=U+n$ with $U\cdot n=0$, $n^2=1$. The tensor is then $T=u\,\ell\otimes\ell$, where $u=T_{\mu\nu}U^\mu U^\nu$ is measured energy density.

Theorem 10.1 (Conditional null-sector functional closure). Under Assumption 8.1, take either a nonzero exact null Maxwell field at a point, or the leading averaged tensor of a single-ray geometric-optics family. In an observer's orthonormal frame, restoring SI units with $x^0 = c\,t$, the same stress tensor has components

$$T^{\hat{\mu}\hat{\nu}} = u \begin{pmatrix} 1 & n^T \\ n & n\,n^T \end{pmatrix}, \quad |n| = 1, \tag{10.4}$$

and consequently

$$S = c\,u\,n, \quad g_{mom} = \frac{S}{c^2} = \frac{u}{c} n, \quad \Pi_{ij} = u\,n_i n_j, \quad \Pi_{nn} = u = I/c. \tag{10.5}$$

Here $I = |S|$, g_{mom} is momentum density and Π is momentum flux. For geometric optics these are leading-order relations. Boundary force is obtained only after the incoming and outgoing fluxes have been specified.

Proof. For geometric optics, (10.3) and the observer normalisation give (10.4). For an exact null field choose the observer's spatial axes so that $E \perp B$ and $|E| = c|B|$. Substitution in

$$u = \frac{\epsilon_0}{2}\left(E^2 + c^2 B^2\right), \quad S = \epsilon_0 c^2 E \times B,$$

$$\Pi_{ij} = u\,\delta_{ij} - \epsilon_0\left(E_i E_j + c^2 B_i B_j\right) \tag{10.6}$$

gives the same rank-one spatial flux. In $x^0 = c\,t$ coordinates, $T^{00} = u$, $T^{0i} = S_i/c = c\,g_{mom,i}$ and $T^{ij} = \Pi_{ij}$. Reading off these components proves (10.5). □

The SI expressions use the vacuum constitutive relation $\epsilon_0 \mu_0 c^2 = 1$ and the standard field normalisation. They do not derive these constitutive assumptions from relational data.

Some texts call $\sigma_{ij} = -\Pi_{ij}$ the Maxwell stress tensor. With that sign convention the Lorentz force density satisfies

$$\partial_t g_{mom} + \nabla \cdot \Pi = -f_{matter}, \quad f_{matter} = \nabla \cdot \sigma - \partial_t g_{mom}. \tag{10.7}$$

The sign convention is fixed here to avoid identifying tensile stress with positive outgoing momentum flux.

10.2. An algebraic diagnostic and counterexamples

For every real classical Maxwell field in a vacuum observer frame,

$$u^2 - \frac{|S|^2}{c^2} = \frac{\epsilon_0^2}{4}\left(E^2 - c^2 B^2\right)^2 + \epsilon_0^2 c^2 (E \cdot B)^2 \geq 0. \tag{10.8}$$

This follows from $|E \times B|^2 = E^2 B^2 - (E \cdot B)^2$. For $u > 0$, equality is equivalent to the two electromagnetic invariants vanishing. Thus $1 - |S|^2/\left(c^2 u^2\right)$ diagnoses departure from a single null field, although its nonzero value depends on the observer.

A static electric or magnetic field has $u > 0$ and $S = 0$. Two incoherent equal counter-propagating beams have total $S = 0$ and $\Pi_{zz} = u$, not $T = u\,\ell \otimes \ell$. A coherent standing wave also has zero time-averaged energy flux and nonzero longitudinal stress; interference affects its local transverse stress. More generally, $T = \sum_a u_a \ell_a \otimes \ell_a$ is rank one only when all occupied directions coincide, for positive u_a. Arbitrary superpositions and non-null electromagnetic configurations therefore fall outside Theorem 10.1. A vanishing averaged flux does not imply zero stored energy or zero force on confining boundaries.

11. Radiation pressure and an independent readout

11.1. Stationary boundaries

Let a beam of intensity I, measured per area perpendicular to its ray, meet a stationary planar surface of area A at angle θ to its normal. The power intercepted per surface area is $I\cos\theta$. By (10.5), a narrow incoming packet of energy dE carries momentum $dE\,n_i/c$.

Let α, r, τ be absorbed, reflected and transmitted fractions of intercepted power, with $\alpha+r+\tau=1$. Assume elastic specular reflection and straight transmission into the same vacuum, with no additional emission. Conservation of momentum gives the vector force per surface area

$$\frac{F}{A}=\frac{I\cos\theta}{c}\left(n_i-r\,n_r-\tau\,n_t\right). \tag{11.1}$$

Since the normal projections of the three directions are $\cos\theta, -\cos\theta, \cos\theta$, respectively,

$$P_\perp=\left(1+r-\tau\right)\frac{I}{c}\cos^2\theta=\left(\alpha+2r\right)\frac{I}{c}\cos^2\theta. \tag{11.2}$$

One cosine accounts for intercepted power and the other for transferred normal momentum. The tangential force in the plane of incidence is $\alpha I\sin\theta\cos\theta/c$ in this specular model.

At normal incidence, complete absorption gives $P=I/c$, ideal reflection gives $P=2I/c$, and lossless straight transmission gives zero net pressure. These are also the jumps of the normal component of momentum flux in (10.6). Perfect reflection reverses momentum without absorbing net beam energy at a stationary ideal mirror, which already separates pressure from heating.

The boundary or material assumptions are indispensable. Diffuse scattering requires its angular distribution; refraction requires momentum accounting for field and medium; heating followed by thermal emission adds a recoil term. The formulas above describe the specified optical momentum transfer, not all forces on an arbitrary illuminated object. Absorption over a finite interval can increase internal energy; steady thermal operation needs an energy-removal model.

11.2. A moving-mirror control

For a prescribed perfectly reflecting plane moving along the incident normal at speed βc, $|\beta|<1$, neglect recoil during the reflection of a packet and work in the incident laboratory frame. Boost the incoming null energy–momentum to the mirror rest frame, reverse its spatial momentum there, and boost back. The reflected packet energy and frequency obey

$$\frac{E_r}{E_i}=\frac{\omega_r}{\omega_i}=q_\beta:=\frac{1-\beta}{1+\beta}. \tag{11.3}$$

With $\gamma_\beta=\left(1-\beta^2\right)^{-1/2}$, for example, $E'=\gamma_\beta\left(1-\beta\right)E_i$ and $E_r=\gamma_\beta\left(1-\beta\right)E'$, which gives this ratio. The number of equal-energy incident packets reaching unit mirror area per laboratory time is $\left(I/E_i\right)\left(1-\beta\right)$. Multiplying it by their momentum change $\left(E_i+E_r\right)/c$ yields

$$P_{lab}=\frac{2I}{c}\frac{1-\beta}{1+\beta}. \tag{11.4}$$

This derivation uses null packet momentum, not a quantisation assumption. The mechanical power density $\beta c\,P_{lab}$ equals $I\left(1-\beta\right)\left(1-q_\beta\right)$, the intercepted energy minus the energy carried away by reflected packets per laboratory time. Omitting the interception factor would give an incorrect force. Acceleration, finite-mass recoil and nonideal optical response require an enlarged model.

11.3. What makes the test independent?

Definition 11.1 (Operational pressure closure residual). At a stationary boundary with known nonzero $b=\left(\alpha+2r\right)\cos^2\theta$, define

$$R_P = \frac{c_{TOF} P_\perp}{b I} - 1, c_{TOF} = L/\Delta t, \tag{11.5}$$

using independently calibrated timing, power and force measurements in the same effective vacuum regime.

The action and boundary model predict $R_P = 0$. Time of flight tests propagation; mechanical recoil tests energy–momentum transfer and the boundary coupling. Their apparatus and systematic errors differ. Pressure is thus an independent operational channel, while its value is mathematically constrained by the assumed functional structure. It is not a logically independent axiom once Maxwell dynamics and the boundary conditions have been imposed.

Radiation-pressure radiometry is an existing physical anchor [42, 43]. Such a power meter uses the pressure relation as its calibration law; comparing it with itself would not test that law. An actual closure test needs an independent power or energy calibration and independently characterised reflection and absorption. No experiment or experimental confirmation is claimed in this manuscript.

12. Directed stress, photon-gas pressure and dimension

Directed surface pressure, electromagnetic stress, momentum flux, isotropic photon-gas pressure and thermodynamic radiation pressure are distinct objects. In particular, I/c for an absorbed directed beam must not be substituted for the equation of state of an isotropic gas.

Proposition 12.1 (Isotropic massless pressure). For a noninteracting massless kinetic sector in d spatial dimensions, with isotropic finite energy distribution and dispersion $E = c|p|$,

$$\Pi_{ij} = \frac{u}{d}\delta_{ij}, p_{iso} = u/d. \tag{12.1}$$

Thermal equilibrium is not needed for this kinetic identity.

Angular proof. Let $w(n)\,d\Omega$ be directional energy density. Each direction contributes $w(n) n_i n_j\, d\Omega$ to momentum flux. Rotational symmetry forces its average to be $C\,\delta_{ij}$; tracing and using $\sum_i n_i^2 = 1$ gives $d\,C = u$. □

Trace proof. The massless kinetic tensor is a positive sum or integral of null outer products, hence $T^\mu{}_\mu = 0$. Isotropy gives $T^{\hat{\mu}\hat{\nu}} = diag(u, p, \ldots, p)$, so $-u + d\,p = 0$. □

These are two derivations of the same constraint, not two statistically independent experiments. For classical Maxwell theory in four spacetime dimensions the trace vanishes directly from (8.3). In general spacetime dimension D, that same action instead gives

$$T^\mu{}_\mu = \gamma(1 - D/4) F_{\alpha\beta} F^{\alpha\beta}. \tag{12.2}$$

One must therefore not justify $p = u/d$ for arbitrary Maxwell fields in arbitrary dimensions by an unqualified tracelessness claim. The massless kinetic argument is valid in any stated d; arbitrary classical Maxwell tracelessness is special to $D = 4$.

For a three-dimensional equilibrium photon gas, two transverse states and zero chemical potential give

$$u = 2\int \frac{d^3k}{(2\pi)^3} \frac{\hbar c|k|}{e^{\hbar c|k|/(k_B T)} - 1} = \frac{\pi^2 (k_B T)^4}{15 \hbar^3 c^3}, p = u/3. \tag{12.3}$$

Equivalently the equilibrium free energy has $p = -(\partial F/\partial V)_T = u/3$ in the thermodynamic limit [44]. Finite cavities, boundary stresses and vacuum contributions need separate terms. Quantum trace anomalies also lie outside the classical trace proof. The ratio u/p can diagnose effective dimension only conditional on isotropy, a known massless dispersion and absence of additional stress. Failure of any of these assumptions can imitate a different value; the relation does not derive the observed spatial dimension.

13. Rest mass, pressure and confined radiation

13.1. The local gravitational source

Radiation can have zero constituent rest mass and nonzero stress. Ideal dust has nonzero rest-mass density and zero pressure in its comoving frame:

$$T^{\hat{\mu}\hat{\nu}}_{rad} = diag(u, u/3, u/3, u/3), T^{\hat{\mu}\hat{\nu}}_{dust} = diag(\rho_m c^2, 0, 0, 0). \tag{13.1}$$

Thus pressure and rest-mass density cannot be identified. In general relativity the source is the full tensor, including energy density, momentum density, isotropic pressure and anisotropic stress:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T^{total}_{\mu\nu}. \tag{13.2}$$

Equation (13.2) is an additional gravitational physical anchor [38], obtained by varying a specified gravitational action together with the matter action. Maxwell metric variation supplies its matter source, but does not derive the gravitational action or G. In a semiclassical treatment the source requires a renormalised expectation value, not an unspecified zero-point sum.

13.2. Composite invariant mass

For an isolated system in special relativity with well-defined total four-momentum,

$$M^2 c^4 = E^2_{tot} - c^2 |P_{tot}|^2. \tag{13.3}$$

For two photons of sharp momenta, energies E_1, E_2 and angle ϑ between their directions, add their four-momenta to obtain

$$M^2 c^4 = 2 E_1 E_2 (1 - \cos\vartheta). \tag{13.4}$$

Parallel momenta give $M = 0$; equal counter-propagating photons of energy E give $M = 2E/c^2$. Each constituent remains massless. For wavepackets, one must specify the total momentum distribution instead of silently assigning a sharp invariant mass to every quantum state.

For a reflecting cavity, the field alone exchanges momentum with the walls. A *closed* cavity system includes wall energy, wall stress and any supporting components. In its rest frame

$$M_{closed} = E_{closed}/c^2. \tag{13.5}$$

Adding retained energy ΔE increases this rest energy by ΔE, when that is the complete energy change of the closed system. Wall deformation or other energy changes must be included in ΔE.

The role of stress can be displayed without a gravitational argument. For a stationary, spatially localised closed tensor in Minkowski space, conservation and vanishing surface terms imply the Laue identity

$$\int T^{ij}_{total} d^3x = \int \partial_k \left(x^i T^{kj}_{total} \right) d^3x = 0. \tag{13.6}$$

The positive integrated radiation pressure is balanced by material stress. Omitting the walls can therefore spoil the transformation properties of integrated subsystem quantities [45]. With a conserved closed tensor, $P^\mu = c^{-1} \int T^{0\mu} d^3x$ is the relevant total four-vector under the usual isolation hypotheses. Radiation energy, composite inertia and gravitational sourcing are connected through these structures; no conversion of an individual photon's rest mass is involved. Global total energy in a general curved spacetime is a separate issue.

14. Lorentzian calibration and the meaning of $E = mc^2$

The invariant one-particle mass relation is

$$E^2 = c^2 |p|^2 + m^2 c^4. \tag{14.1}$$

It comes from the Lorentzian norm of $p^\mu=(E/c,p)$, with $p^\mu p_\mu=-m^2c^2$. For the positive-energy rest state, $p=0$ gives $E=mc^2$. For $m=0$, it gives $E=c|p)$. The factors of c relate temporal and spatial components and restore compatible units; their powers are fixed by the tensor and dimensional structure of these formulas.

No-go statement 14.1 (A square does not supply probability). The occurrence of c^2 in $E=mc^2$ does not imply a probability modulus square or a Born rule.

The mass-shell norm is an indefinite Lorentzian bilinear form; a quantum probability uses a positive Hilbert-space inner product and a state with a specified measurement rule. Neither a Hilbert space nor a probability measure follows from (14.1). In SI c^2 is dimensional; its numerical value changes with units while probabilities do not. In $c=1$ units the relation becomes $E^2=|p)^2+m^2$, with $E=m$ at rest, and the probability postulates remain independent. Any deeper norm-theoretic analogy would require an additional construction.

15. Frequency, wavelength and the quantum line

For a local future-directed phase covector κ, an observer measures $\omega=-cU^\mu\kappa_\mu$ when $U^2=-1$ and coordinates use $x^0=ct$. The spatial projection is the physical wavevector k in that observer's frame. The null condition is

$$\omega=c|k),f=\omega/(2\pi),\lambda=2\pi/|k),f\lambda=c. \tag{15.1}$$

These are local vacuum relations. Global frequency conservation requires a suitable stationary symmetry; Doppler shifts and gravitational redshift compare different observer measurements. In a dispersive medium, phase, group and front velocities require separate treatment.

Physical Anchor 15.1 (Quantised Maxwell sector). In Minkowski vacuum, quantise the two physical transverse modes with the canonical symplectic normalisation fixed by the action. In the usual photon Fock representation the normal-ordered translation generators are

$$H=\sum_{s=\pm1}\int d^3k\,\hbar c|k)a_s^\dagger(k)a_s(k),$$
$$P=\sum_{s=\pm1}\int d^3k\,\hbar k\,a_s^\dagger(k)a_s(k), \tag{15.2}$$

with $\left[a_s(k),a_{s'}^\dagger(k')\right)=\delta_{ss'}\delta^3(k-k')$.

The definitions of the translation generators, equivalently their action on $e^{-i\omega t+ik\cdot x}$, give

$$E=\hbar\omega,p=\hbar k,E=c|p). \tag{15.3}$$

The last relation follows from the first two together with the independent massless dispersion. Quantisation provides $\hbar$ and the operator interpretation; the classical cone alone provides neither. Conversely the existence of unitary translations alone does not force the Maxwell dispersion or helicity content. Representation theory constrains the allowed massless particle sectors after Poincare symmetry is assumed [46, 47].

On the physical Fock space H is self-adjoint and nonnegative and hence generates a strongly continuous unitary group. The same free Maxwell theory has a local field-strength algebra, massless dispersion and stress–energy observables after the appropriate operator definition. This is simultaneous compatibility within one constructed continuum theory. A generic curved, nonstationary spacetime need not admit the same preferred particles or a single Fock representation with unitary time evolution.

The finite scalar-chain result of [36] supplies a different control: finite unitary dynamics can have analytic tails and a microcausal continuum limit. Its lattice dispersion is not exactly linear at finite spacing. It does not supply the Maxwell gauge constraint, transverse states or helicity. No scalar-to-photon inference is made here.

16. Polarisation and helicity delimit closure

The null cone determines allowed wave covectors. It does not determine the dimension or transformation law of the fibre over each covector. For Maxwell theory the gauge quotient of Section 9 provides two transverse amplitudes. For positive-energy massless finite-helicity Poincare representations the little group is locally $ISO(2)$, with its translation part acting trivially on the physical helicity representation; Maxwell theory carries helicities $+1$ and -1 [46]. That restriction is an additional representation choice: the general massless representation problem is broader than the photon sector.

The geometry supplies the null directions and screen bundle; the action and constraints select the physical amplitudes; representation theory specifies their helicities; state preparation chooses their occupation, relative phase and statistical mixture. Geometric optics transports the polarisation class along a ray, with connection data and the transport equation in (10.2). At higher orders, wavelength-dependent polarisation effects may require corrections to the leading ray picture [41].

No-go statement 16.1 (The stress tensor does not recover polarisation). Neither a common null cone nor a specified leading null stress tensor uniquely determines the polarisation state.

For a fixed direction and energy density, orthogonal linear polarisations give the same tensor (10.4); circular polarisations of opposite helicity also have the same leading averaged tensor. A massless scalar on the same Minkowski cone has no photon helicity. Polarimetry and helicity-sensitive interactions therefore test information not contained in the energy–momentum closure alone.

17. Vacuum correlations, commutators and null structure

17.1. Causal response without photon detection

For a real scalar control field with $P=\Box_g - m^2$ in the ¿ convention, set $E=G_{ret}-G_{adv}$. The free field algebra obeys

$$¿\, i\hbar E(f,h) I. \tag{17.1}$$

For this sign of P, E is minus the sine response of ∂_t^2+K in a static splitting, consistently with [36]. Green operators have causal support on a globally hyperbolic spacetime [30]. The sign convention does not affect support. For Maxwell theory one uses gauge-invariant field strengths or the physical test-function quotient, not arbitrary gauge-potential components as independent observables.

For a state ω, the two-point function $W_\omega(f,h)=\omega(\phi(f)\phi(h))$ has the fixed antisymmetric part $W_\omega(f,h)-W_\omega(h,f)=i\hbar E(f,h)$ but a state-dependent symmetric part. In four-dimensional Minkowski space the massless scalar vacuum, in $c=1$ units and canonical normalisation, has

$$W_0(x,y)=\frac{\hbar}{4\pi^2}\frac{1}{|x-y|^2-(t_x-t_y-i0)^2}. \tag{17.2}$$

It is nonzero at spacelike separation even though the commutator there vanishes. Correlation and signalling are different mathematical questions [31, 47].

For example, let a nonselective local operation have operators K_a in a region spacelike to a bounded readout B, with $\sum_a K_a^{¿} K_a = I$. Local commutativity gives

$$\omega'(B)=\sum_a \omega\left(K_a^{¿} B K_a\right)=\omega\left(B\sum_a K_a^{¿} K_a\right)=\omega(B). \tag{17.3}$$

Conditional postselection can change conditional statistics, but knowledge of the selected outcome requires communication. Retarded linear response is likewise controlled by a commutator, not by the symmetric correlation. Thus a detector with no counts has not demonstrated the absence of a causal algebra or of possible response.

17.2. Singularities as conditional geometric data

For a Hadamard scalar state on a specified globally hyperbolic spacetime, the microlocal characterisation is

$$WF(W_\omega)=\{(x,k;y,-k'):(x,k)\sim(y,k'),k\ \text{future null}\}. \tag{17.4}$$

The relation $\sim$ denotes transport along a null geodesic, including the coincident null directions; “future” refers to the metric-dual vector in the already chosen orientation. Radzikowski established the equivalence with the Hadamard condition for the scalar setting [48]. Propagation of singularities follows the Hamilton flow of the principal symbol under its real-principal-type hypotheses [49]. Vector-valued Hadamard constructions require the corresponding bundle and field assumptions [50]; Maxwell observables also require the gauge treatment just stated.

Equation (17.4) identifies the same null directions as the effective wave operator. It is meaningful conditional encoding of conformal null geometry, not a derivation from an unspecified vacuum: the operator, smooth structure and time orientation have already entered the construction. Smooth state-dependent changes in a Hadamard two-point function do not change this wavefront set, but can change renormalised stress–energy. In curved spacetimes, or for massive fields, a causal commutator can have interior tails; its singular directions and its entire support should not be conflated. Exact light-cone-only support in a special flat massless example is not a general Huygens principle.

18. Reconstruction of quadratic and quantum observables

The earlier response theorem does not automatically control quadratic observables or their boundary traces. We now state sufficient additional conditions without assuming equality of the stresses to be proved.

18.1. A quantitative classical criterion

Identify two effective descriptions by a common smooth diffeomorphism, the induced tensor pullback, physical measure and observer frame. After this identification, work on a compact region K with a fixed smooth Lorentzian metric. Use any fixed positive auxiliary tensor norm to define $L^2(K)$; on K the resulting norms are equivalent.

Proposition 18.1 (Stress convergence from calibrated field convergence).

Let F_n and G_n be reconstructed real two-forms with $G_n=s_nF_n+r_n$, where the real constants satisfy $0<s_-\le|s_n|\le s_+<\infty$. Suppose

$$\sum_n\|F_n\|_{L^2(K)}<\infty,\ \|r_n\|_{L^2(K)}\to0,\ \gamma_n^{(2)}s_n^2-\gamma_n^{(1)}\to0, \tag{18.1}$$

with both positive action normalisations uniformly bounded. Then the corresponding Maxwell stresses satisfy

$$\begin{aligned}&\left\|T_{\gamma_n^{(2)}}(G_n)-T_{\gamma_n^{(1)}}(F_n)\right\|_{L^1(K)}\\&\le C_K\left[\left|\gamma_n^{(2)}s_n^2-\gamma_n^{(1)}\right|\|F_n\|_{L^2(K)}^2+\|r_n\|_{L^2(K)}\left(\|F_n\|_{L^2(K)}+\|r_n\|_{L^2(K)}\right)\right)\to0.\end{aligned} \tag{18.2}$$

Uniform convergence of identified metric coefficients, inverses and measure densities, with uniform nondegeneracy, permits the same conclusion with a further term tending to zero.

Proof. For fixed metric the stress is a continuous quadratic map of the six components of a two-form. Denote its associated bilinear map by B_g. Expanding $G_n=s_nF_n+r_n$ separates the difference into $\left(\gamma_n^{(2)}s_n^2-\gamma_n^{(1)}\right)B_g(F_n,F_n)$, the cross term $2\gamma_n^{(2)}s_nB_g(F_n,r_n)$, and $\gamma_n^{(2)}B_g(r_n,r_n)$. On a compact region $|B_g(F,G)|\le C_K|F||G|$. Integration and Cauchy–Schwarz give (18.2). Uniform changes of the coefficients of this bilinear map add their supremum difference times the bounded squared field norms; measure changes are treated in the same way. □

The constants s_n describe a common field readout change and the normalisation condition matches the action, not the desired stress itself. With fixed normalisation, $s_n=1$ is the simplest case. Gauge-fixed

potentials converging strongly in H^1 on a spacetime region give the needed strong convergence of $F = dA$. For uniform-in-time spatial stress convergence one can instead impose uniform L^2 convergence of E and B on the time slices.

Corollary 18.2 (Fluxes and ratios). Under Proposition 18.1, bounded volume-smeared stress and flux readouts converge. Ratios of such readouts converge when the limiting denominator is bounded away from zero. For surface traction and beam pressure the same conclusion holds if the traces of the reconstructed fields on the measurement world surface satisfy the analogous L^2 conditions and the boundary scattering and readout maps agree.

The volume estimate alone does not provide a trace theorem. A trace condition, such as a suitable stronger Sobolev bound and trace convergence, must be proved from a model of the fields and boundary; it is not supplied by cone equality. Noninvertible coarse-graining that removes sectors or boundary observables is likewise outside this equivalence statement.

18.2. An exact counterexample to weaker control

Proposition 18.3 (Strong potentials can hide finite radiation stress).

In flat $c = \gamma = 1$ Maxwell theory, the source-free potentials

$$A_y^{(n)} = \frac{\sin n(z-t)}{n}, A_0^{(n)} = A_x^{(n)} = A_z^{(n)} = 0 \tag{18.3}$$

converge locally uniformly to zero, while their stresses do not converge to the stress of that limiting potential.

Proof. The Lorenz condition and wave equation hold exactly. The fields are $E_y^{(n)} = \cos n(z-t)$ and $B_x^{(n)} = -\cos n(z-t)$. They converge weakly locally to zero, but

$$T_n^{\mu\nu} = \cos^2 n(z-t)\,\ell^\mu \ell^\nu \rightharpoonup 1/2\,\ell^\mu \ell^\nu, \ell = (1,0,0,1). \tag{18.4}$$

The oscillatory cosine term in $\cos^2 = (1+\cos 2n(z-t))/2$ vanishes against smooth compact tests. The limit is nonzero, while $T[0) = 0$. □

All these fields use the same Maxwell operator, cone and normalisation. The failure is missing strong derivative/energy control, not a change of units. It is compatible with the fixed-input strong response results of [36]. The sequence carries a quadratic high-frequency defect that a weak field reconstruction loses. Consequently response convergence alone cannot be promoted to stress convergence for arbitrary moving input sequences.

There is an additional scale obstruction. In four dimensions $S_\gamma[A, \Omega^2 g) = S_\gamma[A, g)$ for fixed two-form F, but $T'_{\mu\nu} = \Omega^{-2} T_{\mu\nu}$ and observer energy density $u' = \Omega^{-4} u$. The cone does not distinguish these representatives. Dimensionless null flux ratios may survive, whereas absolute energy density and pressure require the metric representative and observable calibration. A positive common speed calibration alone is insufficient.

18.3. Quantum transport needs an algebra and state

For exact field rescalings $A_2 = s A_1$ and $\gamma_2 s^2 = \gamma_1$, the classical Maxwell symplectic form is preserved after the same local geometric identification: it is bilinear in A, F and proportional to γ. Therefore the induced map preserves Weyl relations at the same $\hbar$. If it also respects localisation, it preserves commutator support; if the states are identified, their correlation functions of corresponding smeared observables agree. A state-preserving algebra isomorphism is unitarily implemented between its cyclic GNS representations. This does not assert unitary equivalence for arbitrary unmatched QFT states [47].

Distributional convergence of two-point functions alone still does not justify taking coincident derivatives in a renormalised stress tensor. Here is a separate sufficient condition in a fixed scalar control theory.

Proposition 18.4 (Conditional quantum stress convergence). Fix a smooth metric, a common Hadamard subtraction H_{sing} and a common local renormalisation prescription. Let Hadamard two-point functions be $W_n = H_{sing} + w_n$ near the diagonal, where their smooth remainders converge in C^2 on compact neighbourhoods of the diagonal. Then their point-split renormalised stress expectations converge locally uniformly.

Proof. The fixed prescription has the form

$$\langle T_{\mu\nu}(x) \rangle_n = D_{\mu\nu}(x, y) w_n(x, y)\big)_{y=x} + C_{\mu\nu}(g), \tag{18.5}$$

with D a bidifferential operator of total order at most two and fixed smooth coefficients. C^2 convergence makes its diagonal evaluation converge uniformly on compact sets. The common local term $C_{\mu\nu}(g)$ cancels in differences. □

This elementary continuity statement is conditional on the existence of the common Hadamard and renormalisation construction [31]. Maxwell versions need the corresponding physical vector/field-strength construction and control of its subtractions; these are not inferred from the scalar chain theorem. Equality of wavefront sets alone permits different smooth remainders and hence different quantum stress.

19. Functional consistency and operational calibration

19.1. What $c = 1$ simplifies

After a common null calibration, an observer's spatial and temporal increments along a null displacement satisfy $d\ell = d t_O$. Here t_O is the observer-splitting time increment, not the proper time of the null curve. The same representation gives

$$E = |p), \omega = |k), I = u = \Pi_{nn} \text{ in the single null sector}, E_{rest} = m. \tag{19.1}$$

Setting $c = 1$ is enough for $\omega = |k)$; setting $\hbar = 1$ is needed additionally for $E = \omega$ and $p = k$. The classical dispersion does not require a quantum unit convention. The natural-unit equality of represented flux components does not assert that differently dimensioned SI numbers can be equated without conversion.

The conventional statement is that a given common cone may be represented by unit speed. The substantive statements are that the same calibration works for all retained physical sectors and that independent readouts obey the functional relations with one set of action and detector parameters. Maxwell closure tests the latter inside one electromagnetic sector. It does not establish universality with other matter or gravity sectors.

19.2. A loop with independently identifiable inputs

Frequency standards compare phase accumulation with a clock; interferometry compares phase with reconstructed displacement; time of flight compares controlled emission and detection; cavity resonances combine propagation with boundary phase conditions; recoil and pressure compare optical energy with mechanical momentum transfer. An ideal one-dimensional round-trip resonator has $f_m = m c/(2 L)$ when reflection phases have the assumed values. General cavities require their full mode and boundary problem.

Corollary 19.1 (Narrowband photon pressure loop). For a stationary boundary at normal incidence, assume the null dispersion, canonical quantum translation relations, incident photon rate $\dot{N}$ over area A, and characterised fractions with $b = \alpha + 2r > 0$. Then

$$I = \frac{\dot{N} \hbar \omega}{A}, F = b \dot{N} \hbar |k), \frac{c F}{b I A} = \frac{c |k)}{\omega} = 1. \tag{19.2}$$

Proof. The first expression counts incident energy per area and time. The second counts momentum transfer, with a reflected photon contributing twice its incident momentum. Division cancels $\dot{N}$ and $\hbar$, and (15.1) supplies the last equality. □

This cancellation shows both strength and limitation: the pressure-to-power ratio tests null energy–momentum closure, but does not independently determine $\hbar$ or photon number statistics. A coherent state and a number-state preparation can agree in mean energy and pressure while having different fluctuations.

Useful mutually comparable speeds are

$$c_{phase}=f\lambda, c_{TOF}=L/\Delta t, c_{pressure}=\frac{bI}{P_{\perp}} \tag{19.3}$$

with the appropriate angular factor in b. Their dimensionless ratios, rather than an isolated dimensional numerical value, are closure tests. If L is assigned from the same time-of-flight reading using an assumed c, that particular comparison is an identity. If force is calibrated from the same pressure law, that comparison is also an identity. Independent realisation of the relevant standards and control of boundary response are what make the loop operationally informative. This extends the clocks and rods distinction of [28, 36]; it does not put metres and seconds into the neutral microscopic carrier.

19.3. Dependency tables

The following two linked tables constitute the functional consistency graph. An edge means that a mathematical operation produces a specified observable once its extra inputs are supplied. In particular the core branches are field variation followed by gauge reduction, metric variation followed by observer projection, and symmetry analysis followed by charge or quantum-generator construction. Boundary momentum balance joins the second and third branches to a force readout.

Use these reconstruction labels: R_1 is the preceding local operator/response identification; R_2 adds the spatial spectral and clock/phase standards; R_3 adds action normalisation, physical metric, measure and strong field-energy control; R_4 adds surface traces, scattering data and force/power readouts; R_5 adds a local physical algebra, symplectic normalisation and identified state; R_6 adds the Hadamard remainder and renormalisation control needed for quantum stress. These are named extra requirements, not automatic consequences of R_1. They are not a definition that presupposes cone or stress equality.

Table 1. *Functional origins, assumptions and relation to the common cone. The same row identifiers are used in Table 2.*

ID / observable	Mathematical origin	Additional assumptions	Cone and c dependence
P: propagation	Principal part of $\delta_A S=0$, after physical reduction.	Hyperbolic Maxwell sector, gauge quotient, specified background.	Null cone fixes local characteristic speed; orientation and scale remain separate.
F: frequency	Phase derivative contracted with observer time direction.	Clock, phase-bearing state or classical solution, observer.	$\omega=c\lvert k\rangle$ in the nondispersive vacuum sector; cone does not choose ω.
W: wavelength	Spatial phase gradient and physical spatial measure.	Locally resolved phase, rods or interferometric readout.	$f\lambda=c$ locally; cone alone does not set a wavelength.
E: energy	Time-translation generator or stress projection; quantum mode energy.	Action normalisation; symmetry for global energy; quantisation for $\hbar\omega$.	Null modes satisfy $E=c\lvert p\rangle$; amplitude and occupation remain free.
M: momentum	Spatial translation generator or integrated momentum density.	Translation symmetry or local frame; quantisation for $\hbar k$.	Null direction aligns momentum; $\lvert p\rangle=E/c$ is sector dependent.
EF: energy flux	$S_i=cT^{0i}$ from metric variation and projection.	Field amplitude, action and metric normalisation, observer.	$\lvert S\rangle=cu$ in one null sector; inequality otherwise.
MF: momentum flux	$\Pi_{ij}=T^{ij}$, distinct from momentum density.	Same as EF; specified spatial normal for a flux readout.	$\Pi_{nn}=I/c$ for a collimated null beam; cone alone does not fix tensor components.

ID / observable	Mathematical origin	Additional assumptions	Cone and c dependence
RP: beam pressure	Incoming minus outgoing momentum flux.	Material or boundary, angle, reflection, transmission, absorption and motion.	$P = bI/c$ under the stated stationary assumptions; extra factors for moving boundaries.
GP: gas pressure	Angular second moment, or traceless kinetic tensor plus isotropy.	Massless distribution, isotropy, d; equilibrium only for thermodynamics.	$p = u/d$; c cancels in the ratio but enters the spectrum and thermal density.
T: stress–energy	Metric derivative of the full specified matter action.	Metric representative, normalisation, state for expectations.	Same cone admits different magnitudes and non-null fields; T contains more data than the cone.
H: polarisation / helicity	Gauge quotient, screen transport, massless representation.	Maxwell degrees of freedom, helicity representation, preparation.	Cone supplies allowed directions; it does not select helicity ± 1 or a polarisation state.
V: vacuum correlation	State expectation of field products.	Field algebra, normalisation, state; Hadamard condition when imposed.	Null singularities match the assumed operator; spacelike correlations may be nonzero.
C: commutator	Canonical/Peierls structure and causal Green operator.	Physical algebra, symplectic normalisation, hyperbolicity and global conditions.	Causal support uses the same cone; coefficient is not determined by the cone alone.
U: quantum unitarity	Self-adjoint Hamiltonian in a specified representation.	Domain, positive physical inner product, suitable global evolution.	Unitarity alone implies neither a speed nor local commutativity.
G: gravitational coupling	Joint variation of gravitational and matter actions.	Gravitational field equations, G, total source; renormalisation if semiclassical.	Common matter cone does not derive gravity; c^{-4} converts source units in Einstein's equation.

Table 2. *Action dependence, reconstruction requirements, independent readouts and failure modes. Readouts are potential tests, not experiments reported here.*

ID	From the same action?	Survives reconstruction?	Readout and counterexample / failure mode
P	Yes, principal variation and reduction.	R_1, with fixed retained sectors and local limiting operator.	Arrival fronts and intersector comparisons; distinct positive branch speeds survive common rescaling.
F	Dynamics allows phase; its measured value also needs a clock and state.	R_2 with identified clock and solution.	Frequency comparison; observer motion or nonstationary geometry changes measured frequency.
W	Phase solutions come from dynamics; length readout is additional.	R_2, including spatial measure and phase localisation.	Interference or cavity comparison; arbitrary nonlocal isospectral maps lose spatial meaning.
E	Yes for the classical generator; quantum energy adds quantisation.	R_3 classically; R_5 and domain control quantum mechanically, R_6 for local quantum stress.	Independent calorimetry or energy transfer; γS keeps the cone but changes fixed-amplitude energy.
M	Yes for translation charges; quantum relation needs canonical generators.	R_3 with spatial calibration; R_5 for quantum generators.	Recoil; a standing field stores energy with zero net momentum.

ID	From the same action?	Survives reconstruction?	Readout and counterexample / failure mode
EF	Yes, metric variation and observer projection.	R_3 for volume-smearing, R_4 for surface flux; R_6 if quantum.	Power transfer; a static field or opposing beams have energy without directed net flux.
MF	Yes, different components of the same tensor.	R_3, and trace control for surface transfer.	Direction-resolved momentum transfer; strong potentials with oscillating derivatives can retain finite flux.
RP	Maxwell action plus the interaction/boundary model.	R_4; R_1 alone does not identify force or absorption.	Mechanical force versus independently measured power; same beam gives factors 0, 1 or 2 at different boundaries.
GP	Radiation dynamics plus the distribution, and thermodynamic structure if used.	R_3 with angular measure and state; extra equilibrium calibration for temperature.	Equation-of-state comparison; anisotropic stress or a massive component invalidates inference of d.
T	Yes, once the metric dependence of the action is fixed.	Proposition 18.1; R_6 for renormalised quantum expectations.	Full stress projections; conformal rescaling or a quadratic high-frequency defect changes stress without changing the cone.
H	Gauge action fixes physical modes; representation and state fix further content.	Physical bundle/gauge map and R_5 for the prepared quantum state.	Polarimetry and helicity-sensitive transfer; scalar and photon fields share a cone but differ in helicity.
V	Quadratic action constrains bisolutions; it does not select their symmetric state part.	R_5 with state identification; wavefront-set preservation needs stronger control than distributional convergence alone.	Correlation measurements; different Hadamard states share singular directions but have different smooth correlations.
C	Yes after canonical/Peierls quantisation; retarded convention is extra.	R_1 plus local symplectic identification at the same $\hbar$.	Controlled linear response; nonzero spacelike correlations coexist with zero signalling commutator.
U	Hamiltonian follows from the action; self-adjointness and representation must be established.	Specified strong unitary limit as in [36], or a state-preserving dynamical algebra isomorphism.	Reversible quantum evolution; finite harmonic systems are unitary yet have analytic spatial tails.
G	Requires an additional gravitational action, not Maxwell action alone.	Source transport needs R_3 or R_6; convergence of the coupled metric needs another theorem.	Gravitational response to total energy and stress; equal cones leave the gravitational dynamics unspecified.

20. Optional compact-sphere control

On the *prescribed* ultrastatic effective spacetime $R \times S^3_R$ with metric $-c^2 dt^2 + h_R$, the diameter is πR. Geodesic signals can reach every spatial point from a given point within $\pi R/c$, under the ideal preparation and detection assumptions. This is a bound on potential access in the specified geometry; it is neither an actual signal history nor a cosmological horizon.

For the scalar Laplacian,

$$\lambda_l = \frac{l(l+2)}{R^2}, \omega_l = \frac{c}{R}\sqrt{l(l+2)}, l = 0, 1, \ldots. \tag{20.1}$$

The corrected result of [36] is retained: the corresponding half-wave group has strongly almost-periodic orbits, but has no nonzero exact operator period, and $\left\| e^{itc\sqrt{-\Delta_{h_R}}} - I \right) \geq \sqrt{2}$ for $t \neq 0$. Strong recurrence of each vector cannot be promoted to operator-norm recurrence. A lower-order change to $-\Delta_{h_R} + R^{-2}$ instead gives frequencies $c(l+1)/R$ and an exact period $2\pi R/c$, with the same principal cone. Neither compactness nor recurrence chooses c.

These are scalar control statements. Electromagnetic cavity or sphere modes require their transverse vector spectrum and global gauge-sector analysis. No identification of the observable universe with S^3, and no fundamental compact pregeometric assumption, enters the main results.

21. Non-implications and counterexamples

The following failures delimit the dependency graph. They follow from the displayed constructions or the explicitly retained results of [36].

1. $c = 1$ **does not establish intersector universality.** The ratios v_i / v_j in (2.3) survive common rescaling.
2. **A common cone does not determine polarisation.** Scalar and Maxwell theories on Minkowski space provide different fibres over the same characteristic set.
3. **A common cone does not determine metric scale.** g and $\Omega^2 g$ have the same null set, with different calibrated energies, lengths or times unless the readouts are also adjusted.
4. **A common cone does not select the future.** The quadratic cone and $u\ell \otimes \ell$ are unchanged by $\ell \mapsto -\ell$; orientation is extra data.
5. c^2 **in rest energy does not imply probability.** Equation (14.1) is a Lorentzian norm relation, with no Born rule supplied by its notation.
6. **Vacuum correlation does not imply spacelike signalling.** The Minkowski two-point function is nonzero at spacelike separation, while local nonselective operations leave distant readouts unchanged.
7. **No detected photons does not imply no causal structure.** The field algebra and possible retarded response persist in a vacuum state.
8. **Actual signal history does not define the whole causal domain.** Unrealised causal paths remain part of the model's possible propagation.
9. **Pressure does not require nonzero particle rest mass.** A photon gas or reflected beam provides the counterexample.
10. **Rest-mass density does not require pressure.** Ideal dust has $\rho_m > 0$ and $p = 0$ in its rest frame.
11. **Null dust is not general Maxwell theory.** Static fields, standing waves and non-collinear beam mixtures violate single-ray rank-one closure.
12. S^0 **proves no physical ontology.** The two-point lemma concerns a predicate on a mathematical carrier.
13. **Suspension is not dynamical production of physical dimensions.** A topological construction supplies no Hamiltonian or operational measure.
14. **Hopf non-factorisation does not prove physical interaction.** The bundle has no dynamical coupling until one is separately specified.
15. **Compactness does not select the value of** c**.** The same S^3_R supports wave operators with any prescribed positive speed.
16. **A domain of description is not automatically a cosmological horizon.** Domain, past history, future accessibility and expansion scale require different defining data.

17. **Identical homogeneous propagation does not fix energy normalisation.** S and γS give the counterexample of Proposition 8.2 when field amplitudes are held fixed.
18. **Strong potential convergence does not imply stress convergence.** The exact sequence in Proposition 18.3 retains finite quadratic stress.
19. **Equal Hadamard singularities do not fix quantum stress.** Different smooth state remainders survive coincident differentiation.
20. **The ratio p/u does not determine dimension without its hypotheses.** Anisotropy or nonradiative stress can alter it in a fixed spatial dimension.

22. What is proved, assumed, interpreted and open

Status	Content and limit
Retained results	Common calibration criterion, relative defect, bounded-energy reconstruction comparison, spectral error control, scalar unitary continuum limit with its zero mode, and corrected compact recurrence from [36]. Long proofs are not repeated.
Proved here within the control assumptions	Physical Maxwell characteristic quotient; null tensor component closure; stationary and moving boundary momentum balance; isotropic pressure identity; calibrated L^2-to-L^1 stress estimate; the exact oscillatory counterexample; conditional continuity of point-split quantum stress. These include standard identities and elementary proofs, not newly discovered physical laws.
Physical anchors	Maxwell action and constitutive vacuum sector; canonical photon quantisation; relativistic total four-momentum; Einstein coupling when invoked. Existing radiation-pressure metrology anchors an observable channel, not a new experiment performed here.
Assumptions	Effective manifold, retained sector content, gauge reduction, well-posedness, positive normalisation, stated geometric-optics regime, chosen observers and states, boundary laws, and stronger convergence where quadratic or quantum observables require it.
Interpretation	Distinction belongs to a relational carrier; phenomenal presence is not identified with independent substantial self-being. These claims are not proved by the physical calculations.
Open problems	Selection and stability of an electromagnetic sector from neutral relational dynamics; intersector cone universality under interactions; physical clocks and records; stress-preserving microscopic limits with boundaries; geometric scale and gravitational dynamics.

Hypothesis 22.1 (Stable joint functional reconstruction). A specified class of microscopic dynamics and states may admit reconstructions whose physical responses, symplectic structures and calibrated stress observables converge jointly, with one limiting null calibration across the retained physical sectors.

This statement is stronger than either cone convergence or the closure of one assumed Maxwell sector. Symmetry can protect two electromagnetic polarisations without enforcing equal speeds for inequivalent matter representations. Nor does the hypothesis establish attraction to a universality manifold. Any attraction claim must identify the coarse-grained, statistical or dissipative evolution to which it applies and respect the finite conservative recurrence limitations retained from [36].

Open Problem 22.2 (The next technical theorem). For one explicitly specified constrained microscopic electromagnetic model coupled to a force-measuring boundary, prove a joint finite-time response and stress limit with controlled gauge reduction, no unresolved quadratic energy defect and convergent boundary traces. Show that two independently specified admissible reconstructions give the same timing/pressure closure residual. If quantum local stress is claimed, additionally prove the required state and renormalisation convergence. No equality of the desired pressure or cone may be inserted into the reconstruction definition.

This is a concrete next interface theorem. A more fundamental theorem would also have to select such a constrained model, a stable local sector and common calibration for inequivalent physical sectors from neutral relational data, with stability under a stated class of microscopic perturbations. This paper does not claim that selection.

23. Conclusion

A substantial subset of light observables is constrained by a single effective Maxwell functional, through distinct variations, symmetry constructions and observer projections. In a single null sector those operations close propagation, energy flux, momentum flux and longitudinal stress into one tensor structure. Pressure tests that structure through interaction with an independently characterised boundary; isotropic radiation and composite invariant mass provide different controlled consequences.

The closure has precise limits. Polarisation needs gauge and representation data, quantum energies need quantisation, correlations need a state, pressure needs a boundary, and gravity needs an additional action. Reconstruction of quadratic observables requires more than preservation of a cone or strong convergence of potentials. The calibrated field-energy estimate and its counterexample locate this extra requirement explicitly. The resulting programme concerns existence and preservation of a joint functional structure between descriptions, with its assumptions exposed.

Appendix A. Proof of the Spectral Bound

By the min–max principle, the first r eigenvalues of $L = L_0 + E$ are at most ε: it suffices to restrict the quadratic form of L to the r-dimensional $\ker L_0$. Since $E \geq 0$, the remaining eigenvalues are no smaller than the corresponding eigenvalues of L_0 and are therefore at least γ. For $\varepsilon < \gamma/2$ the spectral split is unambiguous.

Let F contain orthonormal eigenvectors spanning the low subspace of L, so that $LF = F\Lambda$ and $0 \leq \Lambda \leq \varepsilon I$. On the orthogonal complement of $\ker L_0$ define $B_0 = (I - P_0)L_0(I - P_0)$, restricted to that complement. Then $B_0 \geq \gamma I$. The matrix $K = (I - P_0)F$ satisfies the Sylvester equation

$$B_0K - K\Lambda = -(I - P_0)EF. \tag{A1}$$

Spectral separation gives the integral representation

$$K = -\int_0^\infty \exp(-tB_0)(I - P_0)EF \exp(t\Lambda)dt. \tag{A2}$$

It can be verified by direct differentiation of the integrand. Therefore

$$\|K\| \leq \varepsilon \int_0^\infty \exp[-(\gamma - \varepsilon)t]dt = \varepsilon/(\gamma - \varepsilon). \tag{A3}$$

For orthogonal projectors of the same finite rank, $\|P - P_0\| = \|(I - P_0)F\|$, which gives Eq. (12). The estimate uses the operator norm. It does not require a separate choice of eigenvector orientation and remains meaningful when the low levels are degenerate. □

Appendix B. Second-Branch Coefficient and Infrared Elimination

In the two-component model, the normalized matrix T has orthonormal eigenvectors

$$z_0 = (\sqrt{a_1},\sqrt{a_2})^T/\sqrt{(a_1 + a_2)}, \quad z_1 = (\sqrt{a_2},-\sqrt{a_1})^T/\sqrt{(a_1 + a_2)}. \tag{B1}$$

The first has eigenvalue 0 and the second eigenvalue g_0^2 from Eq. (25). For $D = \mathrm{diag}(b_1/a_1,b_2/a_2)$,

$$z_0^TDz_0 = (b_1 + b_2)/(a_1 + a_2), \tag{B2}$$

$$z_1^TDz_0 = (a_2b_1 - a_1b_2)/[\sqrt{(a_1a_2)}(a_1 + a_2)]. \tag{B3}$$

Second-order perturbation theory gives $-|z_1^{\mathrm{T}} D z_0|^2 \nu^2 / g_0^2$. Substitution of Eqs. (25) and (B3) yields K in Eq. (27). Equation (37) is the multicomponent analogue of the same calculation. For $r > 1$, D_{00} is diagonalized before applying a nondegenerate correction; within a degenerate sector, the compressed second-order matrix must be diagonalized.

Eliminating gapped coordinates is an infrared change of description. The exact inverse block in Eq. (36) depends on ω and therefore corresponds, in general, to an effective kernel that is nonlocal in time. A local two-derivative action arises only as the beginning of an expansion in ν/g^2 and ω^2/g^2. It cannot be used to determine signal fronts, which require control over arbitrarily high frequencies.

Appendix C. Finite algebraic controls and reproducibility

No simulation or numerical fit is needed for the arguments. The source contains the standard-library Python script reproduce_light_closure.py in a filecontents* block. LaTeX can extract it on compilation; it can also be copied directly from that block and run with Python 3. It compares field-tensor pressure with packet momentum bookkeeping, checks polarisation-independent null stress, calibrated field rescaling, moving-mirror energy balance, two-photon invariant mass, and the oscillatory field-energy example.

The antipodal directions $\{\pm e_i\}_{i=1}^{d}$, each weighted $1/(2d)$, integrate angular second moments exactly. They are a finite quadrature for this moment, not a rotationally invariant gas at all higher moments. The following table consists of analytic values; the embedded code provides small independent algebraic checks.

Control	Result
Normal absorption; normal ideal reflection	$cP/I = 1;\ cP/I = 2.$
Absorption $1/5$, reflection $1/2$, transmission $3/10$ at $\theta = \pi/3$	$cP/I = (1/5+1)/4 = 3/10.$
Receding mirror, $\beta = 1/5$	$q_\beta = 2/3;\ cP/I = 4/3.$
Isotropic second moment in $d = 2, 3, 4, 5$	$p/u = 1/2, 1/3, 1/4, 1/5.$
Two equal-energy photons, $\vartheta = 0, \pi/2, \pi$	$Mc^2/E = 0, \sqrt{2}, 2.$
$A_y = \sin n(z-t)/n$ over a full phase period, $c = \gamma = 1$	$\langle A_y^2 \rangle = 1/(2n^2);\ \langle u \rangle = \langle \Pi_{zz} \rangle = 1/2.$
$F_2 = F_1/\sqrt{7},\ \gamma_2 = 7\gamma_1$	$T_2 = T_1$ component by component.

These calculations are neither experimental evidence nor substitutes for the propositions. All graphs have been replaced by informative tables so the manuscript has no external graphical dependencies.

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